

Unimodality of point-target first-passage laws on the reflecting box: theorems, counter-examples, certified computations, and what remains open

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Companion note R4

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Note. This document is the companion note R4 of the article “Mean and most probable first-passage times of lazy random walks in reflecting hypercubic lattices” ([paper.pdf](#), Section 7; Supplementary Material, Section S8), in the repository <https://github.com/zhouyi-xiaoxiao/master-dissertation>. In the repository, its scripts are in `code/msc_rigorous_unimodality/` and its data in `data/msc_rigorous_unimodality/`; paths written `scripts/...` and `data/...` below refer to these folders; the paths in the `% src` comments of the L^AT_EX source are relative to the root of the repository, and the run logs they cite are in `data/msc_rigorous_*/logs/`. Other run logs, the code of the internal checks and working notes mentioned below are not part of the repository; `notes/VERIFICATION.md` summarises how this note was checked.

Status labels: a *Theorem*, *Lemma*, *Proposition* or *Corollary* has a complete proof here (standard results are cited with author, title and theorem number). A *Computer-assisted Theorem*, *Lemma* or *Proposition* is proved by a finite computation together with lemmas proved here that turn the finite computation into a statement about the whole infinite support (and, in Section 9, about all N). In Sections 7–8 the computation is in exact integer or rational arithmetic; in Section 9 it is in certified ball arithmetic (Arb: every number is a ball that contains the true value, and every decision is an inequality between balls), cross-checked against exact integer arithmetic. No floating-point comparison enters any proof. A *Conjecture* is not proved; the evidence is stated. A *Numerical observation* is a reproducible computation with no claim beyond the range computed. Scripts and data are in the directory of this file (`scripts/`, `data/`); see Appendix A.

Answer in one page

Question. Is the first-passage PMF $f(t)$ (discrete time) or density $g(t)$ (continuous time) of the lazy walk on $\{1, \dots, N\}^d$ with reflecting (move-cancelling) walls and one absorbing site unimodal?

Answer. For the three geometries of the article — corner $\rightarrow$ opposite corner, corner $\rightarrow$ centre, centre $\rightarrow$ corner — *yes, for every N* : in continuous time in every dimension (Theorem 4.12), and in discrete time for every activity $q \leq 4/5$ — the value is $q = 4/5$ — in every dimension $d \geq 2$, and in $d = 1$ for the corner pair and for corner $\rightarrow$ centre (Computer-assisted Theorem 9.13). The bound $4/5$ is sharp for the corner pair: for $d = 1$, $N = 4$ the PMF is unimodal iff $q \leq 4/5$. The restriction $d \geq 2$ for centre $\rightarrow$ corner is necessary: in $d = 1$ that PMF is unimodal at $q = 4/5$ if and only if $N \notin \{5, 7, 9, 11\}$ (Proposition 7.3, Computer-assisted Theorem 7.4). In discrete time with q close to 1 the answer is *no* in general ($d = 1$, $d = 2$), and for arbitrary start/target pairs in $d \geq 2$ the answer is *no* even in continuous time. So the naive statement “the point-target first-passage law is unimodal” is false, and the true statement depends on the pair, on the dimension and on q .

1. *Three geometries, all N , all d* (Theorem 4.12). Corner $\rightarrow$ opposite corner (any $N \geq 2$), corner $\rightarrow$ centre and centre $\rightarrow$ corner (odd N): the continuous-time density is unimodal with a unique mode, and the PMF is unimodal for every $q \leq q_N = 1/(1 + \cos(\pi/N))$, in particular for every $q \leq 1/2$. For the hypercube $N = 2$ this is every $q \in (0, 1]$.
2. *The value* (Theorem 9.13, computer-assisted; Theorems 9.7, 9.11, Lemma 9.10, Proposition 9.12). For every $q \leq 4/5$: the corner $\rightarrow$ opposite corner PMF is unimodal for *every* $N \geq 2$ and *every* $d \geq 1$; the corner $\rightarrow$ centre and centre $\rightarrow$ corner PMFs are unimodal for *every odd* $N \geq 3$ and *every* $d \geq 2$. $4/5$ is sharp in the first statement ($d = 1, N = 4$). Mechanism: the increment $u(t) - u(t - 1)$ of the free propagator is log-concave; this is reduced to the one-dimensional walk, where it is proved for all N by splitting the spectrum into “twisted groups” of 16–17 eigenvalues from a compact two-parameter family that is certified in ball arithmetic.
3. *One-dimensional log-concavity* (Theorem 9.11, Corollary 9.8, computer-assisted; Proposition 9.6). The end-to-end sequence $K_q^{n+1}(1, N) - K_q^n(1, N)$ is log-concave for every $N \geq 9$ when $q \leq 4/5$; for $4 \leq N \leq 501$ it is log-concave *iff* $q \leq q_1(N) = 1/(1 + (2(N - 1))^{-1/2})$ (so the PMF is unimodal for $q \leq q_1(N)$ in every dimension; $q_1(N) \rightarrow 1$). Explicit tail: at $q = 4/5$ the log-concavity inequalities hold for all $n \geq 1 + (N^2/8)(10 \ln N - 8.43)$, every $N \geq 10$.
4. *Criterion for an arbitrary pair*, indeed for any finite Markov chain (Lemma 4.7, Theorem 4.11): if the time-derivative u' of the *free* propagator $u(t) = \mathbb{P}_{x_0}(X_t = a)$ is positive and log-concave, the first-passage density is unimodal; discrete analogue with $u(t) - u(t - 1)$. It rests on the last-exit decomposition $g = u' + r * u'$ with a nonincreasing kernel r .
5. *Location of the mode* for these geometries (Corollary 4.13): $m_u < t^* \leq m(q\alpha_1) \leq m(q/M_{\max})$, with m_u the maximiser of u' and $m(\cdot)$ defined by an explicit one-dimensional equation; numerically the upper bound exceeds the mode by 4.7–24.9% for the corner pair in the tabulated cases ($d = 2, 3, 8 \leq N \leq 64$); for smaller N and in $d = 1$ it is cruder (a factor 1.26–1.94 in the cases computed).
6. *Laziness monotonicity* (Theorem 2.5, Corollary 2.6). For a fixed pair, the number of sign changes of $t \mapsto f_q(t + 1) - f_q(t)$ is nonincreasing as q decreases, and g' has at most as many sign changes as the difference sequence of any discrete PMF. Hence $\{q \in (0, 1] : f_q \text{ unimodal}\}$ is empty or an interval $(0, q^*]$, and if it is non-empty the continuous-time density is unimodal.
7. *Descartes’ rule on the residues cannot work* (Theorem 3.2). If the graph distance between start and target is D , the residue sequence of the spectral expansion has at least $D - 1$ sign changes; in 1D end-to-end it alternates strictly.
8. $d = 1$, *every start and target*: unimodal in continuous time and for $q \leq 1/2$ (Theorem 4.3). $d = 1$ *is not unimodal for q near 1*: for $N \geq 4$ and $q > (2N - 4)/(2N - 3)$, $f(N - 1) > f(N) < f(N + 1)$; for $N = 4$ the threshold $4/5$ is exact (Proposition 4.4, hand proof). Centre $\rightarrow$ corner in $d = 1$, odd $N \geq 7$, $M = (N - 1)/2$: $f(M) > f(M + 1) < f(M + 2)$ for every $q > (N - 3)/(N - 1)$; for $N = 3$ the PMF is unimodal *iff* $q \leq 4/5$ (Proposition 7.3). *Exact thresholds in $d = 1$* (Computer-assisted Theorem 7.4, exact integer arithmetic): end to end, $q^*(N) = (2N - 4)/(2N - 3)$ for $N = 4, N = 20$ and every $22 \leq N \leq 300$, and $q^*(N) < (2N - 4)/(2N - 3)$ for $5 \leq N \leq 19$ and for $N = 21$ (six-digit brackets); centre $\rightarrow$ corner, $q^*(5) = (4 + \sqrt{2})/7$, $q^*(N) = (N - 3)/(N - 1)$ for $N \in \{7, 9, 17, 21\}$ and every odd $23 \leq N \leq 301$, strictly smaller for $N \in \{11, 13, 15, 19\}$.
9. *Completely monotone cases* (Theorem 4.1): stationary start, or start adjacent to a corner or centre target ($q \leq 1/2$ or continuous time). A single start at distance $D \geq 2$ is never a nonnegative mixture of geometric (exponential) laws.

10. *Corner pair, explicit windows valid for every $q \in (0, 1]$* : f is strictly decreasing for $t > T_N = O(N^2 \log N/q)$, explicit (Theorem 5.6); the continuous-time density is nondecreasing on $[0, t_N]$, $t_N \geq d(N-2)/(2q)$ (Theorem 5.10). For $q \leq 1/2$ and in continuous time these are superseded by item 1.
11. *Corner-to-corner, finite range, q beyond $4/5$* (Computer-assisted Theorem 7.1): unimodal for $d = 2$, $2 \leq N \leq 64$, every $q \in (0, 0.998]$; for $d = 3$, $2 \leq N \leq 30$, every $q \in (0, 1]$; for $d = 4$, $N \leq 8$ and $d = 5$, $N \leq 5$, every $q \in (0, 1]$.
12. *Corner-to-corner, $d = 2$, $q = 1$ is not always unimodal*: exactly for $N \in \{9, 11, 13, 15, 17, 18, 21, 24, 26, 38, 51, 61\}$ among $2 \leq N \leq 64$ the PMF has a one-step dip at its peak; $0.9986 \leq q^*(9) < 0.9987$.
13. *Corner $\rightarrow$ centre and centre $\rightarrow$ corner, exact cone certificates* (odd N): unimodal for every $q \leq 4/5$ for $d = 2$, $N \leq 61$ and $d = 3$, $N \leq 27$ (see also item 2 except for $N = 3$, $d \in \{2, 3\}$ and $N \in \{5, 7\}$, $d = 2$, where item 2 relies on these certificates). The centre $\rightarrow$ corner law is not log-concave (Proposition 8.4).
14. *General pairs in $d \geq 2$: unimodality is false*, in continuous time and for every q (Theorem 8.1); exhaustive exact classification of all pairs for $d = 2$, $N \leq 12$ and $d = 3$, $N \leq 6$ (Computer-assisted Theorem 8.2).
15. *Open* (Section 10). (a) Discrete time with $4/5 < q < 1$ for *all* N : for the corner pair it would follow from the conjecture that the one-dimensional sequence of item 3 is log-concave iff $q \leq q_1(N)$ for *every* N (Conjecture 10.4; proved for $N \leq 501$). (b) Log-concavity (monotone hazard rate) of the corner-to-corner first-passage law itself (Conjecture 10.1). (c) The one-dimensional thresholds for all N beyond the certified range (Conjecture 10.2).

1 Setting, notation, two elementary lemmas

1.1 The walk

Let $d \geq 1$, $N \geq 2$, $\Lambda = \{1, \dots, N\}^d$, $n = N^d$. (The code uses 0-based coordinates.) P is the transition matrix of the walk with $q = 1$:

$$P(x, x \pm e_i) = \frac{1}{2d} \text{ if } x \pm e_i \in \Lambda, \quad P(x, x) = \frac{b(x)}{2d},$$

where $b(x)$ is the number of the $2d$ directions that leave Λ at x . P is symmetric and stochastic. For $q \in (0, 1]$, $P_q = (1 - q)I + qP$; put $L = I - P$, so $P_q = I - qL$.

Fix a target $a \in \Lambda$ and a start $x_0 \neq a$; $\Lambda' = \Lambda \setminus \{a\}$. A is the restriction of L to $\Lambda' \times \Lambda'$ and $Q_q = I - qA$ the restriction of P_q . Put $r := A\mathbf{1}$, i.e. $r(x) = P(x, a)$, and $D := |x_0 - a|_1$.

Discrete time. $T = \min\{t \geq 1 : X_t = a\}$, $S_q(t) = \mathbb{P}(T > t) = e_{x_0}^\top Q_q^t \mathbf{1}$ and

$$f_q(t) = S_q(t-1) - S_q(t) = q e_{x_0}^\top Q_q^{t-1} r \quad (t \geq 1), \quad f_q(0) := 0. \quad (1)$$

We write $a_q(t) := f_q(t+1)$, $t \geq 0$, and $\Delta a(t) := a(t+1) - a(t)$.

Continuous time. The walk with generator $-qL$ (each direction at rate $q/(2d)$; cancelled moves are no-ops) has first-passage density

$$g_q(t) = q e_{x_0}^\top e^{-qtA} r, \quad g_q(t) = q g_1(qt). \quad (2)$$

A is symmetric positive definite: for h on Λ' extended by $\tilde{h}(a) = 0$, $\langle h, Ah \rangle = \langle \tilde{h}, L\tilde{h} \rangle = \frac{1}{2d} \sum_{\{u,v\}} (\tilde{h}(u) - \tilde{h}(v))^2$, which vanishes only for $h = 0$.

Lemma 1.1 (spectrum of the box). *For $k \in \{0, \dots, N-1\}^d$ let $\Psi_k(x) = \prod_i \psi_{k_i}(x_i)$, $\psi_k(j) = c_k \cos(k\pi(j - \frac{1}{2})/N)$, $c_0 = N^{-1/2}$, $c_k = (2/N)^{1/2}$ ($k \geq 1$). Then $\{\Psi_k\}$ is an orthonormal eigenbasis of L , $L\Psi_k = \theta(k)\Psi_k$, $\theta(k) = \frac{1}{d} \sum_i (1 - \cos(k_i\pi/N))$. The spectrum of L lies in $[0, 1 + \cos(\pi/N)] \subset [0, 2]$; its smallest non-zero eigenvalue is $\beta_1 := \frac{1}{d}(1 - \cos(\pi/N)) \geq 2/(dN^2)$.*

Proof. For $d = 1$, $(P\psi)(j) = \frac{1}{2}\psi(j-1) + \frac{1}{2}\psi(j+1)$ with the conventions $\psi(0) := \psi(1)$, $\psi(N+1) := \psi(N)$ (a cancelled move is a hold). The function $j \mapsto \cos(k\pi(j - \frac{1}{2})/N)$ satisfies both conventions, so $P\psi_k = \cos(k\pi/N)\psi_k$ by the addition formula; the ψ_k are the DCT-II vectors, an orthonormal basis of $\mathbb{R}^N$. For general d , $P = \frac{1}{d} \sum_i P^{(i)}$ with $P^{(i)}$ acting on coordinate i . Finally $1 - \cos x = 2 \sin^2(x/2) \geq 2x^2/\pi^2$ on $[0, \pi]$. $\square$

1.2 Sign changes and unimodality

For a real sequence c , $S^-(c)$ is the number of sign changes after deleting zeros; for an infinite sequence it is the supremum over finite truncations. A sequence $(a(t))_{t \geq 0}$ is *unimodal* if $a(0) \leq \dots \leq a(m) \geq a(m+1) \geq \dots$ for some m ; a C^1 function g on $[0, \infty)$ is unimodal if $g' \geq 0$ on $(0, m)$ and $g' \leq 0$ on (m, ∞) for some $m \geq 0$. (Since $f_q(0) = 0 \leq f_q(1)$, unimodality of $(f_q(t))_{t \geq 0}$ and of $(a_q(t))_{t \geq 0}$ are the same.)

Lemma 1.2. *Let $a \geq 0$ be summable and not identically zero. Then a is unimodal iff $S^-(\Delta a) \leq 1$. The same holds for a nonnegative integrable C^1 function with “ g' changes sign at most once”.*

Proof. If a is unimodal the nonzero entries of Δa are $+\dots+ - \dots -$. Conversely, if the last sign of Δa were $+$, a would be eventually nondecreasing and positive, contradicting summability; so the nonzero signs are $(+\dots+)(-\dots-)$ with either block possibly empty. $\square$

2 Monotonicity in the laziness; continuous time is the weakest case

2.1 Variation-diminishing lemmas (self-contained)

Lemma 2.1 (bidiagonal matrices). *Let B be an $m \times m$ nonnegative upper bidiagonal matrix, $(Bx)_i = \alpha_i x_i + \beta_i x_{i+1}$ ($x_{m+1} := 0$), or a nonnegative lower bidiagonal one, $(Bx)_i = \alpha_i x_i + \beta_i x_{i-1}$ ($x_0 := 0$). Then for every $x \in \mathbb{R}^m$:*

1. $S^-(Bx) \leq S^-(x)$;
2. if $S^-(Bx) = S^-(x)$ and $Bx \neq 0$, the first nonzero entries of Bx and of x have the same sign, and so have the last nonzero entries.

Proof. Upper case; put $y = Bx$ and $z = (x_1, y_1, x_2, y_2, \dots, x_m, y_m)$. *Claim:* $S^-(z) = S^-(x)$, and the first (last) nonzero entry of z has the sign of the first (last) nonzero entry of x . Remove the y_i from z one at a time. If $y_i = 0$ nothing changes. If $y_i \neq 0$, then, being a nonnegative combination of x_i and x_{i+1} , it has the sign of a nonzero one of them; that entry is adjacent to y_i in z , so deleting y_i does not change the number of sign changes. Hence $S^-(z) = S^-(x)$. If the first nonzero entry of z is some y_i , then $x_1 = \dots = x_i = 0$ and $y_i = \beta_i x_{i+1}$; if the last nonzero entry of z is y_i , then $x_{i+1} = \dots = x_m = 0$ and $y_i = \alpha_i x_i$. This proves the claim.

Since y is a subsequence of z , $S^-(y) \leq S^-(z) = S^-(x)$. If the first nonzero entry of y had the sign opposite to the first nonzero entry of z , then z would have one sign change before that position and at least $S^-(y)$ after it, so $S^-(z) \geq S^-(y) + 1$; this is impossible when $S^-(y) = S^-(z)$. The last entries are treated in the same way, and the lower bidiagonal case follows by reversing the order of the indices. $\square$

Lemma 2.2 (products and limits). *Properties (1), (2) of Lemma 2.1 hold for every finite product M of nonnegative bidiagonal matrices and for every entrywise limit $M = \lim_j M_j$ of such products.*

Proof. For a product, chain the inequalities; if the two ends are equal and $Mx \neq 0$, every intermediate vector is nonzero and every inequality is an equality, so (2) transfers step by step. For a limit put $y_j = M_j x \rightarrow y = Mx$. For large j every nonzero entry of y has the same sign in y_j , so $S^-(y) \leq S^-(y_j) \leq S^-(x)$. If $S^-(y) = S^-(x)$ and $y \neq 0$, then for large j , $S^-(y_j) = S^-(x)$ and the first nonzero entry of y_j has the sign σ of the first nonzero entry of x . Let i_0 be the first index with $y(i_0) \neq 0$. If $\text{sgn } y(i_0) = -\sigma$, then for large j the vector y_j has an entry of sign σ before i_0 , the sign $-\sigma$ at i_0 , and from i_0 on its nonzero entries include those of y with the same signs, so $S^-(y_j) \geq 1 + S^-(y)$: a contradiction. The last entries are symmetric. $\square$

Remark. Lemmas 2.1–2.2 are the elementary core of the variation-diminishing theorem for totally nonnegative matrices (I. J. Schoenberg, *Über variationsvermindernde lineare Transformationen*, Math. Z. 32 (1930) 321–328; S. Karlin, *Total Positivity*, Vol. I, Stanford Univ. Press 1968, Ch. 5, Theorem 3.1). Only matrices with an explicit bidiagonal factorisation are needed below, so the general theorem is not used.

Lemma 2.3 (binomial matrix). *For $p \in (0, 1]$ and $M \geq 1$ let $b_p(t, n) = \binom{t}{n} p^n (1-p)^{t-n}$, $0 \leq t, n < M$. The matrix $(b_p(t, n))$ has properties (1), (2) of Lemma 2.1.*

Proof. For $p < 1$, $(b_p) = D_1 \Pi D_2$ with positive diagonal $D_1 = \text{diag}((1-p)^t)$, $D_2 = \text{diag}((p/(1-p))^n)$ and $\Pi = ((\binom{t}{n}))$. Let H have entries $H(t, t-1) = t$ and zeros elsewhere. Then $\exp(H)(t, n) = H^{t-n}(t, n)/(t-n)! = t(t-1) \cdots (n+1)/(t-n)! = \binom{t}{n}$, so $\Pi = \exp(H) = \lim_j (I + H/j)^j$, and each $I + H/j$ is lower bidiagonal and nonnegative. Apply Lemma 2.2. $\square$

Lemma 2.4 (Descartes–Laguerre rule for power series). *Let $F(\mu) = \sum_{n \geq 0} c_n \mu^n$ converge for all $\mu > 0$, with real c_n not all zero and $S^-(c) = s < \infty$. Then F has at most s zeros in $(0, \infty)$, counted with multiplicity.*

Proof. Induction on s ; $s = 0$ is clear. Let the first sign change occur between indices $j < k$. Fix a real $\kappa \in (j, k)$ and put $G(\mu) = \mu^{-\kappa} F(\mu)$; G has the same positive zeros as F . The power series $\mu^{\kappa+1} G'(\mu) = \sum_n (n - \kappa) c_n \mu^n$ has coefficients of the sign of $-c_n$ for $n \leq j$ and of c_n for $n \geq k$: the sign change between j and k has disappeared and no other has been created. By induction it has at most $s - 1$ positive zeros, and by Rolle's theorem G has at most s . $\square$

(Classical: G. Pólya and G. Szegő, *Problems and Theorems in Analysis II*, Part V, Chapter 1.)

2.2 The thinning theorem

Theorem 2.5 (laziness monotonicity). *Fix d, N, a, x_0 .*

(a) *For $0 < q' \leq q \leq 1$ and $p = q'/q$, for all $t \geq 0$,*

$$a_{q'}(t) = p \sum_{n=0}^t b_p(t, n) a_q(n), \quad \Delta a_{q'}(t) = p^2 \sum_{n=0}^t b_p(t, n) \Delta a_q(n). \quad (3)$$

Consequently $S^-(\Delta a_{q'}) \leq S^-(\Delta a_q)$.

(b) *For every $q_0 \in (0, 1]$, every rate $q > 0$ and $s \geq 0$, with $\mu = qs/q_0$,*

$$g_q(s) = \frac{q}{q_0} \sum_{n \geq 0} e^{-\mu} \frac{\mu^n}{n!} a_{q_0}(n), \quad g'_q(s) = \left(\frac{q}{q_0}\right)^2 e^{-\mu} \sum_{n \geq 0} \frac{\mu^n}{n!} \Delta a_{q_0}(n). \quad (4)$$

Consequently g'_q has at most $S^-(\Delta a_{q_0})$ zeros in $(0, \infty)$, counted with multiplicity.

Proof. (a) $Q_{q'} = (1 - p)I + pQ_q$, so $Q_{q'}^t = \sum_n b_p(t, n)Q_q^n$ and, by (1), $a_{q'}(t) = q'e_{x_0}^T Q_{q'}^t r = p \sum_n b_p(t, n)a_q(n)$. From $b_p(t + 1, n) = (1 - p)b_p(t, n) + p b_p(t, n - 1)$,

$$a_{q'}(t + 1) - a_{q'}(t) = p \sum_n p [b_p(t, n - 1) - b_p(t, n)]a_q(n) = p^2 \sum_n b_p(t, n)[a_q(n + 1) - a_q(n)].$$

The matrix is lower triangular, so the first M entries of $\Delta a_{q'}$ are p^2 times the leading $M \times M$ block applied to the first M entries of Δa_q ; apply Lemma 2.3 and let $M \rightarrow \infty$.

(b) $-qsA = \mu(Q_{q_0} - I)$, so $e^{-qsA} = e^{-\mu} \sum_n \frac{\mu^n}{n!} Q_{q_0}^n$, which with (2) gives the first formula. As a_{q_0} is bounded, the series may be differentiated termwise, $\frac{d}{d\mu}[e^{-\mu} \sum_n \frac{\mu^n}{n!} a(n)] = e^{-\mu} \sum_n \frac{\mu^n}{n!} \Delta a(n)$. Apply Lemma 2.4 to $\sum_n \Delta a_{q_0}(n)\mu^n/n!$. $\square$

Corollary 2.6. *Let $U = \{q \in (0, 1] : (f_q(t))_{t \geq 0} \text{ is unimodal}\}$.*

1. *U is empty or an interval $(0, q^*]$, $q^* \in (0, 1]$.*
2. *If $U \neq \emptyset$, the continuous-time density g_q is unimodal for every rate $q > 0$.*
3. *If the continuous-time density is not unimodal, the PMF is not unimodal for any $q \in (0, 1]$.*
4. *The number of local maxima of f_q is nonincreasing as q decreases.*

Proof. By Lemma 1.2, $q \in U$ iff $S^-(\Delta a_q) \leq 1$; by Theorem 2.5(a), U is closed downwards. $q \notin U$ means that $\Delta a_q(t_1), \Delta a_q(t_2), \Delta a_q(t_3)$ have strictly alternating signs for some $t_1 < t_2 < t_3$; each is a polynomial in q , so the complement of U is open. This gives (1); (2) is Theorem 2.5(b) with Lemma 1.2; (3) is its contrapositive; (4): the nonzero entries of Δa_q end with $-$ (proof of Lemma 1.2). Count the local maxima of a_q as the number of changes $+\rightarrow-$ in the sequence of nonzero entries of Δa_q , plus one if the first nonzero entry is $-$ (a maximum at the left end; this happens only when $x_0 \sim a$). This number is $(S^-(\Delta a_q) + 1)/2$ if the first nonzero entry is $+$ and $S^-(\Delta a_q)/2 + 1$ if it is $-$; in both cases it equals $\lceil (S^-(\Delta a_q) + 1)/2 \rceil$, a nondecreasing function of $S^-(\Delta a_q)$. Apply Theorem 2.5(a). $\square$

The corollary does *not* say that continuous-time unimodality implies discrete unimodality for small q .

3 Spectral representation; why Descartes' rule on the residues is hopeless

Let $\alpha_1 < \dots < \alpha_m$ be the distinct eigenvalues of A and E_j the spectral projections. (For $d = 1$ with an interior target replace Λ' by the connected component Λ'_0 of x_0 .) Then

$$f_q(t) = q \sum_j C_j (1 - q\alpha_j)^{t-1}, \quad g_q(t) = q \sum_j C_j e^{-q\alpha_j t}, \quad C_j := e_{x_0}^T E_j r = \alpha_j e_{x_0}^T E_j \mathbf{1}. \quad (5)$$

Lemma 3.1. *On the component Λ'_0 of x_0 , $P' := I - A$ is symmetric, nonnegative, irreducible and has a positive diagonal entry; consequently the power P'^m , $m := 2(|\Lambda'_0| - 1)$, is entrywise positive. The largest eigenvalue $1 - \alpha_1$ of P' is simple with a positive eigenvector φ_1 ($\|\varphi_1\|_2 = 1$), $C_1 = \alpha_1 \varphi_1(x_0) \langle \varphi_1, \mathbf{1} \rangle > 0$, and every other eigenvalue λ of Q_q satisfies $|\lambda| < 1 - q\alpha_1$, for every $q \in (0, 1]$.*

Proof. Every component of Λ' contains a site w with $b(w) \geq 1$ (for $d \geq 2$, Λ' is connected and contains a wall site; for $d = 1$ each component contains an endpoint), i.e. $P'(w, w) = b(w)/(2d) > 0$. Let $n' := |\Lambda'_0|$. If $n' = 1$, P' is a positive number and there is nothing to prove; let $n' \geq 2$, $m = 2(n' - 1)$.

Positivity of $B := P'^m$. For $x, y \in \Lambda'_0$ there are nearest-neighbour paths in Λ'_0 from x to w and from w to y , each of length at most $n' - 1$; inserting holds at w between them gives a walk from x to y of length exactly m all of whose steps have positive weight under P' . Hence $B(x, y) > 0$.

Perron's theorem for the symmetric positive matrix B (the symmetric case of Perron's theorem; R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge 2013, Section 8.2; the short proof is included to keep the note self-contained). Let ρ_B be the largest eigenvalue of B and v a unit vector with $\langle v, Bv \rangle = \rho_B$. Then $\rho_B \geq \langle |v|, B|v| \rangle \geq \langle v, Bv \rangle = \rho_B$, so $|v|$ also maximises the Rayleigh quotient and is an eigenvector: $B|v| = \rho_B|v|$; as $B > 0$ and $|v| \neq 0$ the left side is entrywise positive, so $|v| > 0$. Equality $\sum B(x, y)|v(x)||v(y)| = \sum B(x, y)v(x)v(y)$ with all $B(x, y) > 0$ forces $v(x)v(y) = |v(x)||v(y)|$ for all x, y , i.e. $v = \pm|v|$. So every eigenvector of B for ρ_B is strictly of one sign; two orthogonal vectors cannot both be of one strict sign, so the eigenspace is one-dimensional.

Back to P' . P' is symmetric; let $\rho := \max |\text{spec } P'|$. Since m is even, the eigenvalues of B are the numbers $\lambda'^m \geq 0$, so $\rho_B = \rho^m$, and every eigenvector of P' for an eigenvalue $\lambda' = \pm\rho$ is an eigenvector of B for ρ_B . Therefore the eigenvalues of modulus ρ have, together, a one-dimensional eigenspace, spanned by a positive vector φ_1 , and $\langle \varphi_1, P'\varphi_1 \rangle > 0$ shows that the eigenvalue is $+\rho$. So $\rho = 1 - \alpha_1$ is a simple eigenvalue of P' with eigenvector $\varphi_1 > 0$, and every other eigenvalue λ' of P' satisfies $|\lambda'| < 1 - \alpha_1$. $E_1 = \varphi_1 \varphi_1^\top$ and $r = A\mathbf{1}$ give $C_1 = e_{x_0}^\top E_1 r = \alpha_1 \varphi_1(x_0) \langle \varphi_1, \mathbf{1} \rangle > 0$. Finally $Q_q = (1 - q)I + qP'$ has the same eigenvectors, with eigenvalues $1 - q + q\lambda'$, and for $\lambda' \neq 1 - \alpha_1$: $|1 - q + q\lambda'| \leq (1 - q) + q|\lambda'| < (1 - q) + q(1 - \alpha_1) = 1 - q\alpha_1$, for every $q \in (0, 1]$. $\square$

Theorem 3.2 (vanishing moments; lower bound on the sign changes). 1. $\sum_j C_j \alpha_j^i = 0$ for $0 \leq i \leq D - 2$, and $(-1)^{D-1} \sum_j C_j \alpha_j^{D-1} = f_1(D) = G(x_0, a)(2d)^{-D} > 0$, where $G(x_0, a)$ is the number of shortest lattice paths from x_0 to a .

2. $S^-(C_1, \dots, C_m) \geq D - 1$.

3. For $d = 1$, $x_0 = 1$, $a = N$: $m = N - 1 = D$, and $\text{sgn } C_j = (-1)^{j-1}$ for all j .

Proof. (1) $\sum_j C_j \alpha_j^i = e_{x_0}^\top A^i r = \sum_{k \leq i} \binom{i}{k} (-1)^k e_{x_0}^\top P'^k r$, and $e_{x_0}^\top P'^k r = f_1(k + 1)$ is the probability that the $q = 1$ walk first hits a at step $k + 1$; it vanishes for $k + 1 < D$ and equals $G(x_0, a)(2d)^{-D}$ for $k + 1 = D$.

(2) Suppose $S^-(C) = s \leq D - 2$. Choose $\xi_1 < \dots < \xi_s$, none equal to an α_j , separating the maximal blocks of constant sign of the nonzero C_j , and $\pi(\alpha) = \pm \prod_l (\alpha - \xi_l)$ with the sign such that $\pi(\alpha_j) C_j \geq 0$ for all j . Some $C_j \neq 0$ and no α_j is a root, so $\sum_j C_j \pi(\alpha_j) > 0$; but $\deg \pi \leq D - 2$, so by (1) the sum is 0.

(3) In 1D the walk lives on $\{1, \dots, N - 1\}$, A is an irreducible Jacobi matrix with $N - 1$ simple eigenvalues, $D = N - 1 = m$, and (2) gives $S^-(C) \geq m - 1$, the maximum for m numbers; $C_1 > 0$. $\square$

Consequence. The generalised Descartes rule for exponential sums bounds the number of positive zeros of $g'_q(t) = -q^2 \sum_j \alpha_j C_j e^{-q\alpha_j t}$ by $S^-(C) \geq D - 1$. It can therefore prove unimodality only for $D \leq 2$, and in the best understood case (1D end-to-end, where the density is unimodal) it returns the worst possible bound $N - 2$.

Numerical observation 3.3. (`scripts/residue_signs.py`, `mpmath`, 60 digits.) Corner-to-corner: $S^-(C) = D - 1$ for $d = 2$, $N = 2, \dots, 6$ and $d = 3$, $N = 2, 3$; but $S^-(C) = 17 > D - 1 = 11$ for $d = 2$, $N = 7$ and $12 > 8$ for $d = 3$, $N = 4$. In every corner-to-corner case $C_2 < 0$; for centre $\rightarrow$ corner the pattern starts $+++---$ ($C_2 > 0$).

4 Cases with complete proofs

4.1 Completely monotone cases

A sequence $\sum_j w_j \lambda_j^{t-1}$ with $w_j \geq 0$, $\lambda_j \in [0, 1)$ is completely monotone, in particular nonincreasing; likewise $\sum_j w_j e^{-\sigma_j t}$.

Theorem 4.1. *Let $N(a)$ be the set of neighbours of a .*

- (a) *Uniform start on Λ' : $f_q^{\text{unif}}(t) = \frac{q}{n-1} \sum_j \alpha_j \|E_j \mathbf{1}\|^2 (1 - q\alpha_j)^{t-1}$ is completely monotone for $q \leq 1/2$; the continuous-time density is completely monotone for every q .*
- (b) *Uniform start on $N(a)$: $f_q^{N(a)}(t) = \frac{2dq}{|N(a)|} \sum_j \|E_j r\|^2 (1 - q\alpha_j)^{t-1}$; same conclusions.*
- (c) *If a group of symmetries of the box fixes a and acts transitively on $N(a)$ (e.g. a a corner, or N odd and a the centre), then $f_q^y = f_q^{N(a)}$ for every $y \in N(a)$. For $d = 1$, f^y is completely monotone ($q \leq 1/2$; in continuous time for every q) for every $y \sim a$.*
- (d) *For a single start with $D \geq 2$ the law is not a nonnegative mixture of geometric (exponential) laws, with any nodes whatsoever: there is no positive Borel measure μ on $[0, 1)$ with $f_q(t) = \int \lambda^{t-1} \mu(d\lambda)$ for all $t \geq 1$ ($0^0 := 1$), and no positive Borel measure μ on $[0, \infty)$ with $g_q(t) = \int e^{-\sigma t} \mu(d\sigma)$ for all $t > 0$. In particular, in the spectral representation (5) some C_j is negative.*

Proof. (a) $e_{\text{unif}} = \mathbf{1}/(n-1)$ and $r = A\mathbf{1}$, so $e_{\text{unif}}^\top E_j r = \alpha_j \|E_j \mathbf{1}\|^2 / (n-1) \geq 0$; the eigenvalues of A lie in $(0, 2)$ (Lemma 1.1; A is a principal submatrix of L), so $1 - q\alpha_j \in [0, 1)$ for $q \leq 1/2$. (b) The uniform law on $N(a)$ is $\frac{2d}{|N(a)|} r$. (c) A symmetry γ fixing a commutes with Q_q and permutes $N(a)$, so $f^{\gamma y} = f^y$; average over the orbit. In $d = 1$ the walk from $y = a - 1$ cannot reach $a + 1$ before a , so $f^y(t) = \frac{q}{2} Q_q^{t-1}(y, y)$. (d) $D \geq 2$ means $r(x_0) = P(x_0, a) = 0$, so by (1), (2): $f_q(1) = q r(x_0) = 0$ and $g_q(0) = q r(x_0) = 0$. On the other hand $f_q(D) \geq (q/(2d))^D > 0$ (follow a shortest path), and $g_q(t) > 0$ for $t > 0$ (by (4), $g_q(t)$ is a Poisson mixture of the numbers $a_{q_0}(n) \geq 0$, and $a_{q_0}(D-1) = f_{q_0}(D) > 0$). If $f_q(t) = \int \lambda^{t-1} \mu(d\lambda)$ with $\mu \geq 0$, then $\mu([0, 1)) = f_q(1) = 0$, so $\mu = 0$ and $f_q \equiv 0$, contradicting $f_q(D) > 0$. If $g_q(t) = \int e^{-\sigma t} \mu(d\sigma)$ for $t > 0$ with $\mu \geq 0$, then by monotone convergence $\mu([0, \infty)) = \lim_{t \downarrow 0} g_q(t) = g_q(0) = 0$, so $\mu = 0$ and $g_q \equiv 0$, a contradiction. (In words: a nonnegative mixture is nonincreasing, whereas $f_q(1) = 0 < f_q(D)$ and $g_q(0) = 0 < g_q(t)$. The argument does not use uniqueness of the representing measure, and it excludes mixtures with arbitrary nodes, not only the spectral ones.) For the last assertion: by Theorem 3.2(2), $S^-(C) \geq D-1 \geq 1$, so the C_j are not all of one sign. (Directly: $\sum_j C_j = e_{x_0}^\top r = 0$ by Theorem 3.2(1) with $i = 0$, and $C_1 > 0$ by Lemma 3.1.) $\square$

(Background: J. Keilson, *Markov Chain Models — Rarity and Exponentiality*, Springer 1979; D. Aldous and J. A. Fill, *Reversible Markov Chains and Random Walks on Graphs*, Ch. 3. A negative spectral residue would not by itself exclude a nonnegative representation with other nodes; the argument above does not rely on one. Sanity check: `verify_thr1d.py` Z9.)

Numerical observation 4.2. In the exhaustive floating-point scans ($d = 2$, $N \leq 10$; $d = 3$, $N \leq 5$; continuous time) the density is monotone decreasing exactly for the pairs with $x_0 \sim a$, including neighbours not covered by Theorem 4.1(c).

4.2 One dimension: every start, every target

Theorem 4.3. *Let $d = 1$, $N \geq 2$, $x_0 \neq a$ arbitrary. (i) For every $q \in (0, 1/2]$ the PMF f_q is unimodal. (ii) For every rate $q > 0$ the continuous-time density g_q is unimodal.*

Proof. By reflection assume $x_0 < a$; the walk stays in $\{1, \dots, a-1\}$ until absorption, and $Q = Q_q$ is the Jacobi matrix with $Q(i, i) = \delta_i$, $\delta_1 = 1 - q/2$, $\delta_i = 1 - q$ ($i \geq 2$), $Q(i, i \pm 1) = q/2$.

Step 1. Put $u_1 = \delta_1$, $u_i = \delta_i - (q/2)^2/u_{i-1}$. By induction $u_i \geq q/2$ when $q \leq 1/2$: $u_1 \geq q/2$, and $u_{i-1} \geq q/2$ implies $u_i \geq 1 - q - q/2 \geq q/2$. Hence $Q = \mathcal{L}\mathcal{U}$ with $\mathcal{L}$ unit lower bidiagonal with subdiagonal $(q/2)/u_{i-1} > 0$ and $\mathcal{U}$ upper bidiagonal with diagonal $u_i > 0$ and superdiagonal $q/2$. So Lemmas 2.1–2.2 apply to Q_q ($q \leq 1/2$) and, since $e^{-sA} = \lim_j Q_{s/j}^j$, to e^{-sA} for every $s \geq 0$.

Step 2. Let $\rho_t = Q^t e_{x_0}$ and $\Delta_t := \rho_{t+1} - \rho_t = Q^t \Delta_0$, $\Delta_0 = -qAe_{x_0}$, which has a negative entry at x_0 , the entry $q/2$ at $x_0 \pm 1$ (when in $\{1, \dots, a-1\}$) and zeros elsewhere. Since $f_q(t+1) = \frac{q}{2}\rho_t(a-1)$,

$$\Delta a_q(t) = \frac{q}{2} \Delta_t(a-1). \quad (6)$$

$\Delta_t \neq 0$ (Q is invertible) and $\sum_x \Delta_t(x) = -f_q(t+1) \leq 0$; so if $S^-(\Delta_t) = 0$, all nonzero entries of Δ_t are negative.

Step 3. By Step 1, $t \mapsto S^-(\Delta_t)$ is nonincreasing and, on any stretch of times on which it is constant, the signs of the first and last nonzero entries of Δ_t are constant. Let ℓ_t be the sign of the last nonzero entry of Δ_t ; $\Delta_t(a-1)$ is 0 or has the sign ℓ_t . If $1 < x_0 < a-1$: Δ_0 has the pattern $(+, -, +)$; $\ell_t = +$ while $S^-(\Delta_t) = 2$, ℓ_t is constant while $S^-(\Delta_t) = 1$, and $\ell_t = -$ once $S^-(\Delta_t) = 0$. If $x_0 = 1 < a-1$: pattern $(-, +)$, $\ell_t = +$ while $S^- = 1$, then $\ell_t = -$. If $x_0 = a-1$: pattern $(+, -)$ or $(-)$, $\ell_t = -$ always. In every case ℓ_t is $+$ up to some time and $-$ afterwards; by (6) and Lemma 1.2, (i) follows. (ii) is the same argument with $\rho_s = e^{-qsA}e_{x_0}$, $\Delta_s = \frac{d}{ds}\rho_s = e^{-qsA}\Delta_0$, $g'_q(s) = \frac{q}{2}\Delta_s(a-1)$, $\sum_x \Delta_s(x) = -g_q(s)$. $\square$

Remark. (1) Part (ii) is known: U. Rösler, *Unimodality of passage times for one-dimensional strong Markov processes*, Ann. Probab. 8 (1980) 853–859; J. Keilson, *On the unimodality of passage time densities in birth–death processes*, Statist. Neerlandica 35 (1981) 49–55. For $x_0 = 1$ and $q \leq 1/2$, T is a sum of independent geometric variables (J. A. Fill, J. Theoret. Probab. 22 (2009) 543–557), so f is log-concave. (2) The proof uses that the heat flow on a path does not increase the number of sign changes; there is no such statement on $\{1, \dots, N\}^d$, $d \geq 2$. (3) The hypothesis $q \leq 1/2$ cannot be dropped (next proposition).

Proposition 4.4. *Let $d = 1$, $x_0 = 1$, $a = N \geq 4$, $\eta := 1 - q$, $\eta_1 := 1 - q/2$. Then*

$$\begin{aligned} f_q(N-1) &= (q/2)^{N-1}, & f_q(N) &= (q/2)^{N-1} [\eta_1 + (N-2)\eta], \\ f_q(N+1) &= (q/2)^{N-1} \left[\eta_1^2 + (N-2)\eta_1\eta + \binom{N-1}{2}\eta^2 + (N-2)\frac{q^2}{4} \right]. \end{aligned}$$

For every $q \in (\frac{2N-4}{2N-3}, 1]$, $f_q(N-1) > f_q(N) < f_q(N+1)$: the PMF is not unimodal. For $N = 4$ the PMF is unimodal iff $q \leq 4/5$.

Proof. A path first reaching N at time $N-1+j$ consists of $N-1$ right steps and j extra steps. $j = 1$: one hold, at site 1 (probability η_1) or at one of the sites $2, \dots, N-1$ (probability η). $j = 2$: two holds (total weight $h_2(\eta_1, \eta, \dots, \eta)$, the complete homogeneous polynomial) or one left step from a site $k \in \{2, \dots, N-1\}$ and the extra right step ($N-2$ paths with the extra factor $(q/2)^2$). $f_q(N) < f_q(N-1)$ iff $\eta(2N-3) < 1$. With $\eta_1 = (1+\eta)/2$ let $\Phi(\eta) := \eta_1^2 + (N-2)\eta_1\eta + \binom{N-1}{2}\eta^2 + (N-2)(1-\eta)^2/4 - \eta_1 - (N-2)\eta$. $\Phi'(\eta) = \eta/2 + (N-2)(3\eta/2 - 1) + (N-1)(N-2)\eta$ is increasing, so on $[0, 1/(2N-3)]$, $\Phi' \leq [\frac{1}{2} + (N-2)(\frac{7}{2} - N)]/(2N-3) < 0$, and $\Phi(\eta) \geq \Phi(1/(2N-3)) = [N^3 - \frac{15}{2}N^2 + \frac{35}{2}N - 13]/(2N-3)^2 > 0$ for $N \geq 4$ (the cubic equals 1 at $N = 4$ and is increasing). For $N = 4$, as a function of q , $\Phi = \frac{19}{4}q^2 - \frac{15}{2}q + 3 > 0$ for all q , so f_q is not unimodal for $q > 4/5$. *Unimodality at $q = 4/5$, by hand.* On $\{1, 2, 3\}$,

$$5Q_{4/5} = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix}, \quad R_t := (5Q_{4/5})^t e_1 :$$

$$\begin{aligned} R_0 &= (1, 0, 0), & R_1 &= (3, 2, 0), & R_2 &= (13, 8, 4), \\ R_3 &= (55, 42, 20), & R_4 &= (249, 192, 104), & R_5 &= (1131, 898, 488). \end{aligned}$$

Since $f(t+1) = \frac{q}{2}\rho_t(3) = 2R_t(3)/5^{t+1}$: $f(1) = f(2) = 0$, $f(3) = f(4) = 8/125$, $f(5) = 208/3125 > f(4)$, $f(6) = 976/15625 < f(5)$. Moreover $R_5 \leq 5R_4 = (1245, 960, 520)$ componentwise, i.e. $\rho_5 \leq \rho_4$; as $Q \geq 0$ entrywise this propagates, $\rho_{t+1} \leq \rho_t$ for all $t \geq 4$ (the cone certificate, Lemma 6.1, whose two-line proof uses nothing else), so $f(t+2) \leq f(t+1)$ for all $t \geq 4$. Hence $f_{4/5}$ is unimodal with mode 5, and Corollary 2.6 extends this to all $q \leq 4/5$. (Computer-assisted Theorems 7.1(f) and 7.4(a) reproduce it.) $\square$

Example ($N = 4$, $q = 1$): $f(3) = 1/8 > f(4) = 1/16 < f(5) = 3/32$.

Numerical observation 4.5. (Float bisection, valid by Corollary 2.6; see also the exact Computer-assisted Theorem 7.4(a).) The 1D end-to-end threshold is $q^*(4) = 0.8$, $q^*(5) \approx 0.8354$, $q^*(6) \approx 0.8719$, $q^*(8) \approx 0.8987$, $q^*(10) \approx 0.9209$, $q^*(16) \approx 0.9630$, and $q^*(N) = (2N-4)/(2N-3)$ to nine digits for $N = 20, 25, 30, 40$. *The extrapolation “ $q^*(N) = (2N-4)/(2N-3)$ for all $N \geq 20$ ” that these four values suggest is false: $N = 21$ is an exception* (Theorem 7.4(a)).

4.3 The hypercube $N = 2$, every dimension

Proposition 4.6. *Let $N = 2$, $d \geq 1$, $x_0 = (1, \dots, 1)$, $a = (2, \dots, 2)$. The PMF is unimodal for every $q \in (0, 2/3]$ and the continuous-time density is unimodal for every rate.*

Proof. Let $k(x)$ be the number of coordinates of x equal to 2. From a site at level k the walk moves to level $k+1$ with probability $(d-k)q/(2d)$, to level $k-1$ with probability $kq/(2d)$, and otherwise stays; so $k(X_t)$ is a birth–death chain on $\{0, \dots, d\}$ started at 0, and T is its hitting time of d . Its restriction $\tilde{Q}$ to $\{0, \dots, d-1\}$ is a Jacobi matrix with diagonal $1 - q/2$ and $\tilde{Q}(k, k-1)\tilde{Q}(k-1, k) = k(d-k+1)q^2/(4d^2) \leq q^2/4$. The pivots $u_0 = 1 - q/2$, $u_k = 1 - q/2 - \tilde{Q}(k, k-1)\tilde{Q}(k-1, k)/u_{k-1}$ satisfy $u_k \geq q/2$ when $q \leq 2/3$, so $\tilde{Q} = \mathcal{L}\mathcal{U}$ and $\tilde{Q}^\top = \mathcal{U}^\top\mathcal{L}^\top$ are products of nonnegative bidiagonal matrices. With $\rho_t = (\tilde{Q}^\top)^t e_0$, $f_q(t+1) = \frac{q}{2d}\rho_t(d-1)$ and $\Delta_0 = (\tilde{Q}^\top - I)e_0 = (-q/2, +q/2, 0, \dots)$; Steps 2–3 of the proof of Theorem 4.3 apply verbatim. $\square$

(Theorem 4.12(ii) below gives more: for $N = 2$ the corner-to-corner PMF is unimodal for every $q \in (0, 1]$ and every d . Proposition 4.6 is kept because its proof — lumping to a birth–death chain — is independent.)

4.4 The three geometries for every N and every d : last-exit decomposition and strong unimodality

This subsection proves, for *all* N and *all* d , that the corner $\rightarrow$ opposite-corner law and (for odd N) the corner $\rightarrow$ centre and centre $\rightarrow$ corner laws are unimodal, in continuous time and in discrete time for $q \leq 1/2$. Two ingredients are used: a *last-exit decomposition* (Lemma 4.7), which writes the first-passage law as the convolution of the time-derivative of the *free* (unkilled) propagator $u(t) = \mathbb{P}_{x_0}(X_t = a)$ with a *nonincreasing* kernel; and the elementary half of Ibragimov’s theorem on strong unimodality (Lemma 4.8): the convolution of a log-concave function with a nonincreasing kernel on $[0, \infty)$ is unimodal. For the three geometries $u = h^d$, where h is an end-to-end propagator of a birth–death chain, hence N^{-1} times the distribution function of a sum of independent exponential (geometric) variables; this makes u' log-concave.

Notation for this subsection. X is the *unkilled* walk on Λ . In discrete time (activity q): $u(t) := P_q^t(x_0, a)$ for $t \geq 0$, $u(-1) := 0$, $\varphi(t) := u(t) - u(t-1)$; $E(k) := \mathbb{P}_a(X_1 \neq a, \dots, X_k \neq a)$, $E(0) = 1$; $\nu(k) := E(k) - E(k+1)$, the probability that the first return to a occurs at time $k+1$. In continuous time (rate q): $u(t) := (e^{-qtL})(x_0, a)$,

$$r(\tau) := q r^\top e^{-q\tau A} \mathbf{1} = q \sum_{y \sim a} P(a, y) \mathbb{P}_y(T_a > \tau), \quad \nu(\tau) := -r'(\tau) = q^2 r^\top e^{-q\tau A} r.$$

Lemma 4.7 (last-exit decomposition). *Let $x_0 \neq a$ be arbitrary.*

(a) Discrete time, any $q \in (0, 1]$. $1 = E(0) \geq E(1) \geq \dots \geq 0$, and for every $t \geq 0$

$$\mathbb{P}_{x_0}(T \leq t) = \sum_{s=0}^t u(s)E(t-s), \quad f_q(t) = \sum_{k=0}^t E(k)\varphi(t-k), \quad (7)$$

$$f_q(t+1) - f_q(t) = \varphi(t+1) - \sum_{k=0}^t \nu(k)\varphi(t-k). \quad (8)$$

(b) Continuous time, any rate $q > 0$. r is C^∞ , positive and nonincreasing, $\nu \geq 0$, $r(0) = q r^\top \mathbf{1}$, and for every $t \geq 0$

$$\mathbb{P}_{x_0}(T \leq t) = u(t) + \int_0^t u(s)r(t-s) ds, \quad g_q = u' + r * u', \quad (9)$$

$$g'_q(t) = u''(t) + r(0)u'(t) - \int_0^t \nu(\tau)u'(t-\tau) d\tau. \quad (10)$$

Proof. (a) E is nonincreasing because the events decrease. Under $\mathbb{P}_{x_0}$, on $\{T \leq t\}$ let $\sigma := \max\{s \leq t : X_s = a\}$. For $0 \leq s \leq t$, $\{\sigma = s\} = \{X_s = a\} \cap \{X_{s+1} \neq a, \dots, X_t \neq a\}$, and the Markov property at time s gives $\mathbb{P}_{x_0}(\sigma = s) = u(s)E(t-s)$. The events $\{\sigma = s\}$ partition $\{T \leq t\}$ ($s=0$ contributes $u(0)=0$); this is the first identity. Subtracting the identity at $t-1$ from that at t , $f_q(t) = \sum_{k=0}^t E(k)u(t-k) - \sum_{k=0}^{t-1} E(k)u(t-1-k) = \sum_{k=0}^t E(k)\varphi(t-k)$ (the term $k=t$ is $E(t)(u(0)-u(-1))=0$). Subtracting once more, $f_q(t+1) - f_q(t) = E(0)\varphi(t+1) + \sum_{k=0}^t (E(k+1) - E(k))\varphi(t-k)$, which is (8).

(b) Order the sites as (Λ', a) ; then $L = \begin{pmatrix} A & -r \\ -r^\top & c_a \end{pmatrix}$ with $c_a := L(a, a) = r^\top \mathbf{1}$ (the row sums of L vanish). For $s > 0$ let $\hat{u}(s) = [(s+qL)^{-1}](x_0, a)$, $\hat{v}(s) = [(s+qL)^{-1}](a, a)$, $\hat{g}(s) = q e_{x_0}^\top (s+qA)^{-1} r$ and $\hat{r}(s) = q r^\top (s+qA)^{-1} \mathbf{1}$; these are the Laplace transforms of u , of $v(t) := (e^{-qtL})(a, a)$, of g_q (by (2)) and of r . The block-inverse (Schur complement) formula gives, with $\sigma(s) := s + qc_a - q^2 r^\top (s+qA)^{-1} r$,

$$\hat{v}(s) = \frac{1}{\sigma(s)}, \quad \hat{u}(s) = \frac{\hat{g}(s)}{\sigma(s)}.$$

Since $r = A\mathbf{1}$, $q(s+qA)^{-1}r = (s+qA)^{-1}((s+qA)-s)\mathbf{1} = \mathbf{1} - s(s+qA)^{-1}\mathbf{1}$, hence $q^2 r^\top (s+qA)^{-1}r = qc_a - s\hat{r}(s)$ and

$$\sigma(s) = s(1 + \hat{r}(s)), \quad \hat{g}(s) = s\hat{u}(s)(1 + \hat{r}(s)).$$

Because $u(0) = 0$, $s\hat{u}(s)$ is the Laplace transform of u' ; so $\hat{g}$ is the transform of $u' + r * u'$, and uniqueness of the Laplace transform (both sides are continuous and bounded) gives $g_q = u' + r * u'$. Integrating over $[0, t]$ gives the first identity of (9). Differentiating $(r * u')(t) = \int_0^t r(\tau)u'(t-\tau)d\tau$ gives $r(t)u'(0) + \int_0^t r(\tau)u''(t-\tau)d\tau$, and an integration by parts turns the integral into $-r(t)u'(0) + r(0)u'(t) + \int_0^t r'(\tau)u'(t-\tau)d\tau$; this is (10). Finally $e^{-q\tau A} = e^{-q\tau} e^{q\tau P'}$ with $P' = I - A \geq 0$ entrywise, so $e^{-q\tau A}$ is entrywise nonnegative; hence $r > 0$, $\nu = q^2 r^\top e^{-q\tau A} r \geq 0$ (using $A\mathbf{1} = r$ once more), and r is nonincreasing. $\square$

Remark. (1) The first identity in (9) is the continuous-time form of (7): the walker has reached a by time t iff it is at a at time t or it left a for the last time at some $s < t$; $r(t-s)$ is the rate of leaving a times the probability of not returning within $t-s$. (2) Nothing in Lemma 4.7 uses the box, the symmetry of P , or the corner: it holds for every finite Markov chain and every pair of states (in (b) with $r(\tau) := \sum_y q(a, y)\mathbb{P}_y(T_a > \tau)$). (3) (8) is the companion of the renewal equation of Proposition 5.11: there the kernel acts on f , here on φ .

A sequence $(\varphi(t))_{t \in \mathbb{Z}}$ is called *log-concave* if $\varphi \geq 0$, its support $\{t : \varphi(t) > 0\}$ is a set of consecutive integers, and $\varphi(t)^2 \geq \varphi(t-1)\varphi(t+1)$ for all t . For such a sequence

$$\varphi(i-1)\varphi(j+1) \leq \varphi(i)\varphi(j) \quad \text{for all integers } i \leq j. \quad (11)$$

(If the left side is zero there is nothing to prove; otherwise $\varphi > 0$ on $\{i-1, \dots, j+1\}$ and $\varphi(l+1)/\varphi(l)$ is nonincreasing there, so $\varphi(j+1)/\varphi(j) \leq \varphi(i)/\varphi(i-1)$.)

Lemma 4.8 (a log-concave function convolved with a nonincreasing kernel is unimodal). (a) Continuous. Let $\varphi \in C^1([0, \infty))$ with $\varphi > 0$ and $\log \varphi$ concave on $(0, \infty)$, and let $r \in C^1([0, \infty))$ be nonnegative and nonincreasing, $\nu := -r'$. Put $G := \varphi + r * \varphi$. Then for $t > 0$

$$G'(t) = \varphi(t) B(t), \quad B(t) := \frac{\varphi'(t)}{\varphi(t)} + r(0) - \int_0^t \nu(\tau) \frac{\varphi(t-\tau)}{\varphi(t)} d\tau, \quad (12)$$

and B is nonincreasing on $(0, \infty)$. Consequently there is $t^* \in [0, \infty]$ with $G' \geq 0$ on $(0, t^*)$ and $G' \leq 0$ on (t^*, ∞) .

(b) Discrete. Let φ be a log-concave sequence with $\varphi(t) = 0$ for $t < 0$, let $1 = \kappa(0) \geq \kappa(1) \geq \dots \geq 0$, $\nu(k) := \kappa(k) - \kappa(k+1)$, and $F(t) := \sum_{k=0}^t \kappa(k)\varphi(t-k)$. Then $F(t+1) - F(t) = \varphi(t+1) - \sum_{k=0}^t \nu(k)\varphi(t-k)$. This vanishes when $t+1$ lies to the left of the support of φ ; as long as $\varphi(t+1) > 0$,

$$F(t+1) - F(t) = \varphi(t+1) \beta(t), \quad \beta(t) := 1 - \sum_{k=0}^t \nu(k) \frac{\varphi(t-k)}{\varphi(t+1)},$$

with β nonincreasing; and if the support of φ ends at $t_1 < \infty$, then $F(t+1) - F(t) \leq 0$ for $t \geq t_1$. Consequently $S^-(\Delta F) \leq 1$, with sign pattern $(+\dots+)(-\dots-)$.

Proof. (a) The computation leading to (10) gives $G' = \varphi' + r(0)\varphi - \nu * \varphi$; divide by $\varphi(t) > 0$. Let $0 < t_1 < t_2$. $\varphi'/\varphi = (\log \varphi)'$ is nonincreasing. For $0 \leq \tau \leq t_1$: $\varphi(t_1 - \tau)\varphi(t_2) \leq \varphi(t_1)\varphi(t_2 - \tau)$ — trivially if $\varphi(t_1 - \tau) = 0$, and otherwise because the increment of the concave function $\log \varphi$ over an interval of length τ is nonincreasing in the position of the interval. Hence

$$\int_0^{t_1} \nu(\tau) \frac{\varphi(t_1 - \tau)}{\varphi(t_1)} d\tau \leq \int_0^{t_1} \nu(\tau) \frac{\varphi(t_2 - \tau)}{\varphi(t_2)} d\tau \leq \int_0^{t_2} \nu(\tau) \frac{\varphi(t_2 - \tau)}{\varphi(t_2)} d\tau,$$

so $B(t_2) \leq B(t_1)$. Take $t^* := \sup\{t : B(t) \geq 0\}$ ($:= 0$ if the set is empty).

(b) The formula for $F(t+1) - F(t)$ is the computation in the proof of Lemma 4.7(a). If $\varphi(t+1) > 0$ and $\varphi(t+2) > 0$, then for each fixed $k \geq 0$, (11) with $i = t+1-k \leq j = t+1$ gives $\varphi(t-k)\varphi(t+2) \leq \varphi(t+1-k)\varphi(t+1)$, i.e. $\varphi(t-k)/\varphi(t+1) \leq \varphi(t+1-k)/\varphi(t+2)$; the number of terms of the sum grows with t ; so $\beta(t+1) \leq \beta(t)$. To the right of the support, $F(t+1) - F(t) = -\sum_k \nu(k)\varphi(t-k) \leq 0$. So the nonzero entries of ΔF are $(+\dots+)(-\dots-)$. $\square$

Remark. Background: I. A. Ibragimov, *On the composition of unimodal distributions*, Teor. Veroyatnost. i Primenen. 1 (1956) 283–288 [Theory Probab. Appl. 1 (1956) 255–260] (a distribution is strongly unimodal iff it is log-concave); for lattice distributions J. Keilson and H. Gerber, *Some results for discrete unimodality*, J. Amer. Statist. Assoc. 66 (1971) 386–389. Only the direction proved above is used, for a kernel with an atom at 0 and a nonincreasing density on $(0, \infty)$.

Lemma 4.9 (end-to-end propagator of a birth–death chain). (a) Continuous time. Let G be the generator of a birth–death process on $\{0, \dots, M\}$, $M \geq 1$, with birth rates $\lambda_0, \dots, \lambda_{M-1} > 0$, death rates $\mu_1, \dots, \mu_M > 0$ and stationary distribution π . Then $-G$ has $M+1$ distinct real eigenvalues $0 = \vartheta_0 < \vartheta_1 < \dots < \vartheta_M$, and

$$p_t(0, M) = \pi_M \mathbb{P}(\xi_1 + \dots + \xi_M \leq t), \quad \xi_k \sim \text{Exp}(\vartheta_k) \text{ independent.}$$

(b) Discrete time. Let K be the transition matrix of a birth-death chain on $\{0, \dots, M\}$ with $K(i, i+1) = p_i > 0$, $K(i, i-1) = q_i > 0$ and stationary distribution π . Its eigenvalues are real and distinct, $1 = \lambda_0 > \lambda_1 > \dots > \lambda_M \geq -1$. If $\lambda_M \geq 0$, then

$$K^n(0, M) = \pi_M \mathbb{P}(G_1 + \dots + G_M \leq n), \quad \mathbb{P}(G_k = j) = (1 - \lambda_k) \lambda_k^{j-1} \quad (j \geq 1), \text{ independent} \\ (G_k \equiv 1 \text{ if } \lambda_k = 0).$$

Proof. G and K are tridiagonal with positive off-diagonal products, hence similar (by $\text{diag}(\pi)^{1/2}$) to symmetric Jacobi matrices with non-zero off-diagonal entries; their eigenvalues are real and simple, and the eigenvalue 0 (resp. 1) has the constant eigenvector. (a) By Cramer's rule $[(s - G)^{-1}](0, M) = (-1)^M \det(\hat{G}) / \det(s - G)$, where $\hat{G}$ is $s - G$ with row M and column 0 deleted. $\hat{G}$ is lower triangular with diagonal entries $-\lambda_0, \dots, -\lambda_{M-1}$, so $[(s - G)^{-1}](0, M) = \prod_i \lambda_i / (s \prod_{k=1}^M (s + \vartheta_k))$. The spectral decomposition of e^{tG} in $\ell^2(\pi)$ shows that the residue of $[(s - G)^{-1}](0, M)$ at $s = 0$ is π_M ; hence $\prod_i \lambda_i = \pi_M \prod_k \vartheta_k$ and $[(s - G)^{-1}](0, M) = \pi_M \frac{1}{s} \prod_k \frac{\vartheta_k}{s + \vartheta_k}$, the Laplace transform of $\pi_M \mathbb{P}(\sum \xi_k \leq t)$. (b) In the same way $\sum_n K^n(0, M) z^n = [(I - zK)^{-1}](0, M) = z^M \prod_i p_i / ((1 - z) \prod_{k=1}^M (1 - \lambda_k z))$, the residue at $z = 1$ gives $\prod_i p_i = \pi_M \prod_k (1 - \lambda_k)$, and $(1 - \lambda)z / (1 - \lambda z)$ is the generating function of the geometric law above when $\lambda \in [0, 1]$; the factor $1/(1 - z)$ forms partial sums. $\square$

(For $d = 1$ end-to-end, (a) is Lemma 5.9 below. The statement is classical: S. Karlin and J. McGregor, *Coincidence properties of birth and death processes*, Pacific J. Math. 9 (1959) 1109–1140; J. Keilson, *Markov Chain Models — Rarity and Exponentiality*, Springer 1979, §5.1.)

Lemma 4.10 (operations preserving log-concavity). (a) Exponential and geometric kernels, primitives and partial sums.

- (i) Let φ be continuous on $[0, \infty)$, positive with $\log \varphi$ concave on $(0, \infty)$, and $\vartheta \geq 0$. Then $\psi(t) := \int_0^t e^{-\vartheta(t-s)} \varphi(s) ds$ is C^1 , positive and log-concave on $(0, \infty)$.
- (ii) Let φ be a log-concave sequence vanishing for $n < n_0$, $\lambda \in [0, 1]$, $c > 0$. Then $\psi(n) := c \sum_{j \geq 1} \lambda^{j-1} \varphi(n - j)$ ($0^0 := 1$) is a log-concave sequence.
- (iii) Consequently the density of a sum of $M \geq 1$ independent exponential variables is positive and log-concave on $(0, \infty)$, and so is its distribution function; the probability mass function of a sum of M independent geometric variables as in Lemma 4.9(b) is a log-concave sequence, and so is its distribution function.

(b) Binomial convolution (Davenport–Pólya). Let $a = (a_k)_{k \geq 0}$ and $b = (b_k)_{k \geq 0}$ be log-concave sequences (extended by zero to negative indices). Then

$$(a \star b)_n := \sum_{k=0}^n \binom{n}{k} a_k b_{n-k}$$

is a log-concave sequence, with support $\text{supp } a + \text{supp } b$.

Proof. (a)(i) $\psi' = \varphi - \vartheta \psi$ and $\psi > 0$ on $(0, \infty)$. $\psi(t)/\varphi(t) = \int_0^t e^{-\vartheta(t-\tau)} \varphi(t-\tau)/\varphi(t) d\tau$ is nondecreasing in t (as in the proof of Lemma 4.8(a)), so $(\log \psi)' = \varphi'/\psi - \vartheta$ is nonincreasing. (ii) If $\lambda = 0$, ψ is a shift of $c\varphi$. Let $\lambda > 0$; then $\psi(n) > 0$ iff $n \geq n_0 + 1$, and $\psi(n+1) = \lambda \psi(n) + c\varphi(n)$. For $n \geq n_0 + 1$ with $\varphi(n) > 0$, $\psi(n)/\varphi(n) = c \sum_{j \geq 1} \lambda^{j-1} \varphi(n-j)/\varphi(n)$ is nondecreasing in n by (11) (with $i = n+1-j$ and n in the rôle of j); if the support of φ ends, $\varphi(n)/\psi(n)$ is 0 from then on. So $\psi(n+1)/\psi(n) = \lambda + c\varphi(n)/\psi(n)$ is nonincreasing for $n \geq n_0 + 1$, which is log-concavity. (iii) $\vartheta_1 e^{-\vartheta_1 t}$ is log-linear; convolving successively with $\vartheta_k e^{-\vartheta_k t}$ and finally with $\vartheta = 0$ (the primitive) preserves positivity and log-concavity by (i). In the discrete case start from the unit mass at 0, convolve with the geometric laws ($c = 1 - \lambda_k$, $\lambda = \lambda_k$) and finally with $\lambda = c = 1$ (shifted partial sums), using (ii).

(b) Write $K_n(k) := \binom{n}{k} b_{n-k}$, so $(a \star b)_n = \sum_k K_n(k) a_k$, and let $(Sa)_k := a_{k+1}$. Pascal's rule gives $(a \star b)_{n+1} = (Sa \star b)_n + (a \star Sb)_n$. Put $c := a \star b$. Then

$$c_{n+1}c_{n-1} - c_n^2 = [(Sa \star b)_n c_{n-1} - (Sa \star b)_{n-1} c_n] + [(a \star Sb)_n c_{n-1} - (a \star Sb)_{n-1} c_n].$$

The first bracket equals $\sum_{k,l} a_{k+1} a_l M(k,l)$ with $M(k,l) := K_n(k)K_{n-1}(l) - K_{n-1}(k)K_n(l) = -M(l,k)$, i.e. $\sum_{k < l} (a_{k+1} a_l - a_k a_{l+1}) M(k,l)$. For $k < l$: $a_{k+1} a_l - a_k a_{l+1} \geq 0$ by (11); and $M(k,l) \leq 0$, because $\binom{n}{k} \binom{n-1}{l} \leq \binom{n-1}{k} \binom{n}{l}$ (for $l \leq n-1$ this is $n/(n-k) \leq n/(n-l)$; for $l \geq n$ the left side is 0) and $b_{n-k} b_{n-1-l} \leq b_{n-1-k} b_{n-l}$ (by (11) with $i = n-l \leq j = n-1-k$). So the first bracket is ≤ 0 , and the second one by symmetry. The statement about supports is clear. $\square$

(H. Davenport and G. Pólya, *On the product of two power series*, Canad. J. Math. 1 (1949) 1–5, for positive sequences; the proof above allows leading zeros.)

Theorem 4.11 (criterion). *Let $x_0 \neq a$ be arbitrary.*

(a) Continuous time. *If $u'(t) = \frac{d}{dt} \mathbb{P}_{x_0}(X_t = a) > 0$ for all $t > 0$ and $\log u'$ is concave on $(0, \infty)$, then the first-passage density g_q is unimodal for every rate: $g'_q = u' \cdot B$ with B as in (12) nonincreasing, and there is a unique $t^* \in [0, \infty)$ with $g'_q > 0$ on $(0, t^*)$ and $g'_q < 0$ on (t^*, ∞) .*

(b) Discrete time. *If for some $q \in (0, 1]$ the sequence $\varphi(t) = P_q^t(x_0, a) - P_q^{t-1}(x_0, a)$ is log-concave, then the PMF f_q is unimodal.*

Proof. (a) Lemma 4.7(b) and Lemma 4.8(a) with $\varphi := u'$. g'_q is a finite sum of exponentials, not identically zero, so its zeros are isolated; since B is nonincreasing, $g'_q = u' \cdot B$ has at most one sign change, from $+$ to $-$, and vanishes at most at one point; $t^* < \infty$ because g_q is integrable. (b) Lemma 4.7(a) and Lemma 4.8(b) with $\kappa := E$, then Lemma 1.2. $\square$

Theorem 4.12 (corner $\rightarrow$ corner, corner $\rightarrow$ centre, centre $\rightarrow$ corner: all N , all d). *Let $d \geq 1$, $N \geq 2$, and let (x_0, a) be one of*

(CC) $x_0 = (1, \dots, 1)$, $a = (N, \dots, N)$;

(C2M) N odd, $c = (N+1)/2$: $x_0 = (1, \dots, 1)$, $a = (c, \dots, c)$;

(M2C) N odd: $x_0 = (c, \dots, c)$, $a = (N, \dots, N)$

(or any image of these under the symmetries of the box). Then:

(i) *for every rate $q > 0$ the continuous-time first-passage density g_q is unimodal, with a unique mode;*

(ii) *for every $q \in (0, q_N]$, $q_N := 1/(1 + \cos(\pi/N)) > 1/2$, the PMF f_q is unimodal. In particular this holds for every $q \leq 1/2$, for all N and d .*

Proof. One-dimensional factor. Let $K_q := (1-q)I + qP^{(1)}$ be the one-dimensional lazy walk on $\{1, \dots, N\}$ and $L^{(1)} = I - P^{(1)}$. In case (CC) this is a birth–death chain on N sites, $(1, N)$ are its two ends, and $\pi(N) = 1/N$. In cases (C2M), (M2C) put $Y := |X - c| \in \{0, \dots, M\}$, $M := (N-1)/2$. The reflection $x \mapsto N+1-x$ commutes with $P^{(1)}$, so $P^{(1)}$ maps functions of $|x-c|$ to functions of $|x-c|$ and Y is again a Markov chain: a birth–death chain with $\tilde{K}(0, 1) = q$, $\tilde{K}(y, y \pm 1) = q/2$ for $0 < y < M$, $\tilde{K}(M, M-1) = q/2$, stationary weight $\tilde{\pi}(0) = 1/N$, and $K_q^n(1, c) = \tilde{K}^n(M, 0)$; $(M, 0)$ are the two ends of the folded chain. By symmetry of K_q and the reflection, $K_q^n(c, N) = K_q^n(1, c)$. The same holds in continuous time. The eigenvalues of K_q are $1 - q(1 - \cos(k\pi/N))$ (Lemma 1.1), those of the folded chain are among them (an eigenvector of the folded chain lifts to one of K_q); all of them are $\geq 1 - q(1 + \cos(\pi/N)) \geq 0$ when $q \leq q_N$.

(i) $L = \frac{1}{d} \sum_i L_i^{(1)}$ is a Kronecker sum, so $e^{-qtL} = \bigotimes_i e^{-(q/d)tL^{(1)}}$ and $u(t) = h(t)^d$, where h is the one-dimensional propagator (rate q/d) between $x_{0,1}$ and a_1 . By the previous paragraph and Lemma 4.9(a), $h = N^{-1}\Gamma$ with Γ the distribution function of a sum of M_1 independent exponential variables ($M_1 = N - 1$ in case (CC), $(N - 1)/2$ otherwise). By Lemma 4.10(a)(iii), h and h' are positive and log-concave on $(0, \infty)$. Hence $u' = dh^{d-1}h'$ is positive and $\log u' = \log d + (d - 1)\log h + \log h'$ is concave on $(0, \infty)$. Apply Theorem 4.11(a).

(ii) $P_q = \frac{1}{d} \sum_i K_q^{(i)}$, where $K_q^{(i)}$ is K_q acting on coordinate i ; these matrices commute. Let $h(n) := K_q^n(x_{0,1}, a_1)$ and $(\delta h)(n) := h(n+1) - h(n)$. By the multinomial theorem and the tensor structure, for $t \geq 1$,

$$\varphi(t) = [P_q^{t-1}(P_q - I)](x_0, a) = \frac{1}{d} \sum_{i=1}^d d^{-(t-1)} \sum_{n_1 + \dots + n_d = t-1} \frac{(t-1)!}{\prod_j n_j!} (\delta h)(n_i) \prod_{j \neq i} h(n_j),$$

and the d values of i contribute equally, so

$$d^{t-1}\varphi(t) = (\delta h \star h \star \dots \star h)_{t-1} \quad (d-1 \text{ factors } h; \text{ for } d=1, \varphi(t) = (\delta h)(t-1)). \quad (13)$$

For $q \leq q_N$, Lemma 4.9(b) gives $h(n) = N^{-1}\mathbb{P}(G_1 + \dots + G_{M_1} \leq n)$; by Lemma 4.10(a)(iii) h is a log-concave sequence and so is δh (it is N^{-1} times the shifted mass function of $G_1 + \dots + G_{M_1}$). By Lemma 4.10(b), applied $d-1$ times, the right side of (13) is a log-concave sequence; multiplying by the geometric sequence $d^{-(t-1)}$ preserves log-concavity, and $\varphi(t) = 0$ for $t \leq 0$. So φ is log-concave, and Theorem 4.11(b) applies. $\square$

For $N = 2$, $q_2 = 1$: the corner-to-corner PMF of the hypercube $\{1, 2\}^d$ is unimodal for *every* $q \in (0, 1]$ and every d (this improves Proposition 4.6). For $N = 3$, $q_3 = 2/3$; $q_N \downarrow 1/2$ as $N \rightarrow \infty$. The range $q_N < q \leq 4/5$ is the subject of Section 9 (Theorem 9.13). By Corollary 2.6, (i) also follows from (ii); the direct proof gives more (Corollary 4.13). In probabilistic terms $N^d u(t) = \mathbb{P}(\max(\tau_1, \dots, \tau_d) \leq t)$ with τ_i independent copies of $\xi_1 + \dots + \xi_{M_1}$: the free propagator is the distribution function of the maximum of d independent hypoexponential times, and the first-passage density is the density of this maximum, smoothed by the “escape” kernel $\delta_0 + r$.

Corollary 4.13 (where the mode is; continuous time). *Let (x_0, a) satisfy the hypothesis of Theorem 4.11(a) and $D \geq 2$. Let $m_u := \sup\{t : u'' \geq 0 \text{ on } [0, t]\}$ (the maximiser of u'), and for $a' > 0$ let*

$$J_{a'}(t) := u'(t) - a' \int_0^t e^{-a'(t-s)} u'(s) ds, \quad m(a') := \inf\{t \geq m_u : J_{a'}(t) \leq 0\} \in (m_u, \infty].$$

Then $J_{a'} \leq 0$ on $[m(a'), \infty)$, $a' \mapsto m(a')$ is nonincreasing, and the mode t^ of g_q satisfies*

$$m_u < t^* \leq m(q\alpha_1) \leq m(a') \quad \text{for every } 0 < a' \leq q\alpha_1,$$

in particular for $a' = q/M_{\max}$, $M_{\max} := \max_x \mathbb{E}_x[T]$ at $q = 1$ ($= \mathbb{E}_{x_0}[T]$ for the corner pair, Lemma 5.1). Moreover $m(a') \leq \inf\{t \geq m_u : u'(t) \leq a'e^{-a't}u(t)\}$.

Proof. u' is positive and log-concave, hence $u'' \geq 0$ on $[0, m_u]$ and $u'' \leq 0$ on $[m_u, \infty)$; $u'(0) = 0$ because $D \geq 2$. *Lower bound.* For $0 < t \leq m_u$ and $0 \leq \tau \leq t$, $u'(t - \tau) \leq u'(t)$, so by (12) $B(t) \geq u''(t)/u'(t) + r(0) - \int_0^t \nu = u''(t)/u'(t) + r(t) > 0$. Hence $g'_q > 0$ on $(0, m_u]$. *Upper bound.* $r(\tau) = \sum_j w_j e^{-q\alpha_j \tau}$ with $w_j = q r^\top E_j \mathbf{1} = (q/\alpha_j) \|E_j r\|^2 \geq 0$ (as $E_j \mathbf{1} = \alpha_j^{-1} E_j A \mathbf{1}$). By (9) and $u'(0) = 0$,

$$g'_q(t) = u''(t) + \sum_j w_j e^{-q\alpha_j t} I_{q\alpha_j}(t), \quad I_b(t) := \int_0^t e^{bs} u''(s) ds = e^{bt} J_b(t)$$

(integrate by parts). I_b is nondecreasing on $[0, m_u]$, positive at m_u , and nonincreasing on $[m_u, \infty)$; so $\{t \geq m_u : I_b(t) \leq 0\} = [m(b), \infty)$. For $b' \geq b > 0$ and $t \geq m_u$: $I_{b'}(t) = \int_0^t e^{(b'-b)s} e^{bs} u''(s) ds \leq$

$e^{(b'-b)m_u}I_b(t)$, because the weight $e^{(b'-b)s}$ is at most $e^{(b'-b)m_u}$ where $u'' \geq 0$ and at least that where $u'' \leq 0$. Hence $I_b(t) \leq 0$ implies $I_{b'}(t) \leq 0$: $m(\cdot)$ is nonincreasing. For $t \geq m(q\alpha_1)$ every $I_{q\alpha_j}(t) \leq 0$ ($\alpha_j \geq \alpha_1$) and $u''(t) \leq 0$, so $g'_q(t) \leq 0$. $\alpha_1 \geq 1/M_{\max}$ is the argument of Lemma 5.4 ($1/\alpha_1 = \|A^{-1}\|_2 \leq \|A^{-1}\|_\infty = \max_x \mathbb{E}_x T$). Finally $e^{-a'(t-s)} \geq e^{-a't}$ gives $J_{a'}(t) \leq u'(t) - a'e^{-a't}u(t)$. $\square$

For the corner pair m_u is the t_N of Theorem 5.10, and $m(q\alpha_1) < \infty$ because u' decays at rate $q\beta_1 > q\alpha_1$ (Lemma 5.5). A formal leading-order evaluation (replace h by its two slowest modes and the integral in J by u) gives $m(q\alpha_1) \approx (q\beta_1)^{-1} \ln(2d\beta_1/\alpha_1)$, which is also the leading term of the two-pole formula of Section 6 of the article; so the corollary is a rigorous upper bound for the mode which, numerically, exceeds the mode by 4.7–24.9% for the corner pair in the cases tabulated below ($d = 2, 3$, $8 \leq N \leq 64$). For smaller boxes and in one dimension it is cruder: $m(\alpha_1)/t^* = 1.49, 1.39, 1.33, 1.27$ for $d = 2$, $N = 3, 4, 5, 7$, and $1.94, 1.69, 1.63, 1.62$ for $d = 1$, $N = 4, 9, 15, 16$; for corner $\rightarrow$ centre $1.61, 1.31, 1.20$ ($(d, N) = (1, 9), (2, 5), (2, 9)$), and for centre $\rightarrow$ corner it lies between 1.54 and 2.73 in the eight cases computed ($d = 1, 2, 3$, $3 \leq N \leq 33$) (`scripts/table_mode_bounds_small.py`; float, illustration only). In every case computed the chain $m_u < t^* \leq m(\alpha_1) \leq m(1/M_{\max})$ held, as it must. (The asymptotic statement is not proved here.)

geometry	d	N	m_u	mode t^*	$m(\alpha_1)$	$m(1/M_{\max})$	$m(\alpha_1)/t^*$
CC	2	8	37.1	84.5	105.5	108.5	1.248
CC	2	16	149.6	381.0	449.5	459.2	1.180
CC	2	32	599.5	1653.2	1884.7	1917.7	1.140
CC	2	64	2399.2	7022.2	7826.5	7940.6	1.115
CC	3	8	70.2	204.5	224.3	226.0	1.097
CC	3	12	158.4	509.7	542.4	544.9	1.064
CC	3	16	282.0	964.5	1010.7	1014.1	1.048
C2M	2	33	159.6	464.5	519.2	526.9	1.118
M2C	2	33	159.6	417.6	645.9	655.2	1.547
C2M	3	9	22.6	73.6	78.7	79.0	1.070
M2C	3	9	22.6	58.7	91.2	91.7	1.554

Remark 4.14 (scope and limits of the method). 1. *The hypothesis of Theorem 4.11 is a property of the free walk only*: no information on the killed spectrum is needed. It fails whenever u overshoots its limit N^{-d} (then u' changes sign), which is what happens for the bimodal pairs of Section 8.

2. *Positivity of u' cannot be weakened to $u' \geq 0$* . For $d = 3$, $N = 4$, $x_0 = (3, 3, 3)$, $a = (2, 2, 2)$: $u = p^3$ with $p(t) = \mathbb{P}_3(X_t = 2) = \frac{1}{4} + \frac{1}{2}[-\cos^2(\frac{3\pi}{8})e^{-\vartheta_1 t} + \frac{1}{2}e^{-\vartheta_2 t} - \cos^2(\frac{\pi}{8})e^{-\vartheta_3 t}]$, $\vartheta_k = \frac{q}{d}(1 - \cos(k\pi/4))$, and

$$p'(t) = \frac{1}{2}(A^{1/2}e^{-\vartheta_1 t/2} - B^{1/2}e^{-\vartheta_3 t/2})^2, \quad A = \cos^2(\frac{3\pi}{8})\vartheta_1, \quad B = \cos^2(\frac{\pi}{8})\vartheta_3$$

(expand the square and use $2(AB)^{1/2} = \vartheta_2/2$ and $\vartheta_1 + \vartheta_3 = 2\vartheta_2$). So $u' = 3p^2 p' \geq 0$ vanishes at exactly one instant, $t_0 = \ln(B/A)/(\vartheta_3 - \vartheta_1)$, and numerically $\log u'$ is concave on each side of t_0 ; yet this pair is the bimodal pair of the $4 \times 4 \times 4$ box (class **B** in Theorem 8.2). A grid test of the hypothesis must therefore also check that u' has no interior local minimum.

3. *The conclusion cannot be strengthened to log-concavity of the first-passage law*: centre $\rightarrow$ corner satisfies the hypothesis but its PMF is not log-concave (Proposition 8.4).

4. *Discrete time needs $q \leq q_N$* . For $q > q_N$ the one-dimensional chain has negative eigenvalues and δh need not be log-concave: in case (CC), $\delta h(N - 2 + j) = (q/2)^{N-1} h_j(\lambda_1, \dots, \lambda_{N-1})$ (h_j the complete homogeneous symmetric polynomial, $\lambda_k = 1 - q + q \cos(k\pi/N)$), and

$$\delta h(N-1)^2 - \delta h(N-2) \delta h(N) = (q/2)^{2(N-1)} e_2(\lambda) = (q/2)^{2(N-1)} \frac{N-2}{4} (2(N-1)(1-q)^2 - q^2),$$

which is negative iff $q > q_1(N) := 1/(1 + (2(N-1))^{-1/2})$. So in $d = 1$ the hypothesis of Theorem 4.11(b) fails for $q > q_1(N)$. That the *conclusion* also fails for q near 1 is Proposition 4.4 ($d = 1$) and Theorem 7.1(c) ($d = 2$). For $q_N < q \leq q_1(N)$ the hypothesis holds when $q \leq 4/5$ (every N ; Theorem 9.11) and when $N \leq 501$ (Corollary 9.8); in general this is Conjecture 10.4.

5. *Cubes*. The proof of Theorem 4.12 uses that all coordinates contribute the same factor h . For a rectangular box, or a target such as a face centre, $u' = \sum_i h'_i \prod_{j \neq i} h_j$ is the density of the maximum of independent, non-identically distributed log-concave times, which need not be log-concave; Theorem 4.11 then has to be checked case by case.

Numerical observation 4.15. (i) (`verify_lastexit.py` W10; grid test with 30-digit arithmetic, not a proof.) The hypothesis of Theorem 4.11(a) holds on a grid of 1560 times for 10 of 24 pairs ($d = 2$, $N = 3$), 7/45 ($N = 4$), 24/144 ($N = 5$), 36/210 ($N = 6$), 80/480 ($N = 7$), 91/630 ($N = 8$), 49/104 ($d = 3$, $N = 3$), 30/252 ($d = 3$, $N = 4$) (targets up to symmetry; lists in `data/pairs_logconcave_hypothesis.json`); typical members are pairs that are far apart. No pair satisfying the hypothesis fails to be unimodal. (ii) (`scripts/check_lc_threshold.py`, exact integer arithmetic.) For $d = 2$, $2 \leq N \leq 20$ and $N = 24$, and $d = 3$, $2 \leq N \leq 10$, the sequence φ of the corner pair is log-concave on $0 \leq t \leq 5dN^2 + 40$ for $q = 3/5$ and $q = 4/5$ (for $q \leq 4/5$ log-concavity on the whole support is Theorem 9.13(i)); for $q = 9/10$ it fails for $d = 2$, $N \in \{4, \dots, 8\}$ (first inequality only) and holds in the other cases.

Sanity checks (`scripts/verify_lastexit.py` $\rightarrow$ `data/verify_lastexit.json`; all pass): W1 Lemma 4.7(a), exact integers (5292 identities for 9 pairs $\times$ 4 activities); W2 Lemma 4.7(b) in the Laplace domain, exact rationals; W3 (10) by quadrature; W4 Theorem 4.12(i): $u' > 0$, $(\log u')'' \leq 0$, B nonincreasing, 20 cases; W5 Lemma 4.9, discrete exact and continuous to 10^{-40} , including the folding; W6 Lemma 4.10, 600 random exact trials; W7 Lemma 4.8, 600 random exact trials; W8 Theorem 4.12(ii) on windows, exact, and identity (13); W9 Remark 4.14(4), exact; W10 Theorem 4.11 on all pairs of small boxes.

5 Corner target, opposite-corner start: what is proved for all N

Status of this section after Theorem 4.12. For $q \leq 1/2$ and in continuous time, unimodality of the corner pair is Theorem 4.12, and the rise on $[0, t_N]$ (Theorem 5.10) is part of Corollary 4.13 ($t_N = m_u$). (For $q \leq 4/5$ unimodality in discrete time is Theorem 9.13.) The results below are still needed for $4/5 < q \leq 1$, where neither theorem applies: the coupling (Lemmas 5.1–5.4, used in Corollary 4.13 and in the tail bound) and the explicit tail time $T_N(q)$ valid for *every* $q \in (0, 1]$ (Theorem 5.6), which confines any failure of unimodality to a bounded window.

In this section $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$, $D = d(N-1)$; α_1, φ_1 are as in Lemma 3.1, β_1 as in Lemma 1.1.

5.1 A monotone coupling

Lemma 5.1. *Let $a = (N, \dots, N)$, $q \in (0, 1]$, $S_t(x) = \mathbb{P}_x(T_a > t)$. If x and $x - e_i$ lie in Λ , then $S_t(x - e_i) \geq S_t(x)$ for all t . Consequently $S_t(x) \leq S_t(x_0)$ and $\mathbb{E}_x[T_a] \leq \mathbb{E}_{x_0}[T_a]$ for all x . The same holds in continuous time.*

Proof. Run two copies X (from x) and X' (from $x - e_i$) with the same instruction at every step (“stay” with probability $1 - q$, otherwise “attempt $\pm e_j$ ”). Invariant: $X'_s \in \{X_s, X_s - e_i\}$. Let $X'_s = X_s - e_i$. An instruction $\pm e_j$, $j \neq i$, is carried out or cancelled by both. Instruction $+e_i$: X' moves; if X is blocked the two coalesce, otherwise both move. Instruction $-e_i$: X moves; if X' is blocked the two coalesce, otherwise both move. When $X' = a$, $X \in \{a, a + e_i\} \cap \Lambda = \{a\}$, so $T \leq T'$ pathwise. $\square$

Lemma 5.2. $\varphi_1(x) \leq \varphi_1(x_0)$ for all $x \in \Lambda'$.

Proof. $S_t = Q_q^t \mathbf{1} = \sum_j (1 - q\alpha_j)^t E_j \mathbf{1}$, so by Lemma 3.1, $S_t(x)/(1 - q\alpha_1)^t \rightarrow \varphi_1(x)\langle \varphi_1, \mathbf{1} \rangle$ with $\langle \varphi_1, \mathbf{1} \rangle > 0$; pass to the limit in Lemma 5.1. $\square$

Lemma 5.3. $C_1 = \alpha_1 \varphi_1(x_0)\langle \varphi_1, \mathbf{1} \rangle \geq \alpha_1$.

Proof. $\langle \varphi_1, \mathbf{1} \rangle = \sum_x \varphi_1(x) \geq \sum_x \varphi_1(x)^2 / \varphi_1(x_0) = 1/\varphi_1(x_0)$ by Lemma 5.2. $\square$

Lemma 5.4. Let $M := \mathbb{E}_{x_0}[T_a]$ for $q = 1$. Then $1/M \leq \alpha_1 \leq 1/(2(n-1))$ and $M \leq 2d^2(N-1)N^d$.

Proof. Upper bound: the Rayleigh quotient of $\mathbf{1}$ is $\sum_x r(x)/(n-1) = \frac{1}{2}/(n-1)$. Lower bound: $A^{-1} = \sum_k P^k \geq 0$ entrywise and $A^{-1}\mathbf{1} = (\mathbb{E}_x T)_x$; for a symmetric matrix the spectral radius is at most the maximal absolute row sum, so $1/\alpha_1 \leq \max_x \mathbb{E}_x T = M$ (Lemma 5.1). The $q = 1$ chain is the random walk on the network with conductance $1/(2d)$ on each edge and loops $b(x)/(2d)$, of total conductance n . By the commute time identity (D. A. Levin and Y. Peres, *Markov Chains and Mixing Times*, 2nd ed., AMS 2017, Proposition 10.7 [1st ed.: Proposition 10.6]) $\mathbb{E}_{x_0} \tau_a + \mathbb{E}_a \tau_{x_0} = n R(x_0 \leftrightarrow a)$, and by Rayleigh's monotonicity law (ibid., Theorem 9.12) $R(x_0 \leftrightarrow a) \leq 2d \cdot d(N-1)$. $\square$

5.2 The symmetric sector

Lemma 5.5. Let $d \geq 2$ and let H' be the space of functions on Λ' invariant under permutations of the coordinates. Then $e_{x_0}, \mathbf{1}, r \in H'$, $AH' \subset H'$, and the eigenvalues $\alpha'_1 \leq \alpha'_2 \leq \dots$ of $A|_{H'}$ satisfy

$$\alpha'_1 = \alpha_1 < \beta_1 \leq \alpha'_2, \quad \alpha'_j \leq 2 - d\beta_1.$$

Hence $|1 - q\alpha'_j| \leq 1 - q\beta_1$ for $j \geq 2$ and every $q \in (0, 1]$, and

$$f_q(t) = q \sum_j c'_j (1 - q\alpha'_j)^{t-1}, \quad c'_j = \alpha'_j \varphi'_j(x_0) \langle \varphi'_j, \mathbf{1} \rangle, \quad (14)$$

with (φ'_j) an orthonormal eigenbasis of $A|_{H'}$.

Proof. Coordinate permutations fix x_0 and a and commute with P . The Perron vector is invariant, so $\alpha'_1 = \alpha_1$. Extension by zero identifies H' with a hyperplane of the space H of invariant functions on Λ , with $\langle h, Ah \rangle = \langle \tilde{h}, L\tilde{h} \rangle$. By the Courant–Fischer theorem (Horn–Johnson, Theorem 4.2.6),

$$\alpha'_2 = \min_{\substack{V \subset H' \\ \dim V = 2}} \max_{h \in V} \frac{\langle h, Ah \rangle}{\langle h, h \rangle} \geq \min_{\substack{V \subset H \\ \dim V = 2}} \max_{h \in V} \frac{\langle h, Lh \rangle}{\langle h, h \rangle} = \kappa_1,$$

the second smallest eigenvalue of $L|_H$. The symmetrisations of the Ψ_k over sorted k form an orthogonal eigenbasis of $L|_H$ with eigenvalues $\theta(k)$; the smallest is 0 and every other is at least $\theta(0, \dots, 0, 1) = \beta_1$, so $\kappa_1 = \beta_1$. $\alpha_1 \leq 1/(2(n-1)) < 2/(dN^2) \leq \beta_1$ because $dN^2 < 4(N^d - 1)$ for $d \geq 2, N \geq 2$. For $\alpha \in [\beta_1, 2 - d\beta_1]$: $1 - q\alpha \leq 1 - q\beta_1$ and $1 - q\alpha \geq 1 - 2q + qd\beta_1 \geq -(1 - q\beta_1)$. $\square$

5.3 Explicit eventual monotone decay

Theorem 5.6 (tail). Let $d \geq 2, N \geq 2, a = (N, \dots, N), x_0 = (1, \dots, 1)$.

(i) For every $q \in (0, 1]$: $f_q(t+2) < f_q(t+1)$ for every integer

$$t > T_N(q) := \frac{\ln(4\sqrt{n-1}/\alpha_1^2)}{\ln((1 - q\alpha_1)/(1 - q\beta_1))}.$$

(ii) For every rate $q > 0$: $g'_q(s) < 0$ for $s > \ln(4\sqrt{n-1}/\alpha_1^2)/(q(\beta_1 - \alpha_1))$.

(iii) Both thresholds are at most

$$\bar{T}_N(q) := \frac{\ln 4 + \frac{1}{2} \ln(n-1) + 2 \ln(2d^2(N-1)N^d)}{q(\beta_1 - \frac{1}{2(n-1)})} = O(N^2 \log N/q).$$

Proof. (i) By (14), $\Delta a_q(t) = -q^2 \sum_j \alpha_j'^2 \varphi_j'(x_0) \langle \varphi_j', \mathbf{1} \rangle \lambda_j^t$, $\lambda_j = 1 - q\alpha_j'$. The term $j = 1$ is at least $\alpha_1'^2 \lambda_1^t$ (Lemma 5.3). By Lemma 5.5 and the Cauchy–Schwarz inequality,

$$\left| \sum_{j \geq 2} \alpha_j'^2 \varphi_j'(x_0) \langle \varphi_j', \mathbf{1} \rangle \lambda_j^t \right| \leq 4(1 - q\beta_1)^t \|e_{x_0}\| \|\mathbf{1}\| = 4\sqrt{n-1} (1 - q\beta_1)^t.$$

So $-\Delta a_q(t)/q^2 \geq \alpha_1'^2(1 - q\alpha_1)^t - 4\sqrt{n-1}(1 - q\beta_1)^t > 0$ iff $t > T_N(q)$. (ii) The same with $e^{-q\alpha s}$ in place of λ^t . (iii) $\ln \frac{1-q\alpha_1}{1-q\beta_1} = \int_{q\alpha_1}^{q\beta_1} \frac{du}{1-u} \geq q(\beta_1 - \alpha_1)$, and Lemma 5.4. $\square$

Remark 5.7. For an arbitrary pair, Lemma 3.1 gives $\Delta a_q(t) = -q^2 \alpha_1 C_1 (1 - q\alpha_1)^t (1 + o(1))$ with $C_1 > 0$: every PMF and density is eventually strictly decreasing, and a failure of unimodality can only occur in a bounded time window.

5.4 Explicit initial monotone increase (continuous time)

For an arbitrary pair let $u(t) := \mathbb{P}_{x_0}(X_t = a)$ and $v(t) := \mathbb{P}_a(X_t = a)$ for the *unkilled* continuous-time walk.

Lemma 5.8 (renewal equations). $u = g_q * v$ and, with $\gamma := -v' \geq 0$,

$$g_q' = u'' + g_q(0) \gamma + \gamma * g_q', \quad g_q(0) = q r(x_0). \quad (15)$$

If $D \geq 2$ and $u'' \geq 0$ on $[0, t_0]$, then $g_q' \geq u'' \geq 0$ on $[0, t_0]$.

Proof. In block form with respect to $\Lambda = \Lambda' \cup \{a\}$, $L(x, a) = -r(x)$; the Schur complement formula gives $(s + qL)^{-1}(x_0, a) = [e_{x_0}^\top (s + qA)^{-1} q r] (s + qL)^{-1}(a, a)$, i.e. $\hat{u} = \hat{g}_q \hat{v}$, so $u = g_q * v$. $v(t) = \sum_k \Psi_k(a)^2 e^{-q\theta(k)t}$ is completely monotone with $v(0) = 1$. Differentiating twice gives $g_q = u' + \gamma * g_q$ and (15). Iterating (15), $g_q' = \sum_{k < K} \gamma^{*k} * \phi + \gamma^{*K} * g_q'$ with $\phi = u'' + g_q(0)\gamma$, and $|\gamma^{*K} * g_q'| (t) \leq \sup_{[0, t]} |g_q'| (\gamma(0)t)^K / K! \rightarrow 0$. Hence $g_q' = \sum_{k \geq 0} \gamma^{*k} * \phi \geq \phi$ on any interval $[0, t_0]$ on which $\phi \geq 0$. $\square$

Lemma 5.9. Let $h(t)$ be the probability that the one-dimensional reflecting walk on $\{1, \dots, N\}$ with jump rate $c/2$ in each direction, started at 1, is at N at time t , and $\vartheta_k = c(1 - \cos(k\pi/N))$. Then $h(t) = \frac{1}{N} \mathbb{P}(E_1 + \dots + E_{N-1} \leq t)$ with independent $E_k \sim \text{Exp}(\vartheta_k)$. The density $\gamma_N = Nh'$ of $\sum_k E_k$ satisfies $\gamma_N(t) = t^{N-2} e^{-\vartheta_{N-1}t} \psi(t)$ with $\psi > 0$ nondecreasing; in particular $h'' \geq 0$ on $[0, (N-2)/(2c)]$.

Proof. $\hat{h}(s) = (s + cL^{(1)})^{-1}(1, N)$. The cofactor is $(-1)^{N+1}$ times the minor obtained by deleting row N and column 1, which is triangular with diagonal entries $-c/2$; so the numerator is $(c/2)^{N-1}$, and the determinant is $s \prod_{k \geq 1} (s + \vartheta_k)$. Letting $s \rightarrow 0$ in $s\hat{h}(s) \rightarrow 1/N$ gives $(c/2)^{N-1} = \frac{1}{N} \prod_k \vartheta_k$, hence $\hat{h}(s) = \frac{1}{N} \frac{1}{s} \prod_k \frac{\vartheta_k}{s + \vartheta_k}$. Write the $\text{Exp}(\vartheta_k)$ density as $e^{-\vartheta_{N-1}t} \phi_k(t)$ with $\phi_k(t) = \vartheta_k e^{(\vartheta_{N-1} - \vartheta_k)t}$ nondecreasing; then $\gamma_N(t) = e^{-\vartheta_{N-1}t} (\phi_1 * \dots * \phi_{N-1})(t)$ and the substitution $\tau_k = t\sigma_k$ in the convolution integral over the simplex gives $(\phi_1 * \dots * \phi_{N-1})(t) = t^{N-2} \int \prod_k \phi_k(t\sigma_k) d\sigma$. So $(\log \gamma_N)' \geq (N-2)/t - \vartheta_{N-1}$ and $\vartheta_{N-1} \leq 2c$. $\square$

Theorem 5.10 (rise). Let $d \geq 1$, $N \geq 2$ with $d(N-1) \geq 2$, $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$, and let h be as in Lemma 5.9 with $c = q/d$. Put

$$m_N := \sup\{t : h'' \geq 0 \text{ on } [0, t]\}, \quad t_N := \sup\{t : (d-1)h'^2 + hh'' \geq 0 \text{ on } [0, t]\} \geq m_N.$$

Then g_q is nondecreasing on $[0, t_N]$, and $t_N \geq m_N \geq d(N-2)/(2q)$.

Proof. $L = \frac{1}{d} \sum_i L_i^{(1)}$ is a Kronecker sum, so $e^{-qtL} = \bigotimes_i e^{-(q/d)tL^{(1)}}$ and $u = h^d$, $u'' = dh^{d-2}[(d-1)h'^2 + hh'']$. Apply Lemmas 5.8 and 5.9. $\square$

For a general pair the same proof shows that g_q is nondecreasing as long as $u(t) = \prod_i h_{x_0, i, a_i}(t)$ is convex.

Proposition 5.11 (discrete version, $q \leq 1/2$). *Let $q \leq 1/2$, $u(t) := P_q^t(x_0, a)$, and suppose $u(t+2) - 2u(t+1) + u(t) \geq 0$ for $0 \leq t \leq t_0$. Then $f_q(0) \leq f_q(1) \leq \dots \leq f_q(t_0+2)$.*

Proof. $u(t) = \sum_{s=0}^t f_q(s)v(t-s)$ with $v(t) = P_q^t(a, a) = \sum_k \Psi_k(a)^2(1-q\theta(k))^t$ nonincreasing ($q \leq 1/2$); put $\gamma(j) = v(j) - v(j+1) \geq 0$ and $\delta(s) = f_q(s+1) - f_q(s)$. Differencing twice gives $\delta(t+1) = [u(t+2) - 2u(t+1) + u(t)] + \sum_{j=0}^t \delta(t-j)\gamma(j)$; with $\delta(0) = f_q(1) \geq 0$, induction gives $\delta(t) \geq 0$ for $t \leq t_0+1$. $\square$

Numerical illustration (floating point, unit rate; not used in any proof).

d	N	m_N	t_N	mode of g	T_N (Thm 5.6(ii))	$\bar{T}_N$
2	8	22.9	37.1	84.5	428	658
2	16	93.5	149.6	380.9	2109	3246
2	32	375.4	599.5	1653	10017	15499
2	64	1502.9	2399.2	7022	46348	72111
3	8	34.4	70.2	204.4	788	1093
3	16	140.3	282.0	964.4	3933	5313
3	30	494.8	992.6	3808.5	16365	21804

$m_N/(dN^2) \rightarrow 0.1835$, $t_N/(dN^2) \approx 0.293$ ($d=2$), 0.368 ($d=3$). So Theorems 5.6 and 5.10 alone show, for all N , that the density increases on roughly the first 25–45% of the way to the mode and decreases from about 4–7 modal times on. In continuous time and for $q \leq q_N$ the window (t_N, T_N) in between is closed by Theorem 4.12 (and narrowed to $(m_u, m(q\alpha_1))$ by Corollary 4.13); for $q_N < q \leq 4/5$ it is closed by Theorem 9.13; for $4/5 < q \leq 1$ it is where an all- N proof is missing (Section 10.2).

6 The cone certificate and the “front property”

Lemma 6.1 (cone certificate). *For any pair and any $q \in (0, 1]$ let $\rho_t := Q_q^t e_{x_0}$. If $\rho_{T+1} \leq \rho_T$ componentwise for some $T \geq 0$, then $\rho_{t+1} \leq \rho_t$ for all $t \geq T$ and $f_q(t+2) \leq f_q(t+1)$ for all $t \geq T$. In continuous time: if $\frac{d}{ds}\rho_s \leq 0$ componentwise at s_0 , then g_q is nonincreasing on $[s_0, \infty)$.*

Proof. $Q_q \geq 0$ entrywise, so $\rho_t - \rho_{t+1} = Q_q^{t-T}(\rho_T - \rho_{T+1}) \geq 0$, and $f_q(t+1) = q r^\top \rho_t$ with $r \geq 0$. $\square$

In exact arithmetic, for $q = q_n/q_d$, $\rho_t = R_t/\mathfrak{D}^t$ with an integer vector R_t and the integer $\mathfrak{D} = 2dq_d/\gcd$; $f_q(t) = F_t/\mathfrak{D}^t$; $\text{sgn}(f_q(j) - f_q(j-1)) = \text{sgn}(F_j - \mathfrak{D}F_{j-1})$; the cone at T is $R_{T+1} \leq \mathfrak{D}R_T$.

Numerical observation 6.2 (front property). In all 282 corner-to-corner certificates of Section 7 with a unimodal verdict ($d = 1, \dots, 5$) the cone holds at $T = \text{mode} - 1$ (257 cases) or $T = \text{mode}$ (25 cases, 24 of them at $d = 2$, $q = 1$): at the moment the flux into the target stops increasing, the occupation of every site has stopped increasing. For the centre geometries the cone time is between 1.5 and 6 modal times.

Proposition 6.3. *Let a be a corner, x_0 the opposite corner, $y_1 \sim a$, $q \in (0, 1)$, $\rho_t = Q_q^t e_{x_0}$. Consider*

(P1') $\rho_{t+1}(x)\rho_t(x+e_i) \leq \rho_{t+1}(x+e_i)\rho_t(x)$ for all t and all $x, x+e_i \in \Lambda'$;

(P1) $\rho_{t+1}(x)\rho_t(y_1) \leq \rho_{t+1}(y_1)\rho_t(x)$ for all t and $x \in \Lambda'$;

(F) for every $t \geq D-1$: $\rho_{t+1}(y_1) \leq \rho_t(y_1)$ implies $\rho_{t+1} \leq \rho_t$ componentwise;

(LC) $f_q(t)^2 \geq f_q(t-1)f_q(t+1)$ for all $t \geq 1$.

Then (P1') $\Rightarrow$ (P1); (P1) $\Rightarrow$ (F); (P1) $\Rightarrow$ (LC); (F) $\Rightarrow$ f_q unimodal; (LC) $\Rightarrow$ f_q unimodal with non-decreasing hazard rate $f_q(t)/\mathbb{P}(T \geq t)$.

Proof. For $q < 1$, $\rho_t(x) > 0$ iff $t \geq |x - x_0|_1$. By symmetry $f_q(t+1) = \frac{q}{2}\rho_t(y_1)$. (P1') $\Rightarrow$ (P1): if $\rho_t(y_1) = 0$ the inequality is trivial; otherwise take a coordinatewise increasing path $x = z_0, \dots, z_k$ in Λ' ending at a neighbour of a ; all $\rho_t(z_l) > 0$ and (P1') gives $\rho_{t+1}(z_l)/\rho_t(z_l) \leq \rho_{t+1}(z_{l+1})/\rho_t(z_{l+1})$. (P1) $\Rightarrow$ (F): $\rho_{t+1}(x)\rho_t(y_1) \leq \rho_{t+1}(y_1)\rho_t(x) \leq \rho_t(y_1)\rho_t(x)$ and $\rho_t(y_1) > 0$. (P1) $\Rightarrow$ (LC): $\rho_{t+1}(y_1)\rho_{t-1}(y_1) = \sum_z Q_q(y_1, z)\rho_t(z)\rho_{t-1}(y_1) \leq \sum_z Q_q(y_1, z)\rho_t(y_1)\rho_{t-1}(z) = \rho_t(y_1)^2$. (F) $\Rightarrow$ unimodal: let t^* be the first $t \geq D-1$ with $\rho_{t+1}(y_1) \leq \rho_t(y_1)$; f_q increases up to t^*+1 and, by (F) and Lemma 6.1, is nonincreasing afterwards. (LC): the ratios $f_q(t+1)/f_q(t)$ are nonincreasing on $\{D, D+1, \dots\}$, and $\mathbb{P}(T \geq t)/f_q(t) = \sum_{k \geq 0} f_q(t+k)/f_q(t)$ is nonincreasing. $\square$

In $d = 1$ (start 1, target N), (P1') is true for $q \leq 1/2$: for every $\kappa > 0$, $z_t := \rho_{t+1} - \kappa\rho_t$ solves $z_{t+1} = Q_q z_t$ and $z_0 = (Q_q - \kappa)e_1$ has at most one sign change, of pattern $(-, +)$; by Lemma 2.2, z_t never has a positive entry to the left of a negative one. In $d \geq 2$ no such principle is available, and (P1'), (P1), (F), (LC) are conjectural (Section 10.1). (Unimodality itself does not depend on them: Theorems 4.12 and 9.13.)

7 Computer-assisted theorem: finite N , all q at once

Computer-assisted Theorem 7.1. “Unimodal” refers to the PMF on its whole support.

- (a) $d = 2$, corner to opposite corner: for every $2 \leq N \leq 64$ and every $q \in (0, 499/500]$ the PMF is unimodal; for every rate the continuous-time density is unimodal.
- (b) $d = 3$, corner to opposite corner: for every $2 \leq N \leq 30$ and every $q \in (0, 1]$ the PMF is unimodal; so is the continuous-time density. The same holds for $d = 4$, $2 \leq N \leq 8$ and $d = 5$, $2 \leq N \leq 5$.
- (c) $d = 2$, $q = 1$: for $2 \leq N \leq 64$ the corner-to-corner PMF is unimodal except for $N \in \{9, 11, 13, 15, 17, 18, 21, 24, 26, 38, 51, 61\}$, where $f(t^* - 1) > f(t^*) < f(t^* + 1) \geq f(t^* + 2) \geq \dots$ with $t^* = 109, 169, 243, 331, 433, 489, 679, 901, 1067, 2369, 4379, 6355$ respectively, the PMF being nondecreasing before $t^* - 1$. For the other 51 values of N the PMF is unimodal for every $q \in (0, 1]$.
- (d) $d = 2$, $N = 9$: the PMF is unimodal for $q = 0.9986$ and not unimodal for $q = 0.9987$ and $q = 0.999$; hence $0.9986 \leq q^*(9) < 0.9987$.
- (e) N odd, $c = (N+1)/2$; corner $(1, \dots, 1) \rightarrow$ centre $(c, \dots, c)$ and centre $\rightarrow$ corner $(N, \dots, N)$: the PMF is unimodal for every $q \in (0, 4/5]$, and the continuous-time density is unimodal, for $d = 2$, odd $3 \leq N \leq 61$, and $d = 3$, odd $3 \leq N \leq 27$.
- (f) $d = 1$, end to end: for $2 \leq N \leq 60$ the PMF is unimodal for every $q \in (0, 4/5]$.

Proof. For each listed case and the single rational q_0 (499/500 in (a); 1 in (b), (c); 4993/5000, 9987/10000, 999/1000 in (d); 4/5 in (e), (f)) the program `scripts/cert_unimodal.py` computes in exact integer arithmetic the F_j , the R_t , the signs of $F_j - \mathfrak{D}F_{j-1}$ for $1 \leq j \leq T+1$ and the first T with $R_{T+1} \leq \mathfrak{D}R_T$. In every “unimodal” case the signs are $(\geq 0) \dots (\geq 0)(\leq 0) \dots (\leq 0)$; Lemma 6.1 and Lemma 1.2 give unimodality at q_0 , and Corollary 2.6 extends it to all $q \leq q_0$ and to continuous time. In every “not unimodal” case the signs contain $+$, $-$, $+$. Records: `data/cert_*.jsonl`, summarised in `data/summary_certificates.json`. $\square$

Size. (a): 63 runs, up to 7035 steps on 4096 sites with 77 129-bit integers, 88 CPU-minutes. (b), $d = 3$: 29 runs, up to 3807 steps on 27 000 sites with 9825-bit integers, 20 CPU-minutes. *Cross-checks with separately written implementations* (`scripts/verify_certificate_code.py`; same project, same machine): the integer stepper agrees with a from-the-definition implementation in rational arithmetic; a second, dictionary-based implementation of the whole certificate reproduces sign runs, cone time and verdict in 13 cases; the modes agree with the separately written floating-point code of the computations of Sections 5–6 of the article in 29 cases; after the cone time the PMF is nonincreasing for a further $3T$ steps.

What the theorem adds to Theorems 4.12 and 9.13, and what it does not cover. By Theorem 4.12 the continuous-time statements, and by Theorem 9.13 the range $q \leq 4/5$, hold for *all* N . The cone certificates are the only proof for $4/5 < q \leq q_0$ in (a)–(d), and there they are limited to $N \leq 64$ ($d = 2$), $N \leq 30$ ($d = 3$), $N \leq 8$ ($d = 4$), $N \leq 5$ ($d = 5$). Parts (e), (f) are used in the proof of Theorem 9.13 only for the small cases in which the increment of the free propagator is not log-concave ($d = 1$, $3 \leq N \leq 8$; centre geometries with $N = 3$, $d \in \{2, 3\}$ and $N \in \{5, 7\}$, $d = 2$). Not covered by anything: $q \in (0.998, 1)$ for $d = 2$ except as stated in (c), (d); $4/5 < q < 1$ for the centre geometries in $d \geq 2$; even N has no centre site. (One dimension is treated separately and exactly in Section 7.1.)

Numerical observation 7.2 (why $q = 1$, $d = 2$ is marginal). At $q = 1$ the parity of $t + |x|_1$ can change only by a cancelled move at a wall. The parity-odd part of f decays like $\cos(\pi/N)^t$ in $d = 2$, and at the modal time its amplitude is of the same order N^{-6} (up to powers of $\log N$) as the second difference of the smooth part of f ; whether a one-step dip appears depends on the position of the continuous mode relative to the integers. For $q < 1$ the parity-odd part decays like $(2q - 1 + O(N^{-2}))^t$. This is a heuristic explanation, not a proof.

7.1 One dimension: the exact thresholds

In one dimension the threshold q^* of Corollary 2.6 (f_q is unimodal iff $q \leq q^*$) can be determined exactly for small and moderate N . Two pairs are considered: *end to end*, $x_0 = 1$, $a = N$ (the corner pair for $d = 1$), and, for odd N , *centre to corner*, $x_0 = c = (N + 1)/2$, $a = N$; put $M := (N - 1)/2$. (Corner $\rightarrow$ centre in $d = 1$ is the end-to-end problem on $\{1, \dots, c\}$, because the walk cannot pass c before hitting it; its threshold is the end-to-end threshold of $(N + 1)/2$.) For the end-to-end pair Proposition 4.4 gives $q^*(N) \leq (2N - 4)/(2N - 3)$, $N \geq 4$. The analogue for the centre:

Proposition 7.3 (centre $\rightarrow$ corner in one dimension: the elementary dip). *Let $d = 1$, $N \geq 3$ odd, $M = (N - 1)/2$, $x_0 = M + 1$, $a = N$, $\eta := 1 - q$. Then $f_q(t) = 0$ for $t < M$ and*

$$f_q(M) = (q/2)^M, \quad f_q(M + 1) = (q/2)^M M\eta, \quad f_q(M + 2) = (q/2)^M \left[\binom{M+1}{2} \eta^2 + M \frac{q^2}{4} \right].$$

Consequently $f_q(M + 1) < f_q(M)$ iff $q > 1 - 1/M$, and $f_q(M + 2) - f_q(M + 1) = (q/2)^M \frac{M}{4} \Psi_M(\eta)$ with $\Psi_M(\eta) := (2M + 3)\eta^2 - 6\eta + 1$. Hence:

(a) $N = 3$: f_q is unimodal iff $q \leq 4/5$.

(b) $N = 5$: $f_q(2) > f_q(3) < f_q(4)$, so f_q is not unimodal, for every $q > (4 + \sqrt{2})/7 = 0.7734590\dots$

(c) $N \geq 7$: $f_q(M) > f_q(M + 1) < f_q(M + 2)$, so f_q is not unimodal, for every $q \in (\frac{N-3}{N-1}, 1]$.

In particular $q^(5) \leq (4 + \sqrt{2})/7$, $q^*(7) \leq 2/3$, $q^*(9) \leq 3/4$, all below $4/5$: at the value $q = 4/5$ the centre $\rightarrow$ corner PMF in one dimension is not unimodal for $N = 5, 7, 9$.*

Proof. The M sites $M + 1, \dots, 2M = N - 1$ between start and target are interior sites ($M + 1 \geq 2$), with hold probability η . A path that first reaches N at time $M + j$ consists of the M forced right

steps and j extra steps. $j = 0$: one path. $j = 1$: one hold, at one of the M sites. $j = 2$: two holds (a multiset of two of the M sites: $\binom{M+1}{2}$ choices, weight η^2), or one left step from a site $k \in \{M+1, \dots, 2M\}$ to $k-1 \geq M \geq 1$, necessarily followed at once by the step back (M paths, each with the extra factor $(q/2)^2$). This gives the three formulas. $f_q(M+1) < f_q(M)$ iff $M\eta < 1$, and

$$\binom{M+1}{2}\eta^2 + \frac{M}{4}(1-\eta)^2 - M\eta = \frac{M}{4}[2(M+1)\eta^2 + (1-\eta)^2 - 4\eta] = \frac{M}{4}\Psi_M(\eta).$$

(a) $M = 1$: $M\eta < 1$ always, and $\Psi_1(\eta) = (5\eta - 1)(\eta - 1) > 0$ iff $\eta < 1/5$; so $f_q(1) > f_q(2) < f_q(3)$ for every $q > 4/5$. At $q = 4/5$: on $\{1, 2\}$, $5Q = \binom{3}{2}$, $R_0 = e_2 = (0, 1)$, $R_1 = (2, 1)$, $R_2 = (8, 5) \leq 5R_1 = (10, 5)$; so $\rho_2 \leq \rho_1$, hence $\rho_{t+1} \leq \rho_t$ and $f(t+2) \leq f(t+1)$ for all $t \geq 1$ (Lemma 6.1), and $f(1) = 2/5 > f(2) = 2/25 (= f(3))$: $f_{4/5}$ is nonincreasing on $t \geq 1$, hence unimodal; Corollary 2.6 gives all $q \leq 4/5$.

(b) $M = 2$: $\Psi_2(\eta) = 7\eta^2 - 6\eta + 1$ has the roots $(3 \pm \sqrt{2})/7$, so $\Psi_2 > 0$ for $\eta < (3 - \sqrt{2})/7 = 0.2265\dots$, and there $M\eta < 1$.

(c) $M = 3$: $\Psi_3(\eta) = (3\eta - 1)^2 > 0$ for $\eta < 1/3$. $M \geq 4$: $\Psi'_M(\eta) = 2(2M+3)\eta - 6 < 0$ for $\eta < 3/(2M+3)$, and $1/M \leq 3/(2M+3)$; so for $0 \leq \eta < 1/M$, $\Psi_M(\eta) > \Psi_M(1/M) = (M-1)(M-3)/M^2 > 0$. Together with $M\eta < 1$ this is the dip, and $1 - 1/M = (N-3)/(N-1)$. $\square$

Remark (general pairs in one dimension). The proof of Proposition 7.3 uses only that the sites strictly between start and target are interior sites. In $d = 1$ every pair reduces, by reflection, to a start x_0 and the target N at the end; if $x_0 \geq 2$ and $D := N - x_0 \geq 2$, the three formulas hold with M replaced by D , so $f_q(D) > f_q(D+1) < f_q(D+2)$ for every $q > (4 + \sqrt{2})/7$ if $D = 2$ and every $q > 1 - 1/D$ if $D \geq 3$. In particular, for $D \in \{2, 3, 4\}$ the PMF is not unimodal at $q = 4/5$: the hypothesis $q \leq 1/2$ of Theorem 4.3 cannot be replaced by $q \leq 4/5$ for general pairs (smallest case $N = 4$, $x_0 = 2$: $f(2) = \frac{4}{25} > f(3) = \frac{8}{125} < f(4) = \frac{44}{625}$). An exact scan at $q = 4/5$ of all 66 starts with $2 \leq N \leq 12$ (`scripts/scan_pairs1d.py`; the certificate of Theorem 7.4, cross-checked on exact windows of $30N^2 + 100$ steps) finds 29 non-unimodal ones, all with $x_0 \geq 2$ and $D \in \{2, 3, 4, 5\}$; for $D = 5$ the failure is a later dip, $f(9) > f(10) < f(11)$ ($x_0 \geq 3$), and $N = 7$, $x_0 = 2$ is unimodal.

Computer-assisted Theorem 7.4 (one-dimensional thresholds; exact arithmetic). *Let $d = 1$.*

(a) End to end ($x_0 = 1$, $a = N$). $q^*(2) = q^*(3) = 1$. For $N = 4$, for $N = 20$ and for every $22 \leq N \leq 300$,

$$q^*(N) = \frac{2N-4}{2N-3} :$$

at this value of q the PMF is unimodal (with the tie $f(N-1) = f(N)$), and it is not unimodal for any larger q . For $5 \leq N \leq 19$ and for $N = 21$ the PMF at $q = (2N-4)/(2N-3)$ is not unimodal, so $q^*(N) < (2N-4)/(2N-3)$; more precisely $k_N \cdot 10^{-6} \leq q^*(N) < (k_N + 1) \cdot 10^{-6} < (2N-4)/(2N-3)$ with

N	5	6	7	8	9	10	11	12
k_N	835384	871885	882823	898653	917913	920871	937309	940937
$\lfloor 10^6 \frac{2N-4}{2N-3} \rfloor$	857142	888888	909090	923076	933333	941176	947368	952380
N	13	14	15	16	17	18	19	21
k_N	952643	956690	960857	962997	965456	969511	969312	974226
$\lfloor 10^6 \frac{2N-4}{2N-3} \rfloor$	956521	960000	962962	965517	967741	969696	971428	974358

For $N = 21$, $q = 38/39$: $f(142) > f(143) < f(144)$.

(b) Centre to corner (N odd, $x_0 = (N + 1)/2$, $a = N$). $q^*(3) = 4/5$; $q^*(5) = (4 + \sqrt{2})/7$; $q^*(N) = (N - 3)/(N - 1)$ for $N \in \{7, 9, 17, 21\}$ and for every odd $23 \leq N \leq 301$. For $N \in \{11, 13, 15, 19\}$ the PMF at $q = (N - 3)/(N - 1)$ is not unimodal, and

$$q^*(11) \in [0.783458, 0.783459), \quad q^*(13) \in [0.830332, 0.830333), \\ q^*(15) \in [0.854370, 0.854371), \quad q^*(19) \in [0.885540, 0.885541).$$

(c) The value in one dimension. At $q = 4/5$ (hence, by Corollary 2.6, for the statement “unimodal for all $q \leq 4/5$ ”): the end-to-end PMF and the corner $\rightarrow$ centre PMF are unimodal for every N ; the centre $\rightarrow$ corner PMF is unimodal if and only if $N \notin \{5, 7, 9, 11\}$. Exact witnesses at $q = 4/5$: $N = 5$: $f(2) = \frac{4}{25} > f(3) = \frac{8}{125} < f(4) = \frac{44}{625}$; $N = 7$: $f(3) = \frac{8}{125} > f(4) = \frac{24}{625} < f(5) = \frac{144}{3125}$; $N = 9$: $f(4) = \frac{16}{625} > f(5) = \frac{64}{3125} < f(6) = \frac{416}{15625}$; $N = 11$: $f(9) = \frac{1216}{78125} > f(10) = \frac{149952}{9765625} < f(11) = \frac{151104}{9765625}$.

Proof. `scripts/cert_thr1d.py`. For a rational $q = q_n/q_d$ let $g := \gcd(q_n, 2(q_d - q_n), 2q_d - q_n, 2q_d)$ and $\mathfrak{D} := 2q_d/g$. Then $R_t := \mathfrak{D}^t \rho_t$ is an integer vector, $R_{t+1} = \mathfrak{M}R_t$ with the integer Jacobi matrix $\mathfrak{M} = \mathfrak{D}Q_q$ (off-diagonal q_n/g , diagonal $2(q_d - q_n)/g$, first diagonal entry $(2q_d - q_n)/g$), and $F_{t+1} := \mathfrak{D}^{t+1}f_q(t+1) = (q_n/g)R_t(N-1)$. The program records the exact signs of $F_j - \mathfrak{D}F_{j-1}$ ($= \mathfrak{D}^j(f_q(j) - f_q(j-1))$) for $j = 1, 2, \dots$ and stops either at the first T with $R_{T+1} \leq \mathfrak{D}R_T$ componentwise — then Lemma 6.1 gives $f_q(t+2) \leq f_q(t+1)$ for all $t \geq T$, and the verdict is “unimodal” iff the recorded signs are $(\geq 0) \dots (\geq 0)(\leq 0) \dots (\leq 0)$ (Lemma 1.2) — or at the first j at which the nonzero signs so far contain $+, -, +$: verdict “not unimodal”, with the exact witness.

(a) $N = 2$: $f_q(t) = \frac{q}{2}(1 - \frac{q}{2})^{t-1}$. $N = 3$, $q = 1$: $2Q_1 = (\frac{1}{1} \frac{1}{0})$, $R_t = (F_{t+1}, F_t)$ (Fibonacci numbers), $f(t+1) = F_t/2^{t+1}$, i.e. $f = 0, 0, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{3}{32}, \dots$, and $R_2 = (2, 1) \leq 2R_1 = (2, 2)$: unimodal at $q = 1$, hence for all q . For $N = 4, 20, 22, \dots, 300$ the verdict at $q = (2N - 4)/(2N - 3)$ (where $\mathfrak{D} = 2N - 3$, off-diagonal $N - 2$, diagonal 1, first diagonal entry $N - 1$) is “unimodal”; in every one of these cases the cone holds at $T = \text{mode} - 1$. By Proposition 4.4 the PMF is not unimodal for $q > (2N - 4)/(2N - 3)$; so $q^*(N) = (2N - 4)/(2N - 3)$. For $5 \leq N \leq 19$ and $N = 21$ the verdict at $q = (2N - 4)/(2N - 3)$ is “not unimodal”, the verdict at $q = k_N/10^6$ is “unimodal” and at $q = (k_N + 1)/10^6$ “not unimodal” (k_N found by bisection; only the two final verdicts matter); Corollary 2.6(1) gives $k_N \cdot 10^{-6} \leq q^*(N) < (k_N + 1) \cdot 10^{-6}$.

(b) $N = 3$: Proposition 7.3(a). For $N \in \{7, 9, 17, 21\}$ and odd $23 \leq N \leq 301$ the verdict at $q = (N - 3)/(N - 1)$ is “unimodal”, and Proposition 7.3(c) gives non-unimodality above. For $N \in \{11, 13, 15, 19\}$: verdicts “unimodal” at the lower and “not unimodal” at the upper end of the stated intervals, and the upper ends are smaller than $(N - 3)/(N - 1)$. $N = 5$: for $q = (4 + \sqrt{2})/7$ one has $q/2 = (4 + \sqrt{2})/14$, $1 - q = (6 - 2\sqrt{2})/14$, $1 - q/2 = (10 - \sqrt{2})/14$; the same recursion is run in the ring $\mathbb{Z}[\sqrt{2}]$ (with $\mathfrak{D} = 14$), the sign of $a + b\sqrt{2}$ being decided exactly from the signs of a , b and of $a^2 - 2b^2$. The cone holds at $T = 6$ and the signs of $f(j) - f(j-1)$, $j = 1, \dots, 7$, are $0, +, -, 0, -, -, -$ (so $f(2) > f(3) = f(4) > f(5) > \dots$): unimodal at $q = (4 + \sqrt{2})/7$; Proposition 7.3(b) gives non-unimodality above.

(c) End to end: Theorem 9.13(i) with $d = 1$. Corner $\rightarrow$ centre is the end-to-end pair on $\{1, \dots, (N + 1)/2\}$. Centre $\rightarrow$ corner: for $N \geq 25$, Theorem 9.13(ii) (the sequence $\varphi_1 = \delta \tilde{h}_q$ is log-concave) and Theorem 4.11(b); for $N = 3$ and odd $13 \leq N \leq 23$, part (b) ($q^* \geq 4/5$: $q^*(13) \geq 0.830332$, $q^*(15) \geq 0.854370$, $q^*(17) = 7/8$, $q^*(19) \geq 0.885540$, $q^*(21) = 9/10$, $q^*(23) = 10/11$); for $N \in \{5, 7, 9\}$, Proposition 7.3; for $N = 11$, $q^*(11) < 0.783459 < 4/5$. The witnesses are exact rationals computed from the definition.

Two separately written implementations were used: plain Python integers, and integer polynomials of python-flint (one step is the multiplication of the polynomial $\sum_x R_t(x)z^x$ by $(q_n/g)(1 + z^2) + (2(q_d - q_n)/g)z$, plus the wall term); they agree (verdict, cone time, mode, witness) on all $4 \leq N \leq 200$ in (a) and all odd $7 \leq N \leq 81$ in (b). Records: `data/cert_thr1d_boundf.jsonl`, `cert_thr1d_bound.jsonl`, `cert_thr1d_bisect.jsonl`, `cert_thr1d_m2cbound.jsonl`, `cert_thr1d_m2c.json`. $\square$

Size. (a): $N = 300$ needs 29 945 steps with integers of 276 116 bits; the flint runs for $4 \leq N \leq 300$ took 1.84 hours of wall-clock time in total on a shared, heavily loaded machine (largest single N : 205 s), the Python-integer runs for $4 \leq N \leq 200$ 0.44 hours. (b): 0.57 hours (largest single N : 167 s).

Remark 7.5 (the exceptional N ; heuristic). (1) For $5 \leq N \leq 19$ and $N = 21$ the failure at $q = (2N - 4)/(2N - 3)$ is a parity zig-zag at the top of the PMF: on a block of consecutive times $t_1 \leq j \leq m$ the differences $f(j) - f(j - 1)$ alternate $-, +, -, +, \dots, -, +$ (one-step dips $f(t - 1) > f(t) < f(t + 1)$ at $t = t_1, t_1 + 2, \dots$); before the block f is nondecreasing, after it strictly decreasing (exact, on a window of $12N^2 + 200$ steps), and the global maximum lies in $[t_1 - 1, m]$. There is one dip for $N \in \{5, 14, 16, 17, 18, 19, 21\}$, two for $N \in \{6, 9, 11, 12, 13, 15\}$ and three for $N \in \{7, 8, 10\}$; the first dip is at $(N: t_1)$ 5: 7, 6: 8, 7: 11, 8: 16, 9: 23, 10: 28, 11: 37, 12: 44, 13: 53, 14: 62, 15: 71, 16: 82, 17: 93, 18: 104, 19: 117, 21: 143 (the mode is near $N^2/3$; `verify_thr1d.py` Z12) — the same phenomenon as for $d = 2$, $q = 1$ (Theorem 7.1(c), Observation 7.2). The parity-odd part of f_q is governed by the most negative eigenvalue of Q_q , $\lambda_- = 1 - q(1 + \cos(2\pi/(2N - 1)))$, which at $q = (2N - 4)/(2N - 3)$ is $-(1 - 2/(2N - 3)) + O(N^{-2})$; at the modal time $t \approx N^2/3$ its relative size is of order $e^{-N/3}$ times an algebraic factor, to be compared with the relative second difference $\approx N^{-4}$ of the smooth part at its maximum. The two cross around $N \approx 20$, and whether a dip appears near the crossing depends on the position of the continuous mode relative to the integers — which is why $N = 20$ is not exceptional while $N = 21$ is. For larger N the parity-odd part is exponentially negligible at the mode. *This explains why the exceptional set should be finite, but it is not a proof:* a proof would also have to control the early times $t - (N - 1) \ll N$, where at $q = (2N - 4)/(2N - 3)$ the PMF is a near-staircase ($f(N - 1 + 2k + 1)/f(N - 1 + 2k) = 1 + O(1/N)$ for fixed k , with exact equality at $k = 0$). See Conjecture 10.2.

(2) q^* is not monotone in N : $q^*(19) < 0.969313 < 0.969511 \leq q^*(18)$ (end to end).

(3) The two centre geometries differ in $d = 1$: corner $\rightarrow$ centre has $q^* \geq 4/5$ for every N , centre $\rightarrow$ corner has $q^*(7) = 2/3$.

Sanity checks (`scripts/verify_thr1d.py` $\rightarrow$ `data/verify_thr1d.json`; code written from the definition, separately from `cert_thr1d.py`; all pass): Z1 (the three formulas of Proposition 7.3 and the identity with Ψ_M , exact, $M \leq 10$, 7 activities), Z2 (the dip for 40 rational q above the threshold, $M \leq 30$, and $\Psi_M(1/M) = (M - 1)(M - 3)/M^2$), Z3 (the hand computations for $N = 4$ end to end and $N = 3$ centre $\rightarrow$ corner), Z4 (for every bracket of (a) and (b): the exact differences on a window of $12N^2 + 200$ steps have the pattern $+$ $-$ at the lower end and contain $+$ $-$ $+$ at the upper end and at the bound; unimodal pattern at the bound for $N = 4, 20, 22, 23, 30, 40, 55$, and $+$ $-$ $+$ just above it), Z5 (the witnesses of (a), (c) as exact rationals; for $N = 21$ the relative dip is $3.5 \cdot 10^{-6}$ and the relative rise $2.5 \cdot 10^{-5}$), Z6 ($q^*(5) = (4 + \sqrt{2})/7$: 80-digit evaluation, $|f(3) - f(4)| < 10^{-70}$, symbolic verification of the tie, dip at $q + 10^{-9}$), Z7 (both certificate engines against definition-level windows of $30N^2 + 100$ steps in 60 random cases with $N \leq 14$ and arbitrary start), Z8 (audit of the records: completeness of the ranges, the exceptional sets, agreement of the two engines, cone at mode -1), Z12 (the shape of the failure in Remark 7.5(1), exact), Z9 (Theorem 4.1(d)), Z10 (Lemma 3.1: $P^{2(n'-1)} > 0$ for 79 components, exact), Z11 (the count in Corollary 2.6(4)).

8 General start/target pairs: unimodality is false in $d \geq 2$

Theorem 8.1 (explicit counter-examples). (a) $d = 3$, $N = 3$, $x_0 = (3, 2, 2)$ (centre of a face), $a = (1, 2, 2)$ (centre of the opposite face). With unit rate,

$$g_1\left(\frac{21}{10}\right) = 0.0135725889\dots > g_1(3) = 0.0135186607\dots < g_1(13) = 0.0171550593\dots,$$

$$\text{and at } q = 1/2: f(2) = \frac{1}{144} > f(6) = \frac{6577}{995328} < f(25) = 0.0086991405\dots$$

(b) $d = 2$, $N = 8$, $x_0 = (6, 6)$, $a = (3, 4)$: $g_1(\frac{21}{2}) > g_1(\frac{49}{2}) < g_1(\frac{59}{2})$
 $(0.0061735\dots, 0.0059260\dots, 0.0059296\dots)$, and at $q = 1/2$: $f(19) > f(46) < f(63)$.

In both cases neither the continuous-time density (any rate) nor the PMF (any $q \in (0, 1]$) is unimodal.

Proof. By (4) with $q_0 = 1$, $e^t g_1(t) = \sum_n \frac{t^n}{n!} a_1(n)$ with exact rationals $a_1(n) \in [0, 1]$. For rational t and $M + 2 > t$, $P_M(t) = \sum_{n \leq M} \frac{t^n}{n!} a_1(n)$ is an exact rational and $0 \leq e^t g_1(t) - P_M(t) \leq \frac{t^{M+1}}{(M+1)!} (1 - \frac{t}{M+2})^{-1}$; for rational $\delta \geq 0$, $\sum_{k \leq K} \frac{\delta^k}{k!} \leq e^\delta \leq \sum_{k \leq K} \frac{\delta^k}{k!} + \frac{\delta^{K+1}}{(K+1)!} (1 - \frac{\delta}{K+2})^{-1}$. With these enclosures the stated inequalities reduce to comparisons of exact rationals, verified by `scripts/cert_continuous.py` ($M = 119$, resp. 167; relative widths below 10^{-66}). The discrete values are exact rationals (`data/exact_examples.json`). The last statement is Corollary 2.6(3). $\square$

Computer-assisted Theorem 8.2 (exhaustive classification for small boxes). *For $d = 2$, $2 \leq N \leq 12$ and $d = 3$, $2 \leq N \leq 6$, consider all ordered pairs (x_0, a) with the target reduced by the symmetries of the box. Each pair is of exactly one of the types*

U the PMF is unimodal for $q = 1/2$ (hence for all $q \leq 1/2$ and in continuous time);

A the PMF is not unimodal at $q = 1/2$ but is unimodal at $q = 2^{-k}$ for some $2 \leq k \leq 6$ (hence the continuous-time density is unimodal);

B the continuous-time density is not unimodal (hence the PMF is not unimodal for any q),

with the following counts.

d	N	pairs	U	A	B	d	N	pairs	U	A	B
2	2	3	3	0	0	2	10	1485	1481	3	1
2	3	24	24	0	0	2	11	2520	2506	4	10
2	4	45	43	2	0	2	12	3003	2994	1	8
2	5	144	142	2	0	3	2	7	7	0	0
2	6	210	209	1	0	3	3	104	103	0	1
2	7	480	479	1	0	3	4	252	251	0	1
2	8	630	628	1	1	3	5	1240	1222	4	14
2	9	1200	1194	2	4	3	6	2150	2134	2	14

In particular, in $d = 2$ the continuous-time density is unimodal for every pair when $N \leq 7$.

Proof. `scripts/cert_pairs.py`: exact cone certificate (Lemma 6.1) at $q = 1/2$; for the pairs with a pattern $+, -, +$, exact cone certificates at $q = 1/4, \dots, 1/64$; for those still multimodal at $q = 1/64$, the exact three-point certificate of Theorem 8.1 at times located in floating point and rounded to multiples of $1/8$. No pair was left undecided. Corollary 2.6 gives the implications in brackets. $\square$

In every one of the 54 pairs of type **B** there is a coordinate in which start and target lie strictly on opposite sides of the mid-plane of the box, and $2 \leq D \leq 8$; g' has exactly three sign changes (float).

Remark 8.3 (mechanism; heuristic). For a start close to the target the density has an early peak of direct arrivals. If the pair straddles the centre, the unkilld propagator $u(t)$ overshoots the uniform value N^{-d} , then undershoots it (the slowest mode $\sum_i \cos(\pi(x_i - \frac{1}{2}))/N$ has opposite signs at x_0 and a) and returns to it from below after reflection at the walls; the first-passage density inherits the undershoot and the recovery. In the corner-to-corner geometry there is no direct peak.

Proposition 8.4 (unimodal but not strongly unimodal). *For $d = 2$, $N = 3$, $x_0 = (2, 2)$, $a = (3, 3)$, $q = 1/2$: $f(4) = \frac{67}{2048}$, $f(5) = \frac{481}{16384}$, $f(6) = \frac{3457}{131072}$, so $f(5)^2 = 231361/2^{28} < f(4)f(6) = 231619/2^{28}$. The PMF is unimodal (Computer-assisted Theorem 8.2) but not log-concave.*

Numerical observation 8.5. (Exact arithmetic, windows of six cone times, $q = 1/2$.) For centre $\rightarrow$ corner log-concavity and hazard monotonicity fail, after the mode, for every odd $N \leq 31$ ($d = 2$) and $N \leq 11$ ($d = 3$). For corner $\rightarrow$ centre both hold on the same windows. A type-A example: $(2, 4) \rightarrow (2, 2)$ in the 4×4 box at $q = 1/2$, $f(4) = \frac{9}{512} > f(5) = \frac{287}{16384} < f(8) = \frac{73899}{4194304}$.

9 Discrete time up to the value: $q \leq 4/5$, every N , every d

Theorem 4.12 covers $q \leq q_N$, and $q_N \downarrow 1/2$. The numerical study of the article uses $q = 4/5$. This section closes the gap up to $q = 4/5$ for *all* N at once:

for every $q \in (0, 4/5]$ the corner-to-corner PMF is unimodal for all $N \geq 2$ and all $d \geq 1$, and the corner $\rightarrow$ centre and centre $\rightarrow$ corner PMFs are unimodal for all odd $N \geq 3$ and all $d \geq 2$ (Theorem 9.13);

and $4/5$ cannot be replaced by a larger number in the first statement ($d = 1$, $N = 4$, Proposition 4.4).

Plan of the proof.

1. By Theorem 4.11(b) it suffices that the increment $\varphi = \Delta u$ of the *free* propagator is log-concave. By (13) φ is a binomial convolution of one-dimensional sequences, and binomial convolution preserves log-concavity. So one-dimensional log-concavity suffices (Proposition 9.1), and dimensions can be combined in blocks (Lemma 9.3).
2. The one-dimensional sequence is, up to a geometric factor, $W_n = ((yI + \text{Adj } P_m)^n)(1, m) = h_{n-m+1}(\lambda_1, \dots, \lambda_m)$, $\lambda_k = y + 2 \cos(k\pi/N)$, $y = 2(1-q)/q$ (Lemma 9.4): a finite exponential sum with coefficients of alternating signs. Its log-concavity for $n \geq n_0$ follows from the two leading terms (tail Lemma 9.5; explicit $n_0 = O(N^2 \log N)$ in Proposition 9.6), and for $n < n_0$ it is a finite computation — done for $N \leq 1001$ (Computer-assisted Theorem 9.7).
3. For $N > 1001$ the spectrum is split into residue classes $k \bmod G$. Each class is a “twisted group” of 16 or 17 points $y + 2 \cos((l + \beta)\delta)$ from a *compact two-parameter family*; the h -sequence of the whole spectrum is the ordinary convolution of the h -sequences of the groups (Lemma 9.9), ordinary convolution preserves log-concavity (Lemma 9.2), and every member of the family is log-concave (Computer-assisted Lemma 9.10; Taylor models with Cauchy bounds in ball arithmetic). This gives all N (Theorem 9.11).
4. For the few small N where the one-dimensional sequence is *not* log-concave at $q = 4/5$, the blocks $d = 2, 3$ (or $3, 4, 5$; or $4, \dots, 7$) are (Computer-assisted Proposition 9.12).

“Computer-assisted” in this section means: certified ball arithmetic (Arb, through python-flint 0.9; every decision is a comparison of balls that is true for all points of the balls), cross-checked against exact integer arithmetic where stated.

9.1 Reduction to one dimension; convolutions; blocks

Notation. $K_q = (1 - q)I + qP^{(1)}$ is the one-dimensional lazy walk on $\{1, \dots, N\}$. $h_q(n) := K_q^n(1, N)$, $\delta h_q(n) := h_q(n+1) - h_q(n)$ (end to end); for odd N , $c = (N+1)/2$: $\tilde{h}_q(n) := K_q^n(1, c)$, $\delta \tilde{h}_q(n) := \tilde{h}_q(n+1) - \tilde{h}_q(n)$ (end to centre). For a dimension d and one of the three geometries, $u_d(t) := P_q^t(x_0, a)$ is the free propagator and $\varphi_d(t) := u_d(t) - u_d(t-1)$ ($t \geq 1$; $\varphi_d(t) := 0$ for $t \leq 0$), as in Section 4.4. “Log-concave sequence” has the meaning of Section 4.4 (nonnegative, support a set of consecutive integers, $\varphi(t)^2 \geq \varphi(t-1)\varphi(t+1)$); (11) holds for such sequences.

Proposition 9.1 (reduction to one dimension). *Let N and $q \in (0, 1]$ be such that the one-dimensional sequence δh_q is log-concave [resp., for odd N , the sequence $\delta \tilde{h}_q$]. Then for every $d \geq 1$ the sequence φ_d of the corner $\rightarrow$ opposite corner pair [resp. of the corner $\rightarrow$ centre and centre $\rightarrow$ corner pairs] of $\{1, \dots, N\}^d$ is log-concave, and the first-passage PMF f_q is unimodal. Moreover, if δh_q [$\delta \tilde{h}_q$] is log-concave, so is $\delta h_{q'}$ [$\delta \tilde{h}_{q'}$] for every $q' < q$.*

Proof. $h_q(n) = \sum_{i < n} \delta h_q(i)$ is log-concave by Lemma 4.10(a)(ii) ($\lambda = c = 1$). Identity (13) holds for every q , so by Lemma 4.10(b) φ_d is log-concave, and Theorem 4.11(b) applies. For the last statement, $K_{q'} = (1 - p)I + pK_q$ with $p = q'/q$, hence, exactly as in the proof of Theorem 2.5(a), $\delta h_{q'}(n) = p \sum_i \binom{n}{i} (1 - p)^{n-i} p^i \delta h_q(i)$, a binomial convolution of the log-concave sequences $(p^i \delta h_q(i))_i$ and $((1 - p)^k)_k$; apply Lemma 4.10(b). $\square$

Lemma 9.2 (ordinary convolution preserves log-concavity). *Let $a = (a_j)_{j \geq 0}$ and $b = (b_j)_{j \geq 0}$ be log-concave sequences (extended by zero to negative indices). Then $c_n := \sum_{k=0}^n a_k b_{n-k}$ is a log-concave sequence. If $a_j > 0$ and $b_j > 0$ for all $j \geq 0$, then $c_j > 0$ for all $j \geq 0$.*

Proof. For $n \geq 1$ consider the $2 \times \infty$ matrix with rows $(a_{n-k})_{k \geq 0}$, $(a_{n+1-k})_{k \geq 0}$ and the $\infty \times 2$ matrix with columns $(b_k)_{k \geq 0}$, $(b_{k-1})_{k \geq 0}$ (only finitely many non-zero products occur). Their product is $\begin{pmatrix} c_n & c_{n-1} \\ c_{n+1} & c_n \end{pmatrix}$, and the Cauchy–Binet formula gives

$$c_n^2 - c_{n-1}c_{n+1} = \sum_{0 \leq k < l} (a_{n-k}a_{n+1-l} - a_{n-l}a_{n+1-k})(b_k b_{l-1} - b_{k-1} b_l).$$

For $k < l$ both factors are ≥ 0 by (11): $a_{n-l}a_{n+1-k} \leq a_{n+1-l}a_{n-k}$ (take $i = n+1-l \leq j = n-k$) and $b_{k-1}b_l \leq b_k b_{l-1}$ (take $i = k \leq j = l-1$). The support of c is the sum of the supports. $\square$

(Classical: the convolution of two PF_2 sequences is PF_2 ; S. Karlin, *Total Positivity*, Vol. I, Stanford Univ. Press 1968, Ch. 8. Sanity check: `verify_lastexit.py` W6.)

Lemma 9.3 (blocks). *Fix N , q and one of the geometries (CC), (C2M)/(M2C).*

(a) *If φ_d is log-concave, so is u_d .*

(b) *If φ_{d_1} and u_{d_2} are log-concave, then $\varphi_{d_1+d_2}$ is log-concave.*

(c) *If φ_d is log-concave for the activity q , it is log-concave for every activity $q' < q$.*

Proof. (a) $u_d(t) = \sum_{s \leq t} \varphi_d(s)$; Lemma 4.10(a)(ii) with $\lambda = c = 1$. (b) By (13), $d^{t-1} \varphi_d(t) = (\delta h \star h^{*(d-1)})_{t-1}$ for every d ($h = h_q$ or $\tilde{h}_q$), and in the same way $d^t u_d(t) = (h^{*d})_t$. The binomial convolution $\star$ is associative and commutative (it is the product of exponential generating functions), so with $d = d_1 + d_2$

$$d^{t-1} \varphi_d(t) = \sum_{k=0}^{t-1} \binom{t-1}{k} d_1^k \varphi_{d_1}(k+1) \cdot d_2^{t-1-k} u_{d_2}(t-1-k).$$

The sequences $(d_1^k \varphi_{d_1}(k+1))_k$ and $(d_2^k u_{d_2}(k))_k$ are log-concave; apply Lemma 4.10(b). (c) $P_{q'} = (1 - p)I + pP_q$ gives $u_{d,q'}(t) = \sum_n b_p(t, n) u_{d,q}(n)$ and, with Pascal's rule as in the proof of Theorem 2.5(a), $\varphi_{d,q'}(t+1) = p \sum_n \binom{t}{n} (1 - p)^{t-n} p^n \varphi_{d,q}(n+1)$; apply Lemma 4.10(b). $\square$

9.2 The one-dimensional sequences: dictionary, tail lemma, direct certificates

For $m \geq 1$ and $y \geq 0$ let $\text{Adj}(P_m)$ be the adjacency matrix of the path with vertices $1, \dots, m$, and

$$\begin{aligned} J &:= yI + \text{Adj}(P_m), & \tilde{J} &:= yI + \text{Adj}(P_m) - e_m e_m^\top, \\ W_n &:= (J^n)(1, m), & \tilde{W}_n &:= (\tilde{J}^n)(1, m) \quad (n \geq 0). \end{aligned}$$

$h_j(x_1, \dots, x_m)$ denotes the complete homogeneous symmetric polynomial of degree j ($h_0 = 1$, $h_j = 0$ for $j < 0$), i.e. $\sum_j h_j z^j = \prod_k (1 - x_k z)^{-1}$.

Lemma 9.4 (dictionary). *Let $q \in (0, 1]$ and $y := 2(1 - q)/q$.*

(a) *End to end. Let $N \geq 2$, $m = N - 1$, $\lambda_k := y + 2 \cos(k\pi/N)$ ($k = 1, \dots, m$). Then for all $n \geq 0$*

$$\delta h_q(n) = (q/2)^{n+1} W_n,$$

$$W_n = h_{n-m+1}(\lambda_1, \dots, \lambda_m) = \sum_{k=1}^m c_k \lambda_k^n, \quad c_k = (-1)^{k-1} \frac{2}{N} \sin^2 \frac{k\pi}{N}.$$

$W_n = 0$ for $n < m - 1$, $W_{m-1} = 1$, and $W_n > 0$ for all $n \geq m - 1$ if $y > 0$.

(b) *End to centre. Let $N = 2M + 1$, $m = M$, $\tilde{\lambda}_k := y + 2 \cos(2k\pi/N)$ ($k = 1, \dots, M$). Then for all $n \geq 0$*

$$\delta \tilde{h}_q(n) = (q/2)^{n+1} \tilde{W}_n,$$

$$\tilde{W}_n = h_{n-M+1}(\tilde{\lambda}_1, \dots, \tilde{\lambda}_M) = \sum_{k=1}^M \tilde{c}_k \tilde{\lambda}_k^n, \quad \tilde{c}_k = (-1)^{k-1} \frac{4}{N} \sin \frac{2k\pi}{N} \sin \frac{k\pi}{N}.$$

(c) *First inequality. $W_m^2 - W_{m-1}W_{m+1} = e_2(\lambda) = (m-1)(my^2/2 - 1)$ and $\tilde{W}_M^2 - \tilde{W}_{M-1}\tilde{W}_{M+1} = e_2(\tilde{\lambda}) = \frac{1}{2}(M-1)(My^2 - 2y - 2)$. So the first inequality holds iff $q \leq q_1(N) := 1/(1 + (2(N-1))^{-1/2})$ in case (a), and iff $q \leq \tilde{q}_1(N) := 1/(1 + \xi_M)$, $\xi_M := (1 + (1 + 2M)^{1/2})/(2M)$, in case (b). At $q = 4/5$ ($y = 1/2$) these read $N \geq 9$ and $N \geq 25$, with equality exactly for $N = 9$ resp. $N = 25$.*

Consequently $\delta h_q [\delta \tilde{h}_q]$ is a log-concave sequence iff $W_n > 0$ for all $n \geq m - 1$ and $W_n^2 \geq W_{n-1}W_{n+1}$ for all $n \geq m$ [same with $\tilde{W}$].

Proof. (a) By the computation in the proof of Lemma 4.9(b) (K_q is a birth-death chain on $N = m + 1$ sites with $\prod_i p_i = (q/2)^m$, whose eigenvalues other than 1 are $\mu_k = 1 - q + q \cos(k\pi/N) = (q/2)\lambda_k$, Lemma 1.1), $\sum_n h_q(n)z^n = (q/2)^m z^m / ((1-z) \prod_k (1 - \mu_k z))$. Since $h_q(0) = 0$, multiplying by $(1-z)/z$ gives $\sum_n \delta h_q(n)z^n = (q/2)^m z^{m-1} \prod_k (1 - \mu_k z)^{-1}$, i.e. $\delta h_q(n) = (q/2)^m h_{n-m+1}(\mu) = (q/2)^{n+1} h_{n-m+1}(\lambda)$. On the other hand J is a Jacobi matrix with unit off-diagonal entries; its eigenvectors are $\psi_k(x) = (2/(m+1))^{1/2} \sin(k\pi x/(m+1))$ with eigenvalues λ_k , and by Cramer's rule (as in Lemma 4.9) $[(I - zJ)^{-1}](1, m) = z^{m-1} / \det(I - zJ) = z^{m-1} \prod_k (1 - \lambda_k z)^{-1}$. So $W_n = h_{n-m+1}(\lambda)$. The spectral decomposition gives $W_n = \sum_k \psi_k(1)\psi_k(m)\lambda_k^n$, and $\psi_k(m) = (-1)^{k-1}\psi_k(1)$ because $\sin(k\pi m/(m+1)) = \sin(k\pi - k\pi/(m+1))$. Finally J is entrywise non-negative, and for $y > 0$, $n \geq m - 1$ the walk $1 \rightarrow 2 \rightarrow \dots \rightarrow m$ followed by $n - m + 1$ holds has positive weight.

(b) By the proof of Theorem 4.12, $\tilde{h}_q(n) = \tilde{K}^n(M, 0)$ for the folded birth-death chain $\tilde{K}$ on $\{0, \dots, M\}$; the product of its transition probabilities along $M \rightarrow M - 1 \rightarrow \dots \rightarrow 0$ is $(q/2)^M$, and its eigenvalues other than 1 are those eigenvalues of K_q whose eigenvector is symmetric under $j \mapsto N + 1 - j$, i.e. $\tilde{\mu}_k = 1 - q + q \cos(2k\pi/N) = (q/2)\tilde{\lambda}_k$, $k = 1, \dots, M$ (Lemma 1.1: ψ_k is symmetric iff k is even). As in (a), $\delta \tilde{h}_q(n) = (q/2)^{n+1} h_{n-M+1}(\tilde{\lambda})$. The vectors $\tilde{\psi}_k(x) = (2/\sqrt{N}) \sin(x\phi_k)$, $\phi_k = 2k\pi/N$, satisfy $\tilde{\psi}_k(0) = 0$ and $\tilde{\psi}_k(M + 1) = -\tilde{\psi}_k(M)$ (because $\sin((M + 1)\phi) + \sin(M\phi) = 2 \sin((M + \frac{1}{2})\phi) \cos(\phi/2)$ and $(M + \frac{1}{2})\phi_k = k\pi$), hence $\tilde{J}\tilde{\psi}_k = (y + 2 \cos \phi_k)\tilde{\psi}_k$; they are orthonormal ($\sum_{x=1}^M \sin^2(x\phi_k) = M/2 + 1/4 = N/4$). So $\tilde{J}$ has the eigenvalues $\tilde{\lambda}_k$, $\tilde{W}_n = h_{n-M+1}(\tilde{\lambda})$ by Cramer's rule, and $\tilde{\psi}_k(1)\tilde{\psi}_k(M) = (4/N) \sin \phi_k \sin(M\phi_k)$ with $\sin(M\phi_k) = \sin(k\pi - \phi_k/2) = (-1)^{k-1} \sin(\phi_k/2)$.

(c) $h_1^2 - h_0 h_2 = e_2 = (e_1^2 - p_2)/2$ (p_2 the second power sum). In case (a) $e_1 = my$ and $p_2 = my^2 + 2(m - 1)$ ($\sum_{k=1}^{N-1} \cos(k\pi/N) = 0$, $\sum_{k=1}^{N-1} \cos^2(k\pi/N) = (N - 2)/2$). In case (b) $e_1 = My - 1$ and $p_2 = My^2 - 2y + 2M - 1$ ($\sum_{k=1}^M \cos(2k\pi/N) = -\frac{1}{2}$, $\sum_k \cos^2(2k\pi/N) = \frac{M}{2} - \frac{1}{4}$). $\square$

Lemma 9.5 (tail lemma). *Let $a(n) = \sum_{i=1}^r C_i \nu_i^n$ ($n \geq 0$) with real C_i, ν_i . Then for $n \geq 1$*

$$a(n)^2 - a(n-1)a(n+1) = - \sum_{i < j} C_i C_j (\nu_i \nu_j)^{n-1} (\nu_i - \nu_j)^2. \quad (16)$$

Suppose $\nu_1 > \nu_2 > 0$, $|\nu_i| < \nu_2$ for $i \geq 3$, and $C_1 > 0 > C_2$. Put $p_i(n) := |C_i| |\nu_i|^{n-1}$,

$$L_n := p_1(n) p_2(n) (\nu_1 - \nu_2)^2,$$

$$\text{Tot}_n := \sum_{i < j} p_i(n) p_j(n) (\nu_i - \nu_j)^2 = \left(\sum_i p_i \right) \left(\sum_i p_i \nu_i^2 \right) - \left(\sum_i p_i \nu_i \right)^2.$$

(i) *If $2L_{n_0} > \text{Tot}_{n_0}$ for some $n_0 \geq 1$, then $a(n)^2 > a(n-1)a(n+1)$ for every $n \geq n_0$.*

(ii) *If $C_1 \nu_1^{n_0} > \sum_{i \geq 2} |C_i| |\nu_i|^{n_0}$, then $a(n) > 0$ for every $n \geq n_0$.*

Proof. (16): $a(n)^2 - a(n-1)a(n+1) = \sum_{i,j} C_i C_j (\nu_i \nu_j)^{n-1} (\nu_i \nu_j - \nu_j^2)$; symmetrising in (i, j) replaces $\nu_i \nu_j - \nu_j^2$ by $-\frac{1}{2}(\nu_i - \nu_j)^2$. The second expression for Tot_n is the same symmetrisation read backwards. (i) The term $(1, 2)$ of (16) equals $+L_n$, and the others are bounded in absolute value by the corresponding terms of $\text{Tot}_n - L_n$. For $(i, j) \neq (1, 2)$ one has $|\nu_i \nu_j| \leq \nu_1 \nu_2$, so each ratio $p_i(n) p_j(n) / (p_1(n) p_2(n))$ is nonincreasing in n ; hence $(\text{Tot}_n - L_n) / L_n$ is nonincreasing, and $a(n)^2 - a(n-1)a(n+1) \geq L_n - (\text{Tot}_n - L_n) > 0$ for $n \geq n_0$. (ii) $a(n) \geq \nu_1^n (C_1 - \sum_{i \geq 2} |C_i| (|\nu_i|/\nu_1)^n)$ and the bracket is nondecreasing in n . $\square$

Proposition 9.6 (explicit tail time at $q = 4/5$; every N). *Let $y = 1/2$, $N \geq 10$, $m = N - 1$. Then $W_n^2 > W_{n-1} W_{n+1}$ for every*

$$n \geq n_{\text{tail}}(N) := 1 + \frac{N^2}{8} (10 \ln N - 8.43).$$

So for each N the log-concavity of $\delta h_{4/5}$ is a statement about the finitely many n with $N - 1 \leq n < n_{\text{tail}}(N)$.

Proof. Lemma 9.5 with $\nu_k = \lambda_k = \frac{1}{2} + 2 \cos(k\theta)$, $\theta = \pi/N$, $C_k = c_k$. Put $\Lambda_3 := \max_{k \geq 3} |\lambda_k| = \max(\lambda_3, -\lambda_m)$. Every $(i, j) \neq (1, 2)$ has $|\lambda_i \lambda_j| \leq \lambda_1 \Lambda_3$, $|\lambda_i - \lambda_j| \leq 4$, and $\sum_k |c_k| = \frac{2}{N} \sum_k \sin^2(k\pi/N) = 1$, so $\text{Tot}_n - L_n \leq 8(\lambda_1 \Lambda_3)^{n-1}$; thus $2L_n > \text{Tot}_n$ as soon as

$$(\lambda_2/\Lambda_3)^{n-1} > \frac{8}{|c_1 c_2| (\lambda_1 - \lambda_2)^2}.$$

With $\sin x \geq 2x/\pi$ on $[0, \pi/2]$ (all arguments below are $\leq \pi/2$ because $N \geq 10$): $|c_1| = \frac{2}{N} \sin^2 \theta \geq 8/N^3$, $|c_2| = \frac{2}{N} \sin^2 2\theta \geq 32/N^3$, $\lambda_1 - \lambda_2 = 4 \sin(3\theta/2) \sin(\theta/2) \geq 12/N^2$, so the right side is at most $N^{10}/4608$. For the left side: $\lambda_2 - \lambda_3 = 4 \sin(5\theta/2) \sin(\theta/2) \geq 20/N^2$ and $\lambda_2 \leq 5/2$ give, when $\lambda_3 > 0$, $\ln(\lambda_2/\lambda_3) \geq (\lambda_2 - \lambda_3)/\lambda_2 \geq 8/N^2$; and $-\lambda_m = 2 \cos \theta - \frac{1}{2} < \frac{3}{2}$, $\lambda_2 \geq \frac{1}{2} + 2 \cos(\pi/5) > 2.118$ give $\ln(\lambda_2/(-\lambda_m)) > 0.34 \geq 8/N^2$. Hence $\ln(\lambda_2/\Lambda_3) \geq 8/N^2$ (and $\lambda_1 > \lambda_2 > \Lambda_3$, $c_1 > 0 > c_2$, so the hypotheses of Lemma 9.5 hold), and the condition is implied by $(n-1) \cdot 8/N^2 > 10 \ln N - \ln 4608$; $\ln 4608 = 8.4355 \dots$ $\square$

(The certified tail times of Theorem 9.7 are about ten times smaller: $n_0 \approx 0.157N^2$.)

Computer-assisted Theorem 9.7 (direct certificates). (a) $y = 1/2$ ($q = 4/5$), end to end: for every $10 \leq N \leq 1001$, $W_n^2 > W_{n-1} W_{n+1}$ for all $n \geq N - 1$.

(b) $y = 1/2$, end to centre: for every odd $27 \leq N \leq 2001$, $\tilde{W}_n > 0$ for all $n \geq M - 1$ and $\tilde{W}_n^2 > \tilde{W}_{n-1} \tilde{W}_{n+1}$ for all $n \geq M$.

(c) At the threshold, $y = (2/(N-1))^{1/2}$ ($q = q_1(N)$), end to end: for every $4 \leq N \leq 501$, $W_n^2 \geq W_{n-1} W_{n+1}$ for all $n \geq N - 1$, with equality only for $n = N - 1$.

(d) At the threshold, $y = (1 + (1 + 2M)^{1/2})/M$ ($q = \tilde{q}_1(N)$), end to centre: for every odd $7 \leq N \leq 1001$, $\tilde{W}_n > 0$ for $n \geq M - 1$ and $\tilde{W}_n^2 \geq \tilde{W}_{n-1}\tilde{W}_{n+1}$ for all $n \geq M$, with equality only for $n = M$.

In particular ($N = 9$ is the case $m = 8$, $y = \frac{1}{2}$ of (c); $N = 25$ is the case $M = 12$, $y = \frac{1}{2}$ of (d)): $\delta h_{4/5}$ is log-concave for $9 \leq N \leq 1001$ and $\delta \tilde{h}_{4/5}$ for odd $25 \leq N \leq 2001$.

Proof. `scripts/cert_1c1d.py`. For each m : (T) the hypotheses of Lemma 9.5 are verified in ball arithmetic and an n_0 with $2L_{n_0} > \text{Tot}_{n_0}$ and $c_1\lambda_1^{n_0} > \sum_{k \geq 2} |c_k||\lambda_k|^{n_0}$ is found (so the inequalities hold for $n \geq n_0$). (W) For $m \leq n < n_0$ the three numbers W_{n-1}, W_n, W_{n+1} , normalised by λ_1^n , are evaluated from the spectral sum of Lemma 9.4 with a working precision of $1.7m + 320$ bits at the start (the sum has a cancellation of about $1.3m$ bits at $n = m$), and the balls of W_n and of $W_n^2 - W_{n-1}W_{n+1}$ are required to be strictly positive. To save time the precision is lowered and modes k whose term has become negligible are dropped at checkpoints; a dropped mode is bounded for all later n by its absolute value at the time of dropping ($|\lambda_k/\lambda_1| \leq 1$), and the sum of these bounds is added to every later value as an error ball; so the enclosures remain rigorous. (X) For $n \leq 3m + 60$ the numbers W_n are *also* computed exactly — integers for $y = \frac{1}{2}$ (powers of $I + 2 \text{Adj}$), elements of $\mathbb{Z}[\sqrt{2m}]$ resp. $\mathbb{Z}[\sqrt{1 + 2M}]$ at the thresholds — and it is checked that the exact value lies in the ball and that the exact sign of $W_n^2 - W_{n-1}W_{n+1}$ is $+$ (resp. 0 at $n = m$ in the threshold cases, which is how the equality statement is proved). In case (a) positivity of W_n is Lemma 9.4(a). Records: `data/cert_1c1d_*.jsonl`, summary `data/summary_1c1d.json`. $\square$

Size. (a) 992 values of N , $5.2 \cdot 10^7$ certified inequalities, $n_0/N^2 \in [0.140, 0.157]$, smallest relative margin $(W_n^2 - W_{n-1}W_{n+1})/W_n^2 = 2.2 \cdot 10^{-10}$ (at $N = 1001$, $n = n_0 - 1$); 26 CPU-minutes. (b) 988 values, $5.2 \cdot 10^7$ inequalities, 26 minutes. (c), (d) 498 values each, $5.3 \cdot 10^6$ inequalities each.

Computer-assisted Corollary 9.8 (thresholds for $N \leq 501$). *For every $4 \leq N \leq 501$ the sequence δh_q is log-concave iff $q \leq q_1(N)$; for every odd $7 \leq N \leq 1001$, $\delta \tilde{h}_q$ is log-concave iff $q \leq \tilde{q}_1(N)$. Hence, for these N and every $d \geq 1$, the corner-to-corner PMF is unimodal for all $q \leq q_1(N)$ and the corner $\rightarrow$ centre and centre $\rightarrow$ corner PMFs for all $q \leq \tilde{q}_1(N)$.*

Proof. “If”: Theorem 9.7(c),(d) and the monotonicity in q of Proposition 9.1. “Only if”: Lemma 9.4(c). The rest is Proposition 9.1. $\square$

($q_1(N) \rightarrow 1$: $q_1(101) = 0.934$, $q_1(501) = 0.969$. Conjecture 10.4 states that “iff” holds for all N .)

9.3 Twisted groups: the one-dimensional statement for every N

Throughout this subsection $y = 1/2$. For an integer $n \geq 1$ and real β, δ put

$$A_n(\beta, \delta) := (y + 2 \cos((l + \beta)\delta))_{l=0, \dots, n-1}, \quad H_j(\beta, \delta) := h_j(A_n(\beta, \delta)), \quad C_j := H_j^2 - H_{j-1}H_{j+1}.$$

Lemma 9.9 (residue classes). (a) *Let $N \geq 992$, $G := \lfloor N/16 \rfloor$, $\rho := N/G$, $\delta := \pi/\rho$. Then $G \geq 62$, $16 \leq \rho < 16 + 16/G \leq 16 + 16/62$, and the family $(y + 2 \cos(k\pi/N))_{k=1}^{N-1}$ is the disjoint union over $g = 0, \dots, G-1$ of the families $A_{n_g}(\beta_g, \delta)$, where $\beta_g = g/G$ for $g \geq 1$, $\beta_0 = 1$, and $n_g = \lceil \rho - \beta_g \rceil \in \{15, 16, 17\}$. Moreover $n_g = 17$ iff $\beta_g < \rho - 16$ (then $\beta_g < 16/62$), and $n_g = 15$ iff $g = 0$ and $\rho = 16$; in that case the group is $(y + 2 \cos(l\pi/16))_{l=1}^{15}$, the spectrum $\lambda_1, \dots, \lambda_{15}$ of Lemma 9.4(a) for $N = 16$.*

(b) *Let $N = 2M + 1 \geq 1985$ be odd, $G := \lfloor N/32 \rfloor$, $\rho := N/(2G)$, $\delta := \pi/\rho$. Then $G \geq 62$, $16 < \rho < 16 + 16/G$, and $(y + 2 \cos(2k\pi/N))_{k=1}^M$ is the disjoint union over $g = 0, \dots, G-1$ of $A_{n_g}(\beta_g, \delta)$ with the same β_g , $n_g = \lceil \rho - \beta_g \rceil \in \{16, 17\}$, and $n_g = 17$ iff $\beta_g < \rho - 16$.*

(c) *In both cases $\sum_j h_j(\lambda) z^j = \prod_g \sum_j h_j(A_{n_g}(\beta_g, \delta)) z^j$: the sequence $(h_j(\lambda))_j$ is the ordinary convolution of the G sequences $(h_j(A_{n_g}(\beta_g, \delta)))_j$.*

Proof. (a) Split $\{1, \dots, N-1\}$ by the residue g of k modulo G . For $g \geq 1$ the class is $\{g + Gl : l \geq 0, g + Gl \leq N-1\}$ and $(g + Gl)\pi/N = (l + g/G)\delta$; the number of elements is $\lfloor (N-1-g)/G \rfloor + 1 = \lceil (N-g)/G \rceil = \lceil \rho - \beta_g \rceil$. For $g = 0$ the class is $\{G(l+1) : l \geq 0, G(l+1) \leq N-1\}$, the angles are $(l+1)\delta$ and the number of elements is $\lfloor (N-1)/G \rfloor = \lceil \rho - 1 \rceil$. $16G \leq N < 16(G+1)$ gives the bounds on ρ , and $\rho - \beta_g \in [15, 16 + 16/62]$, with value 15 only if $\beta_g = 1$ and $\rho = 16$. (b) The same with $2k\pi/N = (l + g/G) \cdot (2G\pi/N)$; $k \leq M$ is equivalent to the angle being $< \pi$, i.e. to $l + \beta_g < \rho$; $32G \leq N < 32(G+1)$ and N odd give $16 < \rho < 16 + 16/G$. (c) $\prod_k (1 - \lambda_k z)^{-1}$ factorises over the classes. $\square$

Computer-assisted Lemma 9.10 (every twisted group is log-concave). *Let $y = 1/2$ and $\delta \in [0.19322, 0.19635]$ (this interval contains $[\pi/(16 + 16/62), \pi/16]$).*

(F1) *For $n = 16$ and every $\beta \in [0, 1]$: $H_j(\beta, \delta) > 0$ and $C_j(\beta, \delta) > 0$ for all $j \geq 1$.*

(F2) *For $n = 17$ and every $\beta \in [0, 0.2585]$ (note $16/62 < 0.2585$): $H_j(\beta, \delta) > 0$ and $C_j(\beta, \delta) > 0$ for all $j \geq 1$.*

(In (F2) the 17th point may lie beyond the end of the cosine arc: $(16 + \beta)\delta > \pi$ on part of the rectangle. This is harmless — the statement and its proof only need the nodes λ_l to be distinct, which is certified — and Lemma 9.9 uses (F2) only for $\beta < \rho - 16$, where $(16 + \beta)\delta < \pi$.)

Proof. scripts/cert_family.py. The parameter rectangle is covered by boxes $B = \{|\beta - \beta_c| \leq w_1, |\delta - \delta_c| \leq w_2\}$ with $w_1 = 1/20$, $w_2 = 313/400000$ (20 boxes for (F1), 6 for (F2), the latter covering $\beta \in [0, 0.3]$). In the stated ranges all angles $((k+l)/2 + \beta)\delta$ ($0 \leq k < l \leq n-1$) lie in $(0, \pi)$, so $\lambda_k - \lambda_l = 4 \sin(((k+l)/2 + \beta)\delta) \sin((l-k)\delta/2) > 0$: the $\lambda_l = y + 2 \cos((l + \beta)\delta)$ are distinct and decreasing in l , and $H_j = \sum_k c_k \lambda_k^{j+n-1}$ with $c_k = \prod_{l \neq k} (\lambda_k - \lambda_l)^{-1}$, $\text{sgn } c_k = (-1)^k$ (partial fractions).

(T) *Tail, uniformly on a box.* B is covered by 10×4 sub-boxes. On each sub-box, with β, δ entered as balls, the program certifies: all the above differences are > 0 ; $\lambda_1 > 0$; $\lambda_1 > |\lambda_k|$ for $k \geq 2$; and, for an index $j_0 \in \{30, 40, 50, 60\}$ chosen per box,

$$|c_0 c_1| (\lambda_0 - \lambda_1)^2 > \sum_{(k,l) \neq (0,1)} |c_k c_l| \left(\frac{|\lambda_k \lambda_l|}{\lambda_0 \lambda_1} \right)^{j_0+n-2} (\lambda_k - \lambda_l)^2, \quad |c_0| > \sum_{k \geq 1} |c_k| \left(\frac{|\lambda_k|}{\lambda_0} \right)^{j_0+n-1},$$

with the ratios replaced by certified upper bounds. By Lemma 9.5 (with $a(\cdot) = H_{-n+1}$), $C_j > 0$ and $H_j > 0$ for all $j \geq j_0$ and all parameters in B .

(W) *Window $1 \leq j < j_0$: Taylor models.* Write $\beta = \beta_c + r_1 u$, $\delta = \delta_c + r_2 v$ with $r_1 = 1/2$, $r_2 = 313/40000$; B is $\{|u| \leq t, |v| \leq t\}$, $t = 1/10$. H_j and C_j are entire functions of $(u, v) \in \mathbb{C}^2$. For complex $|u|, |v| \leq 1$ the angle $(l + \beta)\delta$ lies in the disc of radius $\varepsilon_l := r_1 \delta_c + (l + \beta_c) r_2 + r_1 r_2$ about $a_l := (l + \beta_c) \delta_c$, and from $\cos(a + \eta) = \cos a \cos \eta - \sin a \sin \eta$, $|\cos \eta - 1| \leq \cosh |\eta| - 1$, $|\sin \eta| \leq \sinh |\eta|$:

$$|\lambda_l| \leq \mu_l := |y + 2 \cos a_l| + 2 |\cos a_l| (\cosh \varepsilon_l - 1) + 2 |\sin a_l| \sinh \varepsilon_l.$$

h_j has nonnegative coefficients, so $|H_j| \leq M_j := h_j(\mu_0, \dots, \mu_{n-1})$ and $|C_j| \leq M_j^2 + M_{j-1} M_{j+1}$ on the closed unit polydisc. By Cauchy's estimates every Taylor coefficient of H_j (of C_j) in (u, v) is bounded by M_j (by $M_j^2 + M_{j-1} M_{j+1}$); so for real $|u|, |v| \leq t$ the Taylor polynomial of total degree p differs from the function by at most the bound times

$$R(t, p) := \sum_{k > p} (k+1) t^k = \frac{(p+2)t^{p+1} - (p+1)t^{p+2}}{(1-t)^2} \quad (= 1.8 \cdot 10^{-14} \text{ for } t = 1/10, p = 14).$$

The Taylor polynomials of total degree $p = 14$ are computed by truncated power-series arithmetic with ball coefficients (320 bits) from the exact centre: first those of λ_l (from the series of $\cos$, $\sin$ in $\eta = r_1 \delta_c u + (l + \beta_c) r_2 v + r_1 r_2 uv$), then those of $H_0, \dots, H_{j_0}$ by the recursion “multiply

the generating function by $(1 - \lambda_l z)^{-1}$, then that of C_j as the truncated product. A polynomial $\sum c_{ab} u^a v^b$ is bounded below on B by $c_{00} - \sum_{(a,b) \neq (0,0)} |c_{ab}| t^{a+b}$. The certificate is that for every box and every $1 \leq j < j_0$ the lower bounds

$$\begin{aligned} c_{00}(H_j) - \sum' |c_{ab}(H_j)| t^{a+b} - M_j R(t, p) &> 0, \\ c_{00}(C_j) - \sum' |c_{ab}(C_j)| t^{a+b} - (M_j^2 + M_{j-1} M_{j+1}) R(t, p) &> 0 \end{aligned}$$

hold as inequalities between balls. They do: over all boxes the lower bound of C_j divided by $H_j(\text{centre})^2$ is at least $1.2 \cdot 10^{-4}$ (the true margin C_j/H_j^2 at the centres is at least $6.3 \cdot 10^{-4}$ for $j < j_0$), $M_j/H_j(\text{centre}) \leq 2.5 \cdot 10^4$, and the lower bound of H_j is at least $0.69 H_j(\text{centre})$. Records: `data/cert_family_F1.jsonl`, `data/cert_family_F2.jsonl`; total running time 19 seconds. $\square$

Cross-check (not part of the proof). A second implementation, written separately within the same project and run on the same machine (code not released: own truncated series arithmetic of total degree 12; Lagrange remainder enclosed over the whole box by evaluating the order-13 coefficients with interval base points, instead of Cauchy estimates; own tail criterion on 12×5 sub-boxes; 256-bit balls), also certifies all 26 boxes.

Why 16. The first inequality $C_1 = e_2(A_n(\beta, \delta)) \geq 0$ fails for some β when $\pi/\delta \leq 14$ (float scan, `explore19_twisted_groups.py`), and holds with a comfortable margin for $\pi/\delta \geq 16$; larger groups have a longer window. As for the full path (Conjecture 10.4), in all cases computed a twisted group is log-concave as soon as $e_2 \geq 0$.

Computer-assisted Theorem 9.11 (one-dimensional log-concavity at $q \leq 4/5$, every N). *Let $q \in (0, 4/5]$.*

- (a) δh_q is a log-concave sequence for every $N \geq 9$.
- (b) $\delta \tilde{h}_q$ is a log-concave sequence for every odd $N \geq 25$.

At $q = 4/5$ these ranges are exact: $\delta h_{4/5}$ is not log-concave for $3 \leq N \leq 8$, and $\delta \tilde{h}_{4/5}$ is not log-concave for odd $3 \leq N \leq 23$.

Proof. By the last statement of Proposition 9.1 it suffices to take $q = 4/5$, $y = 1/2$. By Lemma 9.4 we must show that $(h_j(\lambda))_{j \geq 0}$ [resp. $(h_j(\tilde{\lambda}))_{j \geq 0}$] is positive and log-concave. For $9 \leq N \leq 1001$ [odd $25 \leq N \leq 2001$] this is Theorem 9.7. For $N \geq 992$ [odd $N \geq 1985$], Lemma 9.9 writes the sequence as the convolution of G sequences $(h_j(A_{n_g}(\beta_g, \delta)))_j$ with $\delta = \pi/\rho \in [\pi/(16 + 16/62), \pi/16] \subset [0.19322, 0.19635]$: for $n_g = 16$ it is positive and log-concave by Lemma 9.10 (F1) ($\beta_g \in (0, 1]$), for $n_g = 17$ by (F2) ($\beta_g < 16/62$), and for $n_g = 15$ it is the sequence $(h_j(\lambda_1, \dots, \lambda_{15}))_j$ of $N = 16$, positive and log-concave by Theorem 9.7(a). Lemma 9.2, applied $G - 1$ times, gives the claim. The last sentence is Lemma 9.4(c) ($e_2 < 0$ for $2 \leq m \leq 7$, resp. $2 \leq M \leq 11$); for $N = 3$ in (b), $\tilde{W}_n = (y - 1)^n = (-\frac{1}{2})^n$ is not of one sign. $\square$

Remark (what the theorem says about walks). $(q/2)^{-(n+1)} \delta h_q(n) = W_n$ is the number of walks of length n from one end to the other of a path with $N - 1$ vertices, where a hold at a vertex has weight $y = 2(1 - q)/q$. So (a) says: *for holding weight $y \geq 1/2$ and at least 8 vertices, the weighted number of end-to-end walks of length n is a log-concave function of n .* For $y \geq 2 \cos(\pi/N)$ this is classical (J is then totally nonnegative and W_n is a Pólya frequency sequence); for $y < 2 \cos(\pi/N)$ the matrix has negative eigenvalues and the statement is a collective property of the spectrum.

9.4 Small boxes: blocks of dimensions

For the finitely many N not covered by Theorem 9.11 the one-dimensional factor is not log-concave at $q = 4/5$, but its binomial convolutions are.

Computer-assisted Proposition 9.12 (blocks). *Let $q = 4/5$. The sequence φ_d is log-concave in the following cases:*

- *corner* $\rightarrow$ *opposite corner*: $3 \leq N \leq 8$, $d \in \{2, 3\}$;
- *corner* $\rightarrow$ *centre* (equivalently *centre* $\rightarrow$ *corner*): *odd* $9 \leq N \leq 23$, $d \in \{2, 3\}$; $N \in \{5, 7\}$, $d \in \{3, 4, 5\}$; $N = 3$, $d \in \{4, 5, 6, 7\}$.

It is not log-concave for the centre geometries with $(N, d) \in \{(3, 2), (5, 2), (7, 2), (3, 3)\}$ (yet the PMF is unimodal there: Theorem 7.1(e)).

Proof. `scripts/cert_blocks.py`. By Lemma 1.1, $\varphi_d(t+1) = \sum_K C_K \nu_K^t$ over multisets $K = \{k_1 \leq \dots \leq k_d\}$ of one-dimensional mode indices, with $\nu_K = 1 - q + \frac{q}{d} \sum_i \cos(k_i \pi / N)$ and $C_K = \text{mult}(K) \prod_i w_{k_i} (\nu_K - 1)$, $w_k = \psi_k(x_{0,1}) \psi_k(a_1)$ ($= (-1)^k c_k^2 \cos^2(k\pi/2N)$ for the corner pair; $= (-1)^{k/2} c_k^2 \cos(k\pi/2N)$ for even k and 0 for odd k for the centre pair). The two leading modes are $K_1 = \{0, \dots, 0, k^*\}$ and $K_2 = \{0, \dots, 0, k^*, k^*\}$ ($k^* = 1$ resp. 2) with $C_{K_1} > 0 > C_{K_2}$; the program certifies $\nu_{K_1} > \nu_{K_2} > |\nu_K|$ for all other K , finds t_0 as in Lemma 9.5, and certifies positivity and the log-concavity inequalities on the window $D-1 \leq t < t_0$ both in ball arithmetic and in exact integer arithmetic (the exact value of $\varphi_d(t)$ $(10d)^t$ from the walk on the box). 38 cases, $t_0 \leq 231$, smallest relative margin $6.5 \cdot 10^{-5}$. The failures are exact integer computations (`verify_lcid.py` X9): the first inequality $\varphi(D+1)^2 \geq \varphi(D)\varphi(D+2)$ fails. Records: `data/cert_blocks.jsonl`. $\square$

9.5 The result

Computer-assisted Theorem 9.13 (the value). *Let $q \in (0, 4/5]$.*

- (i) *Corner $\rightarrow$ opposite corner. The sequence φ_d is log-concave for all $N \geq 9$, $d \geq 1$, and for all $2 \leq N \leq 8$, $d \geq 2$. The first-passage PMF f_q is unimodal for **every** $N \geq 2$ **and every** $d \geq 1$.*
- (ii) *Corner $\rightarrow$ centre and centre $\rightarrow$ corner (N odd). The sequence φ_d is log-concave for all $N \geq 25$, $d \geq 1$; for $9 \leq N \leq 23$, $d \geq 2$; for $N \in \{5, 7\}$, $d \geq 3$; and for $N = 3$, $d \geq 4$. The first-passage PMF f_q of both pairs is unimodal for **every odd** $N \geq 3$ **and every** $d \geq 2$.*
- (iii) *Sharpness. For $d = 1$, $N = 4$ the corner-to-corner PMF is not unimodal for any $q > 4/5$ (Proposition 4.4). So $(0, 4/5]$ is exactly the set of activities for which the corner-to-corner PMF is unimodal for all $N \geq 2$ and all $d \geq 1$.*

Proof. (i) $N \geq 9$: Theorem 9.11(a) and Proposition 9.1. $N = 2$: $q_2 = 1$, Theorem 4.12 (the one-dimensional eigenvalue $1 - q$ is ≥ 0 , so δh_q is log-concave and Proposition 9.1 applies). $3 \leq N \leq 8$, $d \geq 2$: at $q = 4/5$, φ_2 and φ_3 are log-concave (Proposition 9.12), so u_2 is (Lemma 9.3(a)), and by Lemma 9.3(b) and induction φ_{2+2k} and φ_{3+2k} are log-concave for all $k \geq 0$; Lemma 9.3(c) extends this to $q \leq 4/5$, and Theorem 4.11(b) gives unimodality. $3 \leq N \leq 8$, $d = 1$: Theorem 7.1(f) and Corollary 2.6.

(ii) $N \geq 25$: Theorem 9.11(b) and Proposition 9.1. $9 \leq N \leq 23$: as in (i) from φ_2, φ_3 . $N \in \{5, 7\}$: $\varphi_3, \varphi_4, \varphi_5$ are log-concave, hence u_3 is, hence φ_{d+3} is whenever φ_d is: all $d \geq 3$. $N = 3$: $\varphi_4, \dots, \varphi_7$ and u_4 are log-concave, hence all φ_d , $d \geq 4$. The remaining cases $N = 3$, $d \in \{2, 3\}$ and $N \in \{5, 7\}$, $d = 2$ are in Theorem 7.1(e) ($q \leq 4/5$). In all log-concave cases Lemma 9.3(c) and Theorem 4.11(b) give unimodality for $q \leq 4/5$.

(iii) Proposition 4.4 and (i). $\square$

Remarks. (1) *What is computer-assisted.* Theorem 9.13 rests on: the finite certificates of Theorem 9.7(a),(b) ($N \leq 1001$, resp. 2001); Lemma 9.10 (26 boxes); Proposition 9.12 (38 small cases); and, for the listed small cases only, the exact cone certificates of Theorem 7.1(e),(f). Everything else — the last-exit decomposition, the reduction to one dimension, the convolution lemmas, the tail lemma, the splitting into residue classes — is proved in the text. (2) *One dimension, centre geometries.* For $d = 1$, corner $\rightarrow$ centre on $\{1, \dots, N\}$ is the end-to-end problem on $\{1, \dots, c\}$ (the walk cannot pass c before hitting it), so it is unimodal for $q \leq 4/5$

by (i). Centre $\rightarrow$ corner in $d = 1$ is unimodal for $q \leq 4/5$ when $N \geq 25$ (Theorem 9.11(b), Proposition 9.1), when $N = 3$ (Proposition 7.3(a)) and when $13 \leq N \leq 23$ (Theorem 7.4(b)), and for $q \leq 1/2$ for every N (Theorem 4.3). It is *not* unimodal at $q = 4/5$ for $N \in \{5, 7, 9, 11\}$ (Proposition 7.3, Theorem 7.4(c)). So the hypothesis $d \geq 2$ in (ii) cannot be dropped, and in $d = 1$ the statement “unimodal for all $q \leq 4/5$ ” holds for centre $\rightarrow$ corner exactly when $N \notin \{5, 7, 9, 11\}$. (3) *Even* N has no centre site; the article uses odd N for the centre geometries. (4) *Beyond* $4/5$. In $d \geq 2$ the PMF stays unimodal well beyond $q = 4/5$ (Theorem 7.1, Corollary 9.8); in $d = 1$ the thresholds are known exactly on a finite range (Theorem 7.4); see Section 10.2 for what is open. (5) *Log-concavity of φ is sufficient, not necessary*: the four exceptional centre cases of Proposition 9.12 are unimodal without it.

Sanity checks. `scripts/verify_lc1d.py` $\rightarrow$ `data/verify_lc1d.json`: X1–X2 (Lemma 9.4, exact: 1771 identities between the one-dimensional chain and the Jacobi-matrix recursion), X3 (spectral formulas: exact values lie in the Arb balls), X4 (identity (16) and the variance form of Tot_n), X5 (brute-force exact scan $n \leq 40m^2$, $m = 3, \dots, 14$, four values of y , both families: log-concave iff the first inequality holds; exactly one equality at the threshold), X6 (the elementary bounds and the tail time of Proposition 9.6 for $N = 10, \dots, 400$), X7 (Proposition 9.1 in action, exact, $d = 2, 3$), X8 ($d = 5$, $N = 9$, float), X9 (the four exceptional cases of Proposition 9.12, exact), X10 (the three-number identity of Section 10.2, exact), X11 (the block identity in the proof of Lemma 9.3(b), exact). `scripts/verify_family.py` $\rightarrow$ `data/verify_family.json`: Y1 (Lemma 9.9 for 21 values of N up to 100003, exact: the classes partition the index set, the group sizes are $\lceil \rho - \beta \rceil$, all parameters lie in the certified regions), Y2 (product identity, 60 digits), Y3 (Taylor models: at random real points the error is below the remainder bound — observed error/bound $\leq 3.3 \cdot 10^{-14}$; at random complex points $|H_j| \leq M_j$), Y4 (negative controls: the procedure refuses two groups with $e_2 < 0$), Y5 (separate brute-force evaluation at 400 random parameter points, $j \leq 600$), Y6 (direct certificates of the type of Theorem 9.7 for $N = 1024, 1500, 2000, 4001$ and odd $N = 2049, 3001, 4001$, beyond the range used in the proof: all certified, in agreement with Theorem 9.11), Y7 (for $N = 1237, 5003, 40009$ and odd $N = 2003, 30011$ *every* residue-class group — 3888 groups — is certified directly at its own parameter point, without Taylor models: window $j < 100$ and the tail criterion at $j_0 = 100$). `scripts/summarize_lc1d.py` checks that all ranges are complete.

10 What remains open: precise conjectures and the exact gap

After Theorems 4.12 and 9.13 the unimodality question for the three geometries is settled in continuous time and in discrete time for $q \leq 4/5$, for all N (and all $d \geq 2$; all $d \geq 1$ for the corner pair; in $d = 1$ the centre $\rightarrow$ corner pair fails at $q = 4/5$ exactly for $N \in \{5, 7, 9, 11\}$, Theorem 7.4(c)). What remains open: the stronger shape properties — log-concavity, monotone hazard rate, the front property — of the corner-to-corner law (Section 10.1), and discrete time with $4/5 < q < 1$ beyond the certified finite ranges (Section 10.2).

10.1 Log-concavity and the front property; thresholds; general pairs

Conjecture 10.1 (corner-to-corner, all N). *Let $d \geq 2$, $N \geq 2$, $a = (N, \dots, N)$, $x_0 = (1, \dots, 1)$. In continuous time, and in discrete time for $q \leq 1/2$, property (P1') of Proposition 6.3 holds: for all t , $\rho_{t+1}(x)/\rho_t(x)$ (resp. $\partial_t \log \rho_t(x)$) is coordinatewise nondecreasing in x . Consequently (Proposition 6.3) the first-passage law is log-concave — hence has a nondecreasing hazard rate — and the cone of Lemma 6.1 holds from the mode on.*

Unimodality itself is not part of the conjecture (Theorems 4.12, 9.13). The conjecture is strictly stronger: the centre $\rightarrow$ corner law is unimodal but not log-concave (Proposition 8.4), so log-concavity of the first-passage law cannot follow from the last-exit argument alone. (Do not confuse the log-concavity of the first-passage PMF f , conjectured here, with that of the increment φ of the free propagator, proved in Sections 4.4 and 9.)

Evidence (exact integer arithmetic on finite windows, `scripts/check_conjectures.py`): (P1'), log-concavity and monotone hazard hold on $0 \leq t \leq 6 \cdot \text{mode}$ for $d = 2$, $N = 2, \dots, 14, 16, 18, 20$ and $d = 3$, $N = 2, \dots, 8$ at $q = 1/2$, and for $d = 2$, $N \leq 8$ at $q = 1/8$; in floating point for $d = 2$, $N \leq 24$ and $d = 3$, $N \leq 10$.

The obstruction. In $d = 1$, (P1') is what the variation-diminishing property of the heat flow on a path gives (Theorem 4.3, Proposition 6.3). In $d \geq 2$ the natural sufficient condition — stochastic monotonicity of the time-reversed chain with rates $\rho_t(y)/\rho_t(x)$, implied by log-supermodularity of ρ_t — fails near the target, because the killed site is the top of the lattice: $\rho(a)\rho(a - e_i - e_j) = 0 < \rho(a - e_i)\rho(a - e_j)$.

Conjecture 10.2 (thresholds). (i) For $d = 2$ corner-to-corner, $\inf_N q^*(N) = q^*(9) \in [0.9986, 0.9987)$, and $q^*(N) = 1$ for infinitely many N and < 1 for infinitely many N .

(ii) For $d \geq 3$ corner-to-corner, $q^*(N) = 1$ for all N .

(iii) For $d = 1$ end to end, $q^*(N) = (2N - 4)/(2N - 3)$ for all $N \geq 22$.

(iv) For $d = 1$ centre $\rightarrow$ corner, $q^*(N) = (N - 3)/(N - 1)$ for all odd $N \geq 21$.

Remark. The natural extension of (iii) to $N \geq 20$ is false: $N = 21$ is an exception ($q^*(21) \in [0.974226, 0.974227)$, while $38/39 = 0.974359 \dots$; exact witness $f(142) > f(143) < f(144)$ at $q = 38/39$; Theorem 7.4(a)). The true statement on the range that has been certified is Theorem 7.4.

Status. Proved, for the corner pair: $q^*(N) \geq 4/5$ for all N and d , with equality for $d = 1$, $N = 4$ (Theorem 9.13); $q^*(N) \geq q_1(N)$ for $N \leq 501$ and all d (Corollary 9.8); $q^*(2) = 1$ in every dimension; the finite ranges of Theorem 7.1. In $d = 1$: the inequalities $q^*(N) \leq (2N - 4)/(2N - 3)$ ($N \geq 4$, Proposition 4.4) and $q^*(N) \leq (N - 3)/(N - 1)$ (centre $\rightarrow$ corner, odd $N \geq 7$, Proposition 7.3) hold for all N ; equality — clauses (iii), (iv) — is proved for $N = 20$, $22 \leq N \leq 300$ resp. odd $21 \leq N \leq 301$ (and $N = 4$; resp. $N = 7, 9, 17$), and is known to fail for $5 \leq N \leq 19$, $N = 21$ resp. $N = 11, 13, 15, 19$ (Theorem 7.4). For $N > 300$ resp. 301 clauses (iii), (iv) are open: what is missing is a proof, for all large N , that the PMF at $q = (2N - 4)/(2N - 3)$ (resp. $(N - 3)/(N - 1)$), which has an exact tie at the first two points of its support, has no second dip. Remark 7.5 explains heuristically why none is expected at the mode; the log-concavity method of Section 9 cannot give it, because at these values of q the increment of the free propagator is not log-concave ($(2N - 4)/(2N - 3) > q_1(N)$ for $N \geq 4$).

Conjecture 10.3 (general pairs). In continuous time, for every d , N and pair, g' has at most three sign changes (the density has at most two local maxima). If $x_0 \sim a$ the density is decreasing.

Evidence. The exhaustive scans of Theorem 8.2 and Observation 4.2. A separately written check within the same project, on the same machine (not released): for 268 adjacent pairs at $q = 1/32$ the PMF is nonincreasing, and no pair of the scans has more than three sign changes.

10.2 Discrete time with $4/5 < q < 1$

What is known. By Corollary 2.6 the set of q for which f_q is unimodal is an interval $(0, q^*]$. By Theorem 9.13, $q^* \geq 4/5$ for the three geometries (all N ; $d \geq 2$, and $d \geq 1$ for the corner pair). $q^* < 1$ does occur ($d = 1$, $N \geq 4$: Proposition 4.4, with the exact values of Theorem 7.4; $d = 2$, $N = 9, 11, \dots$: Theorem 7.1(c)), so some restriction on q is necessary, and $q^* = 4/5$ exactly for $d = 1$, $N = 4$. Beyond $4/5$: $q^* \geq 0.998$ ($d = 2$, $N \leq 64$), $q^* = 1$ ($d = 3$, $N \leq 30$; $d = 4$, $N \leq 8$; $d = 5$, $N \leq 5$) by Theorem 7.1, and $q^* \geq q_1(N)$ resp. $\tilde{q}_1(N)$ for $N \leq 501$ resp. odd $N \leq 1001$ in every dimension by Corollary 9.8.

Conjecture 10.4 (one-dimensional log-concavity up to the first inequality). (a) For every $N \geq 3$ the sequence δh_q (end to end) is log-concave if and only if $q \leq q_1(N) = 1/(1 + (2(N - 1))^{-1/2})$ (equivalently $2(N - 1)(1 - q)^2 \geq q^2$).

(b) For every odd $N \geq 5$ the sequence $\delta\tilde{h}_q$ (end to centre) is log-concave if and only if $q \leq \tilde{q}_1(N)$.

Status. “Only if” is proved (Lemma 9.4(c)). “If” is proved for $N \leq 501$ in (a) and odd $N \leq 1001$ in (b) (Corollary 9.8), and for every N when $q \leq 4/5$ (Theorem 9.11). Open: $N > 501$ (resp. 1001) and $4/5 < q \leq q_1(N)$ (resp. $\tilde{q}_1(N)$). By Proposition 9.1 the conjecture would give unimodality of the corner-to-corner PMF for all $q \leq q_1(N)$ in every dimension, with $q_1(N) \rightarrow 1$.

Why the method of Section 9.3 does not prove it. For a fixed y the splitting into twisted groups works for all large N at once, with groups of a fixed size (16 for $y = \frac{1}{2}$). At the threshold $y = (2/(N-1))^{1/2}$ the whole spectrum is exactly critical ($e_2 = 0$), so no proper sub-group can be log-concave with room to spare, and the group size needed for a fixed y grows like y^{-2} . A proof for all N at the threshold needs a different idea.

Equivalent forms and evidence. With $m = N - 1$: W_n counts walks of length n from one end to the other of the path with m vertices, with holding weight y . Log-concavity of $s_j := h_j(\lambda)$ means that all two-row Schur functions $s_{(b,a)}(\lambda) = h_a h_b - h_{a-1} h_{b+1}$ ($b \geq a \geq 1$; Jacobi–Trudi) are nonnegative, and the conjecture says that this is so as soon as $s_{(1,1)} = e_2 \geq 0$. For arbitrary real λ with $e_2 \geq 0$ this is false (`scripts/explore14_hj_lc.py`, `explore17_general_pairs.py`); it is true for three real numbers of which one is negative (then $h_j(a_1, a_2, -b)^2 - h_{j-1} h_{j+1} = (-a_1 a_2 b)^j h_j(1/a_1, 1/a_2, -1/b)$, an alternating sum with decreasing terms when $1/b \geq 1/a_1 + 1/a_2$), and in all cases computed for the spectrum of a path and for twisted groups. For $m \leq 8$ and $j \leq 9$ the polynomial $s_j^2 - s_{j-1} s_{j+1}$, written in the variable $Y = y^2 - 2/m$, has nonnegative coefficients (`explore15_delta_poly.py`). In the formal scaling limit $m \rightarrow \infty$ with $\theta = y(m/2)^{1/2}$ fixed, s_j becomes the coefficient of z^j in $\exp(\theta z + z^2/2)$, and these coefficients are log-concave when $\theta \geq 1$ (E. A. Bender and E. R. Canfield, *Log-concavity and related properties of the cycle index polynomials*, J. Combin. Theory Ser. A 74 (1996) 57–70: the coefficients of $\exp(\sum_j w_j z^j/j)$ are log-concave when $1, w_1, w_2, \dots$ is; here $w = (\theta, 1, 0, \dots)$) and fail the first inequality when $\theta < 1$. The limit is a heuristic, not a proof; for finite m the hypothesis of that theorem fails ($w_j = \text{tr } J^j$ and $w_3^2 < w_2 w_4$).

Numerical observation 10.5 (in $d \geq 2$ the range is larger still). For $d = 2$, $2 \leq N \leq 20$ and $N = 24$, and $d = 3$, $2 \leq N \leq 10$, the sequence φ of the corner pair is log-concave on $0 \leq t \leq 5dN^2 + 40$ also for $q = 9/10$, except for $d = 2$, $N \in \{4, \dots, 8\}$ where the first inequality fails (`scripts/check_lc_threshold.py`, exact). So the binomial convolution (13) smooths, and in $d \geq 2$ log-concavity of φ extends beyond $q_1(N)$.

The exact gap in discrete time. By (8), $f_q(t+1) - f_q(t) = \varphi(t+1) - \sum_{k \leq t} \nu(k) \varphi(t-k)$ with ν the first-return law of the target and φ the increment of the free propagator. What is missing for $4/5 < q < 1$ is a proof, for all N , that this expression changes sign once; log-concavity of φ suffices, and for the corner pair it suffices to prove Conjecture 10.4(a). Any proof must use $q < 1$ quantitatively, because the statement is false for $q = 1$ ($d = 1, 2$). For every $q \leq 1$ Theorem 5.6 confines a possible failure to $t \leq T_N(q)$.

Not attempted here. Asymptotics (location of the mode, shape of the law) are not treated in this note; Corollary 4.13 provides rigorous two-sided bounds for the mode that an asymptotic analysis can start from (m_u and $m(q/M_{\max})$ are defined by the one-dimensional propagator and the mean first-passage time only).

A Files and reproduction

Interpreter: Python 3 with numpy, scipy, mpmath and python-flint 0.9.0 for Arb (`requirements.txt`). In the repository `scripts/` is `code/msc_rigorous_unimodality/` and `data/` is `data/msc_rigorous_unimodality/`.

scripts/unilib.py	exact integer stepper
scripts/cert_unimodal.py	exact cone certificate → data/cert_*.jsonl
scripts/cert_continuous.py	exact three-point certificates (continuous time)
scripts/cert_pairs.py	exhaustive classification → data/pairs_d*_N*.json
scripts/summarize_certs.py	→ data/summary_certificates.json
scripts/verify_lemmas.py	sanity checks V1–V13 of every lemma
scripts/verify_certificate_code.py	cross-checks A–D of the certificate code (separately written implementations)
scripts/check_conjectures.py	exact evidence for Conjecture 10.1
scripts/verify_lastexit.py	sanity checks W1–W10 of Section 4.4 → data/verify_lastexit.json, data/pairs_logconcave_hypothesis.json
scripts/table_mode_bounds.py	table of Corollary 4.13 → data/table_mode_bounds.json
scripts/check_lc_threshold.py	exact window checks behind Observations 4.15(ii), 10.5 → data/check_lc_threshold.json
scripts/cert_lc1d.py	Theorem 9.7: Arb certificate of the one-dimensional log-concavity (tail lemma, window, exact cross-check) → data/cert_lc1d_*.jsonl
scripts/cert_family.py	Lemma 9.10: Taylor-model certificate for the twisted groups → data/cert_family_F1.jsonl, _F2.jsonl
scripts/cert_blocks.py	Proposition 9.12 (Arb and exact integers) → data/cert_blocks.jsonl
scripts/summarize_lc1d.py	completeness of all ranges of Section 9 → data/summary_lc1d.json
scripts/verify_lc1d.py, verify_family.py	sanity checks X1–X11, Y1–Y7 of Section 9
scripts/recheck_headline.py	from-scratch exact re-check of headline claims of Sections 4, 7, 8
scripts/cert_thr1d.py	Theorem 7.4: exact one-dimensional cone certificates (two engines; $\mathbb{Z}[\sqrt{2}]$ for $q^*(5)$) → data/cert_thr1d_*.jsonl, cert_thr1d_m2c.json
scripts/scan_pairs1d.py	remark after Proposition 7.3: all starts, $d = 1$, $N \leq 12$, $q = 4/5$ → data/scan_pairs1d_q4-5.json
scripts/verify_thr1d.py	sanity checks Z1–Z11 of Section 7.1 → data/verify_thr1d.json
scripts/table_mode_bounds_small.py	Corollary 4.13 for small boxes and $d = 1$ (float) → data/table_mode_bounds_small.json
scripts/exact_examples.py	exact rationals quoted in the text
scripts/residue_signs.py, table_windows.py	Observation 3.3; table of Section 5

Sanity checks of the lemmas (verify_lemmas.py; all pass): V1 identity (3) in exact arithmetic; V2 Lemmas 2.1, 2.3, 2.4 on random inputs; V3 (4) against the spectral formula; V4 Theorem 3.2; V5 Lemma 5.1 (exact, including $q = 1$); V6 Lemmas 5.2–5.5; V7 Theorem 5.6; V8 Lemmas 5.8, 5.9 and Theorem 5.10; V9 Theorem 4.1; V10 Theorem 4.3 and Proposition 4.4; V11 Lemma 6.1; V12 Conjecture 10.1 on windows; V13 Proposition 4.6. The checks W1–W10 of Section 4.4 are in verify_lastexit.py; the checks X1–X11 and Y1–Y7 of Section 9 are in verify_lc1d.py and verify_family.py (all pass).

Commands for Section 9 (about 1.2 CPU-hours in total): cert_lc1d.py E2E 1/2 9 1000; cert_lc1d.py FOLD 1/2 13 1000; cert_lc1d.py E2E thr 3 500; cert_lc1d.py FOLD thr 3 500; cert_family.py F1; cert_family.py F2; cert_blocks.py; summarize_lc1d.py; verify_lc1d.py; verify_family.py.

Commands for Section 7.1 (about 2.9 hours of wall-clock time in total, run in several background chunks): cert_thr1d.py boundf 4 300; cert_thr1d.py bound 4 200; cert_thr1d.py bisect 5 6 ... 19 21; cert_thr1d.py m2c; cert_thr1d.py m2cbound 7

301; `verify_thr1d.py`.

B Citations

Theorem numbers: Levin–Peres–Wilmer, 2nd ed., Proposition 10.7 (commute time identity) and Theorem 9.12 (Rayleigh’s monotonicity law) were checked against the authors’ online copy of the second edition. The Horn–Johnson numbers were not checked against the printed edition; Lemma 3.1 does not depend on them (it has a self-contained proof), and the remaining use (the Courant–Fischer min–max theorem, cited by name) is a textbook fact for symmetric matrices.